AGR 201 - Crop Production Technology-I (Kharif Crops)

Lecture-wise Theory Study Material

B.Sc. (Hons) Agriculture - Third Semester - 2025 Syllabus

Tamil Nadu Agricultural University (TNAU)

This material covers all 17 lectures of the AGR 201 syllabus in order. Each crop lecture follows one fixed structure: origin and distribution, economic importance and nutritional value, soil and climatic requirement, varieties, land preparation, seed rate and spacing, nutrient management, water management, weed management, pest and disease management, harvest and yield, and economics. Every lecture carries at least one table, one worked example or one text diagram, and closes with an Exam focus section. Figures are given as the standard textbook ranges used by TNAU and ICAR; where a value is uncertain the range is stated rather than a false precision.

Contents

Lecture 1. Rice - origin, geographic distribution, economic importance, varieties, soil and climatic requirement.

Lecture 2. Rice - cultural practices, yield and economic benefits.

Lecture 3. Special types of rice cultivation - SRI and hybrid rice.

Lecture 4. Maize - origin, distribution, importance, soil and climate, varieties, cultural practices and yield.

Lecture 5. Sorghum and pearl millet - full package.

Lecture 6. Finger millet and minor millets - full package.

Lecture 7. Pigeon pea - full package.

Lecture 8. Green gram, black gram and cowpea; agronomy of rice fallow pulses.

Lecture 9. Mid-semester examination - revision and model paper.

Lecture 10. Groundnut - full package and economics.

Lecture 11. Sesame and soybean - full package.

Lecture 12. Castor - full package and economics.

Lecture 13. Cotton - full package and yield.

Lecture 14. Jute - full package and yield.

Lecture 15. Fodder sorghum and fodder cowpea.

Lecture 16. Fodder maize and Cenchrus (Kollukattai pull).

Lecture 17. Cumbu Napier and Guinea grass; rice ecosystems, climate, weather, seasons and varieties of Tamil Nadu.

Appendix I. Glossary of terms.

Appendix II. Useful formulas.

Appendix III. Abbreviations.

Appendix IV. Consolidated package of practices (all crops).

Appendix V. Model end-semester question paper.

References.

How to use this material

The kharif (monsoon) season runs from June to October and covers the rain-fed and irrigated crops sown with the south-west monsoon. This course treats the major kharif cereals, millets, pulses, oilseeds, fibre crops and fodder crops of Tamil Nadu, with a strong TNAU and Cauvery-delta flavour.

The fixed structure of every crop lecture

Each crop is presented in the same order so that revision is mechanical. The sequence answers, in order, the questions an examiner asks: where the crop came from, why it is grown, where it grows, which variety to sow, how to prepare the field, how much seed and at what spacing, what fertiliser, how to water, how to control weeds, how to manage pests and diseases, when to harvest, and what the farmer earns.

How to use the tables, worked examples and diagrams

- Package-of-practices tables compress a whole crop into one screen; memorise the numbers (seed rate, spacing, NPK dose, duration, yield) first, then the reasons.
- Worked examples show the arithmetic that appears as 5- and 10-mark numerical questions - seed rate and plant population, fertiliser for a given area, cost of cultivation, gross and net return, yield gap, and SRI population.
- Text diagrams (box art) show relationships that are hard to write in words - ecosystems, cropping systems, SRI layout, growth stages.

A note on figures

Fertiliser doses are quoted as N : P2O5 : K2O in kg per hectare. Yields are quoted as t/ha (tonnes per hectare). Variety durations are the standard TNAU figures in days. Where sources differ slightly, the material gives the range (for example, 110-120 days) and says so. Rates of pesticides, where mentioned, are indicative only; the current TNAU Plant Protection Guide and the label must be followed in the field.

Exam focus at the end of each lecture

Every lecture ends with an Exam focus block: 5-8 key points worth remembering, three short-answer questions with model answers, and two long-answer questions with model-answer outlines. Treat these as the minimum you must be able to reproduce.

Course map: the seventeen lectures

No.	Topic	Key deliverable in this material
1	Rice - origin, distribution, importance, varieties, soil and climate	Package of practices; variety table
2	Rice - cultural practices, yield, economics	Season table; cost of cultivation worked example
3	Special methods - SRI and hybrid rice	SRI layout; hybrid seed production; comparison
4	Maize - full package	Growth-stage diagram; fertiliser worked example
5	Sorghum and pearl millet	Comparison table; seed-rate worked example
6	Finger millet and minor millets	Millet comparison table; cost worked example

No.	Topic	Key deliverable in this material
7	Pigeon pea	Pulse agronomy; population worked example
8	Green gram, black gram, cowpea; rice fallow pulses	Pulse comparison; rice-fallow net return
9	Mid-semester examination	Revision tables; model paper
10	Groundnut	Pegging diagram; economics worked example
11	Sesame and soybean	Oilseed comparison; fertiliser worked example
12	Castor	Value chain; economics worked example
13	Cotton	Growth-stage diagram; seed-rate worked example
14	Jute	Retting; fibre yield worked example
15	Fodder sorghum and fodder cowpea	Fodder systems diagram; herd-area worked example
16	Fodder maize and Cenchrus	Pasture diagram; silage-area worked example
17	Cumbu Napier and Guinea grass; rice ecosystems, climate, seasons	Ecosystem diagram; season table

Learning outcomes

- Identify each crop by its botanical name, family, chromosome number and centre of origin.
- State the economic importance and nutritional value of each crop with standard figures.
- Describe the soil and climatic requirement and match the crop to the right season and ecosystem.
- Name the important TNAU and ICAR varieties of each crop with duration and salient features.
- Write the full package of practices - land preparation, seed rate, spacing, NPK, water, weeds, pests and diseases, and harvest.
- Solve numerical problems on seed rate, plant population, fertiliser quantity, cost of cultivation, net return and yield gap.
- Explain the important cropping systems and intercrops of Tamil Nadu, with emphasis on the Cauvery delta.
- Read a package-of-practices table and a growth-stage diagram and use them to answer examination questions.

How marks are usually distributed

- A typical paper carries short answers (2-5 marks), short notes (5 marks) and long answers (10-15 marks), with a compulsory numerical.
- Long answers are most often asked on the package of practices, origin and importance, or a comparison of two crops.
- Numerical questions are almost always drawn from seed rate, plant population, fertiliser quantity, cost of cultivation and yield gap.

How to answer examination questions

- Begin every crop answer with the botanical name, family and chromosome number - it is easy marks and sets the answer on a firm footing.
- Use the fixed structure of this material as the skeleton of your answer: origin, importance, soil and climate, varieties, cultural practices, harvest and yield.

- For a "package of practices" question, write the answer in the same order as the tables - land preparation, seed rate, spacing, NPK, water, weeds, pest and disease, harvest - and add a small table or flow diagram if time allows.
- For numerical questions, always write the formula first, then substitute the values, then give the unit. Show the working; a right answer with no working gets fewer marks.
- For comparison questions, use a two-column table; it is faster to write and easier for the examiner to mark.
- Quote figures with units (kg/ha, t/ha, days, per cent) and use the standard textbook ranges rather than false precision.
- In the Exam focus section of each lecture, the "key points" are the minimum you must reproduce; the short-answer and long-answer questions are the likely questions themselves.

Syllabus unit-wise weightage (typical)

Unit	Lectures	Approximate share
Cereals and millets (rice, maize, sorghum, pearl millet, millets)	1-6	about 45 per cent
Pulses	7-8	about 15 per cent
Oilseeds (groundnut, sesame, soybean, castor)	10-12	about 20 per cent
Fibre crops (cotton, jute)	13-14	about 10 per cent
Fodder crops and ecosystems	15-17	about 10 per cent

Lecture 1 - Rice: Origin, Geographic Distribution, Economic Importance, Varieties, Soil and Climatic Requirement

1.1 Origin and geographic distribution

Rice is the most important food crop of Tamil Nadu and of the Cauvery delta. This lecture fixes the botanical identity, the origin, the areas of production, the nutritional value, the varieties released for Tamil Nadu, and the soil and climate in which the crop performs best.

Botanical identity

- Common name: Rice (English), Nellu (Tamil), Paddy.
- Botanical name: Oryza sativa L. Family: Poaceae (Gramineae). Chromosome number 2n = 24.
- Rice is a semi-aquatic annual grass. The cultivated species are O. sativa (Asia) and O. glaberrima (West Africa). Wild relatives include O. rufipogon and O. nivara.
- Two ecological groups of O. sativa are recognised:
- Indica group - long, slender grains, grown in the tropics and subtropics (India, South-East Asia). It is the group grown in Tamil Nadu.
- Japonica group - short, round, sticky grains, grown in temperate East Asia (China, Japan, Korea).
- Javanica is a third, intermediate group.

Origin

- Rice was domesticated independently in two centres. The indica type originated in the Indo-Burma region (South-East Asia), and the japonica type in China. Vavilov placed the centre of origin of rice in the Indian (Hindustan) and Chinese centres of diversity.

- Archaeological evidence from the Ganga valley and from South India shows rice cultivation in India for several thousand years. The crop is thus native to the Indian sub-continent and to East Asia.

Geographic distribution

- Rice is grown on about 165 million hectares worldwide (roughly 1.6-1.7 million km2), and is the second most important cereal after wheat by harvested area but the most important by number of people fed.
- Leading producers: China, India, Indonesia, Bangladesh, Vietnam, Thailand, Myanmar, the Philippines, Japan. India is the second largest producer and the largest exporter of rice in the world.
- In India the crop is grown in West Bengal, Uttar Pradesh, Punjab, Andhra Pradesh, Telangana, Tamil Nadu, Bihar, Chhattisgarh, Odisha and Karnataka.
- In Tamil Nadu rice is grown in all districts. The main tracts are the Cauvery delta (Thanjavur, Thiruvarur, Nagapattinam, Mayiladuthurai), the Samba belt of the delta, the rice fallow areas of the delta, and the tank-fed and canal-fed tracts of Ramanathapuram, Sivaganga, Madurai, Tirunelveli, Coimbatore and Chengalpattu.

In Tamil Nadu, rice occupies the largest single share of the cropped area and is the pivot of the Cauvery delta cropping system - rice followed by rice followed by pulses, or rice followed by a short-duration pulse on residual moisture.

1.2 Economic importance and nutritional value

Economic importance

- Rice is the staple food of more than half of the world's population and of a very large majority of the people of Tamil Nadu. It supplies the bulk of dietary energy for the State.
- It is a source of rural employment in transplanting, weeding, harvesting, threshing, milling and marketing, and supports a large milling, parboiling and export industry.
- Rice straw and husk are important by-products. Straw is used as cattle fodder and bedding; husk is used as fuel, in board making and as a soil amendment after charring.
- India earns substantial foreign exchange from export of basmati and non-basmati rice. Tamil Nadu supplies fine and medium slender varieties to the domestic market.

Nutritional value

- Milled (polished) rice contains about 78-80 per cent carbohydrate, 6-7 per cent protein, about 0.5-1 per cent fat, and about 0.4-0.6 per cent crude fibre, with an energy value of roughly 345 kcal per 100 g.
- Rice protein (oryzenin) is of good quality but low in lysine. Rice is low in fat and is easily digestible, which is why it is used in weaning and convalescent diets.
- Milling removes the bran, and with it much of the thiamine, riboflavin, niacin and iron. This is why parboiling and brown rice are nutritionally superior, and why rice is a common vehicle for fortification.
- Rice bran oil, extracted from the bran, is a premium edible oil rich in oryzanol.

Nutrient (per 100 g milled raw rice)	Approximate content
Energy	345 kcal
Carbohydrate	78-80 g
Protein	6-7 g
Fat	0.5-1.0 g
Crude fibre	0.4-0.6 g

Nutrient (per 100 g milled raw rice)	Approximate content
Thiamine (B1)	0.05-0.06 mg (higher in parboiled/brown)
Iron	about 0.7-1.0 mg

Brown (unpolished) and parboiled rice retain more B vitamins, minerals and fibre than polished white rice; parboiling also firms the grain and reduces breakage in milling.

1.3 Soil and climatic requirement

Soil

- Rice can be grown on a wide range of soils, from heavy clay to sandy loam, but performs best on soils with good water-holding capacity - clay and clay loam, and on silty loams of river deltas.
- Optimum soil reaction is pH 5.5 to 6.5, though rice tolerates a wider range of pH 4.5 to 8.0 and is the most tolerant cereal to both acidity and alkalinity, and to salinity during the later stages of growth.
- Soils should be able to hold a standing layer of water. A plough pan or a naturally impervious sub-soil reduces percolation losses.
- Rice is tolerant of salinity, which is why it is the crop of choice for the coastal saline soils of Nagapattinam and Ramanathapuram and for the alkali soils of the Cauvery delta.

Climate

- Rice is a crop of the warm, humid tropics and subtropics. It can be grown from sea level up to about 2500 m in the hills.
- Optimum temperature for growth is about 30-32 degrees C by day and 20-22 degrees C by night. Germination requires a minimum of about 10-12 degrees C; tillering is best at 25-31 degrees C; flowering and grain filling are sensitive to low temperature, and temperatures below about 15-17 degrees C at flowering cause spikelet sterility.
- Rice needs abundant water - about 1000 to 1500 mm of water including rainfall and irrigation over the season, and it is normally grown under assured irrigation or in high-rainfall areas.
- Bright sunshine during flowering and grain filling improves yield. Continuous heavy cloud during this period reduces photosynthesis and yield.
- A long day length slightly delays flowering in photosensitive varieties; most modern high-yielding varieties are photo-insensitive and short-day or day-neutral.

Requirement	Optimum range for rice
Soil	Clay to clay loam, good water holding
Soil pH	5.5-6.5 (tolerates 4.5-8.0)
Day temperature	30-32 C
Night temperature	20-22 C
Minimum for germination	10-12 C
Critical low temp. at flowering	below 15-17 C (sterility)
Water need (season)	1000-1500 mm
Relative humidity	70-90 per cent
Altitude	Sea level to about 2500 m

1.4 Varieties

Rice varieties of Tamil Nadu are released by TNAU (ADT, ASD, CO, TKM and TPS series) and by the ICAR and State institutions (BPT, CR, etc.). They are grouped by duration into short (110-120 days), medium (120-135 days) and long (above 135 days). Duration varies a little with season and locality.

Short-duration varieties (about 105-120 days)

- ADT 43 - fine grain, high yielding (about 5.5-6.0 t/ha), the most widely grown short-duration variety of the delta; suits Kuruvai and Samba.
- ADT 45 - very short (about 105-110 days), fine grain, good for late planting and for the rice-fallow window.
- ADT 36, ADT 37 - short duration, fine grain, moderate yield.
- ADT 48, ADT 49, ADT 50, ADT 51, ADT 52 - short-duration releases of TNAU with good grain quality and moderate resistance to pests.
- ASD 16 - short duration, blast tolerant; ASD 18, ASD 19, ASD 20 - short-duration releases with resistance to blast and other diseases.
- CO 47 - short-medium duration, fine grain; CO 48, CO 50, CO 51 - short-duration releases; CO 51 has tolerance to drought.
- TKM 9, TKM 11, TKM 13 - short-duration varieties suited to rain-fed and tank-fed areas.
- TPS 3, TPS 5 - short-duration varieties from the Paddy Breeding Station.

Medium-duration varieties (about 120-135 days)

- ADT 38 - medium duration, long slender grain, high yield.
- ADT 39 - medium duration, fine grain.
- ADT 46 - medium duration, tolerant to brown plant hopper.
- ADT 47 - medium-short duration, fine grain, good cooking quality.
- ASD 17 - medium duration with disease tolerance.
- CO 43 - medium-long duration (about 135 days), tolerant to salinity and to alkalinity; the standard variety for coastal saline tracts.
- CO 49 - medium duration release.

Long-duration varieties (above 135 days)

- White Ponni and Improved White Ponni - long duration (about 135-140 days), short, bold, white grain, excellent cooking quality; the premium variety of the Cauvery delta, sown in the Samba season.
- BPT 5204 (Samba Mahsuri) - long duration (about 135-140 days), fine slender grain of high market value; widely grown in the delta and in Andhra Pradesh.
- CR 1009 and other long-duration varieties suit the deep-water and late-Samba situations.

> Varieties resistant to major pests and diseases are preferred because they reduce pesticide cost and protect the environment. ADT 46 (BPH tolerant), ASD 16 (blast tolerant) and CO 43 (salt tolerant) are textbook examples of problem-area varieties.

1.5 Text diagram - the rice plant growth stages

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Rice growth stages (duration shown for a 120-day variety):
  Vegetative phase (0-60 d)          Reproductive (60-90 d)      Ripening (90-120 d)
  ---------------------------        ----------------------      --------------------
  Germination -> Seedling            Panicle initiation (PI)     Flowering / Anthesis
     -> Tillering (max tillers)         -> Boot leaf                -> Milky -> Dough
     -> Stem elongation (jointing)      -> Booting -> Heading        -> Maturity
     |                                  |                           |
   Transplanting at 20-25 d           Critical stages for water   Harvest at 80-85%
   (max. tillering at 45-60 d)        and N: PI and flowering     straw-yellow stage
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1.6 Worked example - plant population and seed requirement in rice

Problem: A farmer transplants ADT 43 (short duration) on 1 hectare at a spacing of 20 cm x 10 cm with 2 seedlings per hill. Seed of 1000-grain weight 22 g has 90 per cent germination. Calculate (a) the number of hills per hectare, (b) the number of seedlings needed, and (c) the seed rate.

- Hills per hectare = (10000 m2) / (0.20 m x 0.10 m) = 10000 / 0.02 = 5,00,000 hills per hectare.
- Seedlings needed = 5,00,000 hills x 2 seedlings = 10,00,000 seedlings per hectare.
- A well-managed nursery of 1 hectare supplies planting material for about 10 to 12 hectares of main field at this spacing; the standard seed rate for short-duration transplanted rice is 40-60 kg per hectare.
- Using the standard figure, seed rate for short-duration rice is about 40 kg/ha and for medium and long duration about 60 kg/ha. Adjust upward if germination is poor.
- Check: 40 kg x 0.90 germination = 36 kg viable seed = 36,000 g / 0.022 g per seed = about 16.4 lakh viable seeds per hectare, comfortably above the 10 lakh seedlings required. The seed rate therefore includes a safety margin for nursery losses.

1.7 Package of practices at a glance - rice (transplanted)

Item	Recommendation
Season (Tamil Nadu)	Kuruvai, Samba, Thaladi, Navarai (see Lecture 2)
Varieties	ADT 43, ADT 45, ASD 16, CO 43, Improved White Ponni, BPT 5204
Nursery	Wet nursery; 40-60 kg seed/ha; 20-25 day old seedlings
Spacing	20 x 10 cm (short duration); 20 x 15 cm (medium/long)
Seedlings per hill	2 (short duration); 2-3 (long duration)
FYM	12.5 t/ha at land preparation
NPK (kg/ha)	150 : 50 : 50 (high-yielding varieties)
N splits	1/3 basal + 1/3 tillering + 1/3 panicle initiation
ZnSO4	25 kg/ha once in 2-3 seasons on deficient soils
Green manure	Daincha/Sunhemp before transplanting where possible
Water	Keep 5 cm water; drain before harvest
Expected yield	5-6 t/ha (short duration, irrigated)

1.9 Cropping systems and rotations with rice

- Rice is rarely grown alone in a sequence. The crop fits into several cropping systems depending on water availability, soil and season.
- Rice-rice-pulse: the classic delta system - Kuruvai rice, then Thaladi or Samba rice, then a rice-fallow pulse (black gram or green gram). This is the most intensive and most profitable sequence of the Cauvery delta.
- Rice-pulse: one rice crop followed by a short-duration pulse (black gram, green gram or cowpea) in the same year.
- Rice-groundnut: common in the tank-fed and garden-land tracts where the rice season is short.
- Rice-cotton or rice-maize: grown where a longer dry window is available after the rice harvest.
- Rice-vegetables: in garden land near cities, a rice crop is followed by vegetables (brinjal, bhendi, tomato) for high returns.

- Green manure in rotation: daincha or sunhemp is raised in the fallow window and incorporated before the next rice crop, adding about 40-60 kg N/ha and organic matter.

Rice intercropping

- Rice is a poor intercrop because of its dense stand, but rice + azolla (a water fern used as green manure and feed) is a common combination in Tamil Nadu. Azolla fixes nitrogen and adds 20-30 kg N/ha and suppresses weeds.
- Rice + fish (rice-fish farming) and rice + prawn (rice-prawn in the coastal districts) are integrated farming systems that raise total farm income from the same land and water.

1.10 Comparison of important rice varieties

Variety	Duration	Grain type	Salient feature
ADT 43	110-115 d	Fine, short	Most popular short-duration delta variety
ADT 45	105-110 d	Fine, short	Very short; good for late planting
ASD 16	110-115 d	Medium	Blast tolerant
ADT 46	115-120 d	Medium	Brown plant hopper tolerant
CO 43	about 135 d	Medium-long	Salt and alkali tolerant (coastal)
CO 51	110-115 d	Medium	Drought tolerant
Improved White Ponni	135-140 d	Short bold white	Premium delta variety, Samba season
BPT 5204	135-140 d	Fine slender	High market value (Samba Mahsuri)

Field calendar and common problems

Crop calendar (operations schedule)

Operation	Time (from sowing/transplanting)	Remarks
Nursery sowing	0 (25 days before transplanting)	Wet nursery; 40-60 kg seed/ha
Main field preparation	15-25 days before	Dry plough, puddle, level, add FYM
Transplanting	20-25 day old seedlings	2 seedlings/hill at 20 x 10 cm
Basal fertiliser	At transplanting	Full P and K, 1/3 N
First top-dress N	20-25 days	1/3 N at active tillering
Weeding and herbicide	3-5 days (pre-em) and 20-40 days	Pre-emergence herbicide + hand weeding
Second top-dress N	45-50 days	1/3 N at panicle initiation
Drainage	Maximum tillering	For aeration and BPH control
Harvest	105-140 days	At 80-85 per cent straw-yellow

Common field problems, causes and remedies

Problem	Likely cause	Remedy
Chaffy / empty grains	Low temperature at flowering; excess N; pest damage	Balanced N; timely planting; pest control

Problem	Likely cause	Remedy
Lodging	Excess nitrogen; weak stems; heavy wind	Reduce late N; grow sturdy varieties
Khaira (rusty spots)	Zinc deficiency	Apply ZnSO4 25 kg/ha
Hopper burn	Brown plant hopper	Drain field; tolerant variety; need-based insecticide
Stunted, yellow crop	Nitrogen or iron deficiency; waterlogging	Split N; drain; correct drainage

Objective-type practice questions

Choose the correct answer. Answers are given at the end of the set.

1 The botanical name and chromosome number of rice are?
2 Rice belongs to the family?
3 The two centres of domestication of rice are?
4 India's rank in world rice export is?
5 The optimum soil pH for rice is?
6 The optimum day temperature for rice is about?
7 Zinc deficiency in rice is called?
8 The most salt-tolerant TNAU rice variety is?
9 The staple food of more than half the world population is?
10 Milled rice contains about this much carbohydrate?

Answers: 1-Oryza sativa, 2n=24; 2-Poaceae; 3-Indo-Burma (indica) and China (japonica); 4-first (largest exporter); 5-5.5-6.5; 6-30-32 C; 7-khaira; 8-CO 43; 9-rice; 10-78-80 per cent

1.8 Exam focus

Key points to remember

- Botanical name Oryza sativa, family Poaceae, 2n = 24; indica (tropics) and japonica (temperate) groups.
- Origin: Indo-Burma region for indica and China for japonica; Vavilov Indian and Chinese centres.
- India is the second largest producer and largest exporter of rice; rice is the staple food of more than half the world.
- Milled rice: about 78-80 per cent carbohydrate, 6-7 per cent protein, 345 kcal per 100 g; parboiled and brown rice are nutritionally superior.
- Best soils are clay to clay loam with pH 5.5-6.5; rice tolerates pH 4.5-8.0 and salinity.
- Optimum temperature about 30-32 C day and 20-22 C night; water need about 1000-1500 mm; low temperature at flowering causes sterility.
- TNAU varieties: ADT 43 (short, fine), ADT 45 (very short), ASD 16 (blast tolerant), CO 43 (salt tolerant), Improved White Ponni and BPT 5204 (long, premium).
- Standard NPK for high-yielding transplanted rice is 150 : 50 : 50 kg/ha.

Short-answer questions with model answers

1 Q1. Give the botanical name, family and chromosome number of rice.

- Oryza sativa L.; family Poaceae (Gramineae); chromosome number 2n = 24. Two ecological groups of the cultivated Asian rice are indica and japonica.

1 Q2. State the soil and pH requirement of rice.

- Rice performs best on clay and clay loam soils with good water-holding capacity. The optimum pH is 5.5-6.5, but the crop tolerates a range of about 4.5-8.0 and is the most tolerant cereal to salinity and alkalinity, which makes it suitable for coastal saline and delta alkali soils.

1 Q3. List four short-duration TNAU rice varieties with their approximate duration.

- ADT 43 (about 110-115 days, fine grain); ADT 45 (about 105-110 days, very short); ASD 16 (about 110-115 days, blast tolerant); CO 51 (about 110-115 days, drought tolerant). Durations vary slightly with season.

Long-answer questions with model-answer outlines

1 Q1. Describe the origin, distribution and economic importance of rice.

- Outline: (i) botanical identity and the two cultivated species; (ii) the two centres of domestication - Indo-Burma (indica) and China (japonica), and Vavilov's Indian and Chinese centres of diversity; (iii) world area of about 165 million hectares and the leading producers, with India second in production and first in export; (iv) main Indian and Tamil Nadu tracts, with emphasis on the Cauvery delta; (v) economic importance - staple food, rural employment, milling and export industry, by-products (straw, husk, bran oil); (vi) nutritional value - carbohydrate, protein, energy, and the effect of milling and parboiling. Conclude with one line on why rice remains the pivot of the delta cropping system.

1 Q2. Explain the soil and climatic requirements of rice and justify why rice is grown under standing water.

- Outline: (i) soils - clay to clay loam, water-holding, pH 5.5-6.5 with wide tolerance; (ii) the role of the plough pan in reducing percolation; (iii) climate - warm humid tropics, optimum 30-32 C day and 20-22 C night, water need 1000-1500 mm, humidity 70-90 per cent; (iv) reasons for standing water - it suppresses most weeds, keeps the soil reduced and increases the availability of phosphorus and iron, buffers temperature, and supplies water continuously; (v) the penalty of low temperature at flowering (sterility) and of cloudy weather at grain filling; (vi) note that modern short-duration varieties allow rice to be grown with less water (alternate wetting and drying, SRI).

Lecture 2 - Rice: Cultural Practices, Yield and Economic Benefits

2.1 Seasons of Tamil Nadu

Tamil Nadu does not have a single kharif season like the north. It has several named seasons, each linked to a monsoon or to a canal-release period. Recognising the season name is essential, because the variety, the nursery date and the water source all follow from it.

Season (Tamil name)	Months of sowing	Description and main areas
Sornavari (Chithirai)	April-May	Early crop using the south-west monsoon and canal water; delta and tank-fed areas
Kar (Vaikasi)	May-June	Mainly rain-fed crop of the tank-fed and upland tracts
Kuruvai (Aani)	June-July	Short-duration crop of the delta, grown with Cauvery water; short-duration varieties like ADT 43
Samba (Aadi)	August	The main delta season; medium and long duration varieties (White Ponni, BPT 5204)
Thaladi (Purattasi)	September-October	Late crop after Kuruvai; nursery raised in the Samba field

Season (Tamil name)	Months of sowing	Description and main areas
Navarai (Margazhi)	December-January	Winter/spring crop with tank and well water; short-duration varieties
Late Samba	October-November	Long-duration Samba crop, harvested in the following year

The delta's classic triple sequence is Kuruvai (short rice) followed by Thaladi (medium rice) followed by a rice-fallow pulse (black gram or green gram) on residual moisture. This is the most intensive cropping pattern of Tamil Nadu.

2.2 Cropping systems involving rice

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Common rice-based cropping systems of Tamil Nadu:
  Cauvery delta (assured water):
     Kuruvai rice -> Thaladi rice -> Rice-fallow pulse (black gram/green gram)
     or  Rice -> Rice -> Fallow (with daincha green manure)
  Delta with limited water:
     Rice (Samba) -> Rice-fallow pulse  ->  Cotton/Chillies in the dry window
  Tank-fed / upland:
     Rice (Kar) -> Groundnut / Pulses  ->  Millets
  Garden-land (well irrigated):
     Rice -> Rice -> Vegetables  or  Rice -> Banana -> Pulses
```

2.3 Land preparation

- Dry ploughing: the field is ploughed once or twice with an iron plough or disc plough when the soil has friable moisture, to break the soil and bury stubbles.
- Puddling: for transplanted rice, the field is flooded and puddled twice or thrice with a rotavator or cage-wheel puddler to destroy soil structure, create a soft mud, and form a plough pan that reduces water loss.
- Levelling: the field is levelled with a leveller or by a laser leveller. Good levelling saves water, ensures uniform water depth and even crop maturity, and is one of the cheapest yield-boosting practices.
- Application of organic manure: 12.5 t/ha of farmyard manure or compost is incorporated at the time of the last puddling. Where available, green manure crops (daincha, sunhemp) are raised in situ and incorporated before transplanting.
- Formation of bunds and channels: strong bunds are formed to hold water, and irrigation and drainage channels are opened.
- For direct-sown and rain-fed rice, the land is dry-ploughed and harrowed to a fine tilth; puddling is not done.

2.4 Seed treatment and nursery management

- Seed selection: use certified seed. Soak the seed in salt solution (about 10 per cent) or in brine to discard light, immature seed, then wash.
- Seed treatment: treat with a fungicide (for example, carbendazim) to control seed-borne diseases, and with a bio-fertiliser slurry (Azospirillum and phosphobacteria) before sowing.
- Wet nursery: raise a wet nursery on a raised bed near the main field. Sow pre-germinated seed densely; a nursery of about 8-10 cents supplies one hectare.
- Dry nursery: used for rain-fed and for short-duration crops where water is limited; seed is sown on a dry, well-prepared bed.
- Age of seedlings: transplant 20-25 day old seedlings for short duration and 25-30 day old for medium and long duration. Seedlings older than about 30 days lose vigour and tillering capacity.

2.5 Transplanting

- Pull the seedlings carefully, trim the leaf tips to reduce transpiration, and keep the roots moist.
- Transplant 2 seedlings per hill (2-3 for long duration) at a shallow depth of about 2-3 cm, keeping the roots just below the surface.
- Spacing: 20 x 10 cm for short duration; 20 x 15 cm for medium and long duration; wider spacing (25 x 15 cm) where the soil is fertile and tillering is profuse.
- Plant in rows to allow mechanical weeding and better light interception.
- The recommended practice of planting in rows at the right spacing and depth is called 'correct planting' and by itself raises yield.

2.6 Nutrient management

The standard dose for high-yielding transplanted rice in Tamil Nadu is 150 : 50 : 50 kg N : P2O5 : K2O per hectare. Nitrogen is the key nutrient and is applied in splits to match crop demand.

Split application of nitrogen

- Basal (at transplanting): one-third of N, with the full P and K, incorporated or placed near the root zone.
- First top-dress (active tillering, about 20-25 days after transplanting): one-third of N.
- Second top-dress (panicle initiation, about 45-50 days after transplanting): one-third of N.
- Where the leaf colour is pale, a small extra dose may be given at flowering, but excess late nitrogen encourages pests and lodging.

Other nutrients and amendments

- Zinc: apply zinc sulphate 25 kg/ha once in 2-3 seasons where the soil is deficient; zinc deficiency causes 'khaira' (rusty brown spots and stunted growth).
- Green manure and organic manure improve soil health and supply micronutrients.
- On acid soils apply lime; on alkali/saline soils apply gypsum and leach. CO 43 is used on such soils.
- For direct-sown rice, N is given in three splits timed to the seedling, tillering and panicle stages.

Tools for nitrogen management

- Leaf Colour Chart (LCC) and the Green Seeker sensor help time the top-dress of nitrogen to actual crop need, saving fertiliser without loss of yield.
- Site-Specific Nutrient Management (SSNM) adjusts the dose to the yield target and the indigenous nutrient supply of the field.

2.7 Water management

- Maintain a shallow standing water layer of about 5 cm from transplanting to the panicle initiation stage.
- Drain the field for a few days at maximum tillering ('drainage for aeration') to encourage deep rooting and reduce unproductive tillers and the build-up of the brown plant hopper.
- Maintain water at flowering and grain filling, the most critical stages; water stress at these stages causes severe yield loss.
- Alternate Wetting and Drying (AWD): allow the field to dry to about 15 cm below the surface before re-flooding. This saves about 20-25 per cent of irrigation water without reducing yield, and is the standard water-saving practice.
- Stop irrigation about 10-15 days before harvest to allow uniform maturity and easy harvest.
- In the delta, water is supplied on a rotational 'turn' system (pattam/thaladi turns); the farmer must transplant and top-dress to match the canal turns.

2.8 Weed management

- The major weeds of rice are grasses (Echinochloa colonum and E. crus-galli, barnyard grass; Cyperus rotundus, purple nutsedge; Cynodon dactylon), sedges and broad-leaved weeds (Monochoria, Eclipta, Ludwigia).
- Standing water itself suppresses many weeds; good puddling and levelling reduce weed pressure.
- Hand weeding: two hand weedings, at about 20-25 days and 40-45 days after transplanting.
- Mechanical weeding: use a rotary weeder or cono-weeder between rows; this also aerates the soil.
- Chemical control: apply a pre-emergence herbicide such as butachlor or pretilachlor within 3-5 days of transplanting, on moist soil. Where needed, apply a post-emergence herbicide such as bispyribac-sodium or a tank mix for the sedge and broad-leaved weeds, following the current TNAU Plant Protection Guide.
- Integrated Weed Management (IWM) - combining puddling, water management, hand/mechanical weeding and need-based herbicide - is the recommended approach.

2.9 Pest and disease management

Major insect pests

- Brown Plant Hopper (BPH, Nilaparvata lugens) - the most serious pest of the delta; it causes 'hopper burn' (circular patches of dried plants). Management: drain the field, avoid excess nitrogen, grow tolerant varieties (ADT 46), conserve natural enemies, and use need-based insecticide.
- Stem borer (yellow stem borer, Scirpophaga incertulas) - causes 'dead heart' in the vegetative stage and 'white ear head' at flowering. Management: remove egg masses, use light traps and pheromone traps, release Trichogramma, and apply insecticide only at the economic threshold.
- Leaf folder (Cnaphalocrocis medinalis) - folds the leaves and scrapes the green tissue. Management: use sex pheromone traps and need-based insecticide.
- Gall midge, green leaf hopper (vector of tungro virus), rice bug, thrips and caseworm are other pests of local importance.

Major diseases

- Blast (Pyricularia oryzae) - leaf and neck blast, most severe with high nitrogen and cool humid weather. Management: grow resistant varieties (ASD 16), balance nitrogen, treat seed, and use a fungicide at the neck-emergence stage.
- Bacterial leaf blight (Xanthomonas oryzae) - water-soaked to yellow lesions from the leaf tip; severe in the delta after storms. Management: resistant varieties, balanced fertiliser, avoid excess nitrogen, drain and dry the field.
- Sheath blight (Rhizoctonia solani), sheath rot, brown spot (Helminthosporium oryzae) and tungro virus (spread by green leaf hopper) are other important diseases.

Follow Integrated Pest Management (IPM): grow resistant varieties, follow the economic threshold level, conserve natural enemies, use pheromone and light traps, and apply pesticide only as the last resort, at the recommended dose.

2.10 Harvest and yield

- Harvest when the grains in the lower part of the panicle have reached the hard-dough stage and about 80-85 per cent of the grains have turned straw-yellow. Harvesting too early gives immature, chaffy grain; too late causes shattering and breakage.
- Drain the field 10-15 days before harvest.

- Harvest by sickle or by a combine harvester. Thresh, winnow and dry the grain to about 12-14 per cent moisture for safe storage.
- Average yield: about 5-6 t/ha for irrigated high-yielding short-duration varieties, 6-7 t/ha for well-managed medium and long duration varieties, and about 2.5-3.5 t/ha for rain-fed rice. The national average is much lower than the potential because of rain-fed area and management gaps.
- Rice equivalent yield is often used to compare a rice crop with a rice-fallow pulse: rice yield + (pulse yield x price ratio).

Component of yield	Standard value for irrigated rice
Panicles per m2	250-350
Grains per panicle	100-150
Test weight (1000 grains)	20-25 g
Irrigated yield (short duration)	5-6 t/ha
Irrigated yield (medium/long)	6-7 t/ha
Rain-fed yield	2.5-3.5 t/ha
Straw yield	roughly equal to grain yield

2.11 Worked example - cost of cultivation and net return of rice

Problem: A farmer grows 1 hectare of ADT 43 transplanted rice in the Kuruvai season and obtains 5.5 t/ha of paddy. Paddy sells at Rs. 2,300 per quintal. Calculate the gross return, the cost of cultivation and the net return, using the standard cost items below.

Cost item (per hectare)	Approximate amount (Rs.)
Land preparation and puddling	6,000
Seed (40 kg) and nursery	2,000
Fertiliser (150:50:50) and Zn	8,500
FYM 12.5 t	6,000
Transplanting labour	8,000
Weeding and herbicide	3,500
Plant protection	3,000
Irrigation charges	1,500
Harvesting, threshing, transport	7,500
Total cost of cultivation	46,000

- Gross return = 5.5 t/ha x 1000 kg/t / 100 kg per quintal x Rs. 2,300 = 55 quintals x Rs. 2,300 = Rs. 1,26,500 per hectare.
- Net return = gross return - cost of cultivation = Rs. 1,26,500 - Rs. 46,000 = Rs. 80,500 per hectare.
- Benefit-cost ratio = gross return / cost of cultivation = 1,26,500 / 46,000 = about 2.75.
- If straw is sold at Rs. 1,500 per tonne and 5 tonnes of straw are obtained, an extra Rs. 7,500 is added, raising the net return to about Rs. 88,000 and the B:C ratio to about 2.9.
- Note: the cost items above are indicative textbook values; actual costs vary with locality, labour rates and season. The method of calculation is what the examination tests.

2.12 Worked example - yield gap

Problem: The potential (demonstration) yield of a rice variety is 6.5 t/ha and the actual average yield obtained by farmers in a block is 4.0 t/ha. Calculate the yield gap and the yield gap index.

- Yield gap = potential yield - actual yield = 6.5 - 4.0 = 2.5 t/ha.
- Yield gap index (per cent) = (yield gap / potential yield) x 100 = (2.5 / 6.5) x 100 = about 38.5 per cent.
- The gap is closed by better variety choice, timely transplanting, balanced fertiliser, correct water management, timely weed and pest control, and good levelling. Yield gap analysis is the first step in any technology-transfer or demonstration programme.

2.14 Post-harvest operations and value addition

- Drying: the harvested paddy is dried to about 12-14 per cent moisture for safe storage and milling. Sun drying on a clean floor is common; mechanical dryers are used in large farms and mills.
- Cleaning and grading: the grain is winnowed, cleaned and graded by size and quality before storage or sale.
- Storage: store paddy in dry, aerated godowns or in bags on wooden pallets. Fumigate against storage pests (rice weevil, lesser grain borer) as recommended. Tamil Nadu's regulated markets and the Food Corporation of India (FCI) procure paddy at the minimum support price.
- Milling: paddy is milled to rice. Modern mills give a milling outturn of about 65-68 per cent (rice from paddy) and a head-rice recovery of about 50-55 per cent.
- Parboiling: soaking, steaming and drying the paddy before milling firms the grain, reduces breakage, drives vitamins into the endosperm and gives a higher head-rice recovery. Most rice in Tamil Nadu is parboiled.
- By-products: rice bran (used for bran oil and cattle feed), rice husk (fuel, board making, silica), and broken rice (used for starch and feed). Rice bran oil is a premium edible oil rich in oryzanol.
- Value addition: puffed rice, flaked rice (aval), rice flour, idli/dosa batter, and branded rice for export raise the farmer's share of the consumer rupee.

2.15 Organic and mechanised rice cultivation

- Organic rice: replaces inorganic fertiliser with compost, green manure, Azolla, azospirillum and phosphobacteria, and replaces pesticides with botanical and biological control. It commands a price premium but may yield slightly less; it is suited to high-value and export markets.
- Mechanisation: drum seeders for direct-sown rice, rice transplanters, cono-weeders, combined harvesters and laser levellers reduce labour cost and improve timeliness. Mechanisation is important where labour is scarce and costly.
- Direct Seeded Rice (DSR): sowing pre-germinated or dry seed directly in the field, without transplanting. DSR saves about 20-25 per cent of water and a large amount of labour, but needs good weed management (a pre-emergence herbicide and timely weeding) because there is no puddling to suppress weeds.
- Aerobic rice: rice grown like an upland crop without standing water, with irrigation only to keep the soil moist. It saves water but yields less; suited to water-scarce areas.

Field calendar and common problems

Crop calendar (operations schedule)

Operation	Time	Remarks
Field preparation	Before transplanting	Puddle, level, incorporate green manure

Operation	Time	Remarks
Azolla incorporation	1-2 weeks after transplanting	Adds N and suppresses weeds
Drainage for aeration	Maximum tillering	Reduce unproductive tillers and BPH
AWD cycles	Vegetative stage	Re-flood when water falls 15 cm below surface
Final irrigation stop	10-15 days before harvest	Uniform maturity
Harvest and threshing	105-140 days	Dry to 12-14 per cent moisture
Parboiling and milling	After harvest	For better recovery and nutrition

Common field problems, causes and remedies

Problem	Likely cause	Remedy
High weed pressure	Poor puddling/levelling; no pre-em herbicide	Level well; apply pre-emergence herbicide
Water shortage	Weak bunds; poor levelling; high percolation	Repair bunds; laser level; AWD
Poor milling recovery	Harvesting too late; uneven moisture	Harvest at right stage; dry uniformly
Storage pests	High moisture; poor storage	Dry to 12-14 per cent; fumigate
Low benefit-cost ratio	High labour cost	Mechanise; grow short-duration varieties

Objective-type practice questions

Choose the correct answer. Answers are given at the end of the set.

1 The short-duration delta rice season of June-July is?
2 The standard NPK dose for high-yielding transplanted rice is?
3 Nitrogen in rice is applied in this many splits?
4 The water-saving rice practice is?
5 Hopper burn is caused by?
6 Dead heart and white ear head are caused by?
7 Rice is harvested when the grain is?
8 The rice crop is drained before harvest by about?
9 Bacterial leaf blight is caused by?
10 The delta sequence is?

Answers: 1-Kuruvai; 2-150:50:50; 3-three; 4-alternate wetting and drying (AWD); 5-brown plant hopper; 6-stem borer; 7-80-85 per cent straw-yellow; 8-10-15 days; 9-Xanthomonas oryzae; 10-Kuruvai-Thaladi-rice fallow pulse

2.13 Exam focus

Key points to remember

- Tamil Nadu seasons: Sornavari, Kar, Kuruvai, Samba, Thaladi, Navarai and Late Samba; the delta sequence is Kuruvai - Thaladi - rice-fallow pulse.
- Land preparation for transplanted rice includes dry ploughing, puddling, levelling, incorporation of 12.5 t/ha FYM and bund formation.

- Seed rate: 40 kg/ha (short duration) and 60 kg/ha (medium/long); transplant 20-25 day old seedlings, 2 per hill, at 20 x 10 cm.
- Standard NPK 150 : 50 : 50 kg/ha with N in three splits (basal, tillering, panicle initiation); zinc sulphate 25 kg/ha on deficient soils.
- Water: 5 cm standing water, drainage at maximum tillering, AWD to save water, stop irrigation 10-15 days before harvest.
- Major pests: brown plant hopper (hopper burn), stem borer (dead heart, white ear head), leaf folder. Major diseases: blast, bacterial leaf blight, sheath blight.
- Harvest at 80-85 per cent straw-yellow grain; irrigated yield 5-6 t/ha (short) and 6-7 t/ha (medium/long).
- Cost of cultivation of irrigated rice is roughly Rs. 45,000-50,000 per hectare with a benefit-cost ratio of about 2.5-2.9 at present prices.

Short-answer questions with model answers

1 Q1. What is the standard fertiliser dose for high-yielding transplanted rice, and how is nitrogen applied?

- The standard dose is 150 : 50 : 50 kg N : P2O5 : K2O per hectare. The full P and K are given basally; nitrogen is given in three equal splits - one-third basal at transplanting, one-third at active tillering, and one-third at panicle initiation. This matches nitrogen supply to crop demand and reduces losses.

1 Q2. Explain Alternate Wetting and Drying (AWD) and its advantage.

- In AWD the field is allowed to dry naturally until the water level falls to about 15 cm below the soil surface, then re-flooded to about 5 cm. It saves about 20-25 per cent of irrigation water without reducing yield, improves rooting and aeration, and reduces methane emission from paddy fields.

1 Q3. Distinguish 'dead heart' from 'white ear head'.

- Both are symptoms caused by the stem borer (Scirpophaga incertulas). 'Dead heart' appears in the vegetative stage, when the central shoot of a tiller dries and can be pulled out easily, because the larva bores into the growing point. 'White ear head' appears after flowering, when the panicle emerges but remains empty and white because the stem is cut below the panicle.

Long-answer questions with model-answer outlines

1 Q1. Write the cultural practices for a high-yielding transplanted rice crop in the Cauvery delta.

- Outline: (i) season and variety choice (Kuruvai with ADT 43; Samba with White Ponni/BPT 5204); (ii) land preparation - dry ploughing, puddling, levelling, FYM 12.5 t/ha; (iii) seed rate, seed treatment and nursery; (iv) transplanting - age, spacing, seedlings per hill, depth; (v) nutrient management - 150:50:50, split nitrogen, zinc, green manure; (vi) water management - standing water, drainage at tillering, AWD, stop before harvest; (vii) weed management - two hand weedings plus pre-emergence herbicide; (viii) pest and disease management in IPM mode; (ix) harvest at 80-85 per cent yellowing and post-harvest drying; (x) expected yield and the economics in one line.

1 Q2. Justify rice as the pivot of the Cauvery delta cropping system and explain the economics that sustain it.

- Outline: (i) assured canal water and heavy delta soils make rice the most reliable crop; (ii) the short-duration window allows two rice crops and a pulse; (iii) rice supplies the staple for the State and supports milling, parboiling and export; (iv) by-products (straw, husk, bran) add income; (v) a worked net return of about Rs. 80,000 per hectare and a B:C ratio of about 2.7-2.9 at current prices; (vi) risks - water shortage in the Cauvery, pest outbreaks (BPH), and rising labour cost - and the responses (short-duration varieties, SRI, mechanisation, rice-fallow pulses).

Lecture 3 - Special Types of Rice Cultivation: SRI and Hybrid Rice Cultivation

3.1 Why special methods are needed

Conventional transplanted rice uses 40-60 kg of seed, a flooded field, close spacing and heavy nitrogen. It needs large amounts of water and labour, and yields have plateaued. Two special methods address this: the System of Rice Intensification (SRI), which changes the plant and water management, and hybrid rice, which raises the genetic yield ceiling. Both are examined as separate topics.

3.2 System of Rice Intensification (SRI)

Concept and origin

- SRI is a set of agronomic practices developed in Madagascar in the 1980s by Fr. Henri de Laulanie. It is based on the principle of giving each plant more space, air and light, and keeping the soil moist rather than continuously flooded, so that the plant expresses its full tillering potential.
- SRI is a method, not a variety. It can be used with any high-yielding variety, but short and medium duration varieties with strong tillering respond best.

The six core principles of SRI

1. Young seedlings: transplant 8-12 day old seedlings (2-3 leaves) instead of 20-30 day old seedlings, to preserve tillering potential.
2. Single seedling per hill: plant one seedling per hill instead of 2-3, to reduce competition.
3. Wide spacing: plant in a square pattern, commonly 25 x 25 cm, sometimes 30 x 30 cm or wider, to give each plant room.
4. Careful transplanting: transplant shallow (1-2 cm), keeping the root tip and the growing point intact, and avoid inverting the root.
5. Alternate wetting and drying: keep the soil moist but not continuously flooded during the vegetative phase, and drain the field at tillering; the soil is kept aerobic rather than reduced.
6. Mechanical weeding and organic matter: use a rotary weeder (cono-weeder) two or three times to control weeds and aerate the soil; add organic manure/compost rather than relying only on inorganic fertiliser.

SRI layout - text diagram

```
SRI planting layout (25 x 25 cm square planting):
  25 cm
  |<---->|
  o------o------o------o------o   o = one seedling per hill
  |      |      |      |      |   rows 25 cm apart
  o------o------o------o------o   plants 25 cm apart in the row
  |      |      |      |      |
  o------o------o------o------o   Weeder (cono-weeder) runs between
  |      |      |      |      |   rows to control weeds and aerate soil
  o------o------o------o------o
  Compare conventional: 20 x 10 cm with 2-3 seedlings per hill.
  SRI gives each hill ~2.5 to 4 times more area.
```

Advantages and disadvantages of SRI

- Advantages: saves seed (only 5-8 kg/ha against 40-60 kg/ha), saves water (about 25-40 per cent), saves the cost of nursery raising and early transplanting labour, gives more tillers and longer panicles, and often yields 10-30 per cent more than conventional methods where water and labour are well managed. It is labour-intensive for weeding but suited to organic and water-scarce situations.

- Disadvantages: needs more careful weed management and frequent shallow irrigation (alternate wetting and drying) rather than continuous flooding; it needs discipline; in very heavy clay soils and where water is abundant, the gain over well-managed conventional rice is smaller.

3.3 Worked example - SRI plant population and seed requirement

Problem: A farmer adopts SRI at a spacing of 25 cm x 25 cm with one seedling per hill. Calculate (a) the number of hills per hectare, (b) the plant population, and (c) the nursery area and seed needed if 8 kg of seed is required per hectare and a nursery of 100 m2 supplies one hectare.

- Hills per hectare = 10000 m2 / (0.25 m x 0.25 m) = 10000 / 0.0625 = 1,60,000 hills per hectare.
- Plant population = 1,60,000 plants per hectare (one seedling per hill).
- This is only about one-third of the conventional population of 5,00,000 hills at 20 x 10 cm; the wide spacing is compensated by profuse tillering (each SRI hill may produce 30-50 tillers against 10-15 in the conventional crop).
- Seed needed = 8 kg per hectare (SRI uses very little seed). Nursery area = 100 m2 (about 2.5 cents) per hectare at this seed rate. The small nursery saves land and water.
- Check on tillering: 1,60,000 hills x 30 productive tillers = 48 lakh panicle-bearing tillers per hectare, compared with about 50 lakh for a good conventional crop - showing that fewer hills can give the same or a higher productive tiller number.

3.4 Hybrid rice cultivation

Concept

- Hybrid rice is the first-generation (F1) progeny of a cross between two genetically diverse parents. It shows heterosis or hybrid vigour, giving a yield advantage of about 15-20 per cent over the best inbred variety.
- Hybrid seed is produced by a three-line system (A, B and R lines) or a two-line system (using a photoperiod- or temperature-sensitive genic male sterile line).
- The A line (cytoplasmic male sterile, CMS line) is the female parent; the B line (maintainer) maintains the sterility of the A line; the R line (restorer) restores fertility in the F1 hybrid.

Steps in hybrid seed production

- Isolation: keep an isolation distance of at least 100 m from other rice to avoid foreign pollen.
- Planting ratio: plant the A line and R line in rows, commonly 2 rows of A : 1 row of R, or as the local package directs, with staggered sowing so that both flower together.
- Supplementary pollination: use a rope or a bamboo stick to shake the pollen from the R line onto the A line at flowering, or use a leaf blower.
- GA3 (gibberellic acid) is sprayed to improve panicle exsertion in the CMS line.
- Rogue out off-types and harvest the seed from the A line rows as hybrid seed.

Hybrid varieties and management

- Tamil Nadu and India grow public and private hybrids (for example, from the TNAU, IRRI and private seed companies). Hybrids are usually medium to long duration and high yielding.
- Hybrid seed is costly, so the seed rate is kept low (about 15-20 kg/ha). The crop must be managed well - balanced fertiliser, timely water and pest control - because hybrids have a higher yield potential that is realised only with good management.
- Hybrid rice needs a slightly higher nitrogen dose and careful pest management, especially for the brown plant hopper.

Comparison table

Feature	Conventional rice	SRI	Hybrid rice
Seed rate	40-60 kg/ha	5-8 kg/ha	15-20 kg/ha
Seedling age	20-30 days	8-12 days	20-25 days
Seedlings/hill	2-3	1	1-2
Spacing	20 x 10 cm	25 x 25 cm	20 x 15 cm
Water	Continuous flooding	Alternate wetting/drying	Continuous flooding
Weed control	Herbicide + hand weeding	Rotary weeder (2-3 times)	Herbicide + hand weeding
Yield potential	5-6 t/ha	6-8 t/ha (well managed)	6.5-8 t/ha
Main cost	Labour, water	Weeding labour	Cost of hybrid seed

3.5 Worked example - comparison of returns

Problem: Compare the gross margin of SRI and conventional rice on 1 hectare. Conventional: yield 5.5 t/ha, cost Rs. 46,000. SRI: yield 6.5 t/ha, seed cost lower by Rs. 1,500, water cost lower by Rs. 1,500, but weeding cost higher by Rs. 3,000. Paddy at Rs. 2,300 per quintal.

- Conventional gross return = 55 quintals x Rs. 2,300 = Rs. 1,26,500. Net return = 1,26,500 - 46,000 = Rs. 80,500.
- SRI adjusted cost = 46,000 - 1,500 - 1,500 + 3,000 = Rs. 46,000 (the savings and extra cost roughly cancel in this example).
- SRI gross return = 65 quintals x Rs. 2,300 = Rs. 1,49,500. Net return = 1,49,500 - 46,000 = Rs. 1,03,500.
- Extra net return from SRI = Rs. 1,03,500 - Rs. 80,500 = Rs. 23,000 per hectare, an increase of about 29 per cent, driven mainly by the higher yield from more tillers.
- Note: SRI results vary with soil, water and labour; on light soils and with assured water the gain may be smaller.

3.7 Other special methods and a summary

Direct Seeded Rice (DSR)

- In DSR, seed is sown directly in the main field (dry or pre-germinated) without raising and transplanting seedlings. It saves water (about 20-25 per cent), labour and the cost of nursery raising, and shortens the crop duration slightly. The main problem is weeds, which must be controlled with a pre-emergence herbicide and timely weeding. DSR is suited to areas with scarce labour.

Aerobic rice and water-saving methods

- Aerobic rice is grown without standing water, keeping the soil moist like an upland crop. It saves 40-50 per cent of water but yields about 20-30 per cent less than flooded rice. Alternate Wetting and Drying (AWD) is the more widely adopted middle path.

Organic rice

- Organic rice uses compost, green manure, Azolla and bio-fertilisers instead of inorganic fertiliser, and biological and botanical control instead of chemical pesticides. It suits export and premium domestic markets and is often combined with SRI.

Summary of the three special methods

Feature	SRI	Hybrid rice	DSR
Main idea	Change plant/water management	Raise genetic yield	Avoid transplanting
Seed	5-8 kg/ha	15-20 kg/ha	20-30 kg/ha
Water saving	High (25-40 per cent)	None	High (20-25 per cent)
Main benefit	Seed, water, tillering	15-20 per cent more yield	Labour and water saving
Main problem	Weeding labour	Cost of hybrid seed	Weed control
Best fit	Water-scarce, disciplined farms	High-input irrigated farms	Labour-scarce farms

Field calendar and common problems

Crop calendar (operations schedule)

Operation	Time	Remarks
Nursery (SRI)	8-12 days before planting	Very young seedlings, 2-3 leaves
Transplanting (SRI)	8-12 day seedlings	1 seedling/hill at 25 x 25 cm, shallow
Rotary weeding	10, 20, 30 days after planting	2-3 times; also aerates the soil
AWD irrigation	Vegetative phase	Keep moist; flood at flowering
Hybrid seed production	At flowering	GA3 spray; rope/leaf-blower pollination
Harvest	105-140 days	As per variety

Common field problems, causes and remedies

Problem	Likely cause	Remedy
Heavy weeds in SRI	Wide spacing; alternate wetting	Rotary weeder 2-3 times; keep moist, not flooded
Few tillers in SRI	Old seedlings; deep planting	Use 8-12 day seedlings; plant shallow
Poor hybrid seed set	Non-synchronised flowering	Staggered sowing; GA3; supplementary pollination
High hybrid seed cost	F1 seed price	Use seed rate of 15-20 kg/ha; save only F1
SRI yield not higher	Heavy soil; abundant water; poor weeding	Reserve SRI for water-scarce, well-managed fields

Objective-type practice questions

Choose the correct answer. Answers are given at the end of the set.

1 SRI was developed in?

2 The seedling age used in SRI is?

3 The SRI spacing is commonly?

4 The number of seedlings per hill in SRI is?

5 SRI saves seed to about?

6 Heterosis means?

7 The female parent in hybrid rice is the?

8 The maintainer line in hybrid rice is the?

9 The restorer line in hybrid rice is the?

10 Hybrid rice gives this much more yield than inbreds?

Answers: 1-Madagascar; 2-8-12 days; 3-25 x 25 cm; 4-one; 5-5-8 kg/ha; 6-hybrid vigour; 7-A line (CMS line); 8-B line; 9-R line; 10-15-20 per cent

3.6 Exam focus

Key points to remember

- SRI was developed in Madagascar; its principles are young seedlings (8-12 days), one seedling per hill, wide square spacing (25 x 25 cm), shallow planting, alternate wetting and drying, and mechanical weeding with organic matter.
- SRI saves seed (5-8 kg/ha) and water (25-40 per cent) and can yield 10-30 per cent more where well managed; its weakness is weed management.
- Hybrid rice is F1 seed showing heterosis, giving about 15-20 per cent more yield than inbreds; seed is produced with the A, B and R (three-line) system.
- Hybrid seed rate is low (15-20 kg/ha) but the seed cost is high; the crop needs good management to realise its potential.
- SRI population at 25 x 25 cm is about 1.6 lakh hills per hectare against about 5 lakh at 20 x 10 cm.
- Both methods are answers to specific problems - SRI to water and seed scarcity, hybrid rice to the yield ceiling.

Short-answer questions with model answers

1 Q1. State the six principles of SRI.

- Young seedlings (8-12 days), single seedling per hill, wide square spacing (25 x 25 cm or wider), careful shallow transplanting, alternate wetting and drying instead of continuous flooding, and mechanical weeding with application of organic matter.

1 Q2. What are A, B and R lines in hybrid rice seed production?

- A line is the cytoplasmic male sterile (CMS) line, used as the female parent. B line is the maintainer, which maintains the sterility of the A line when crossed with it. R line is the restorer, which restores fertility in the F1 hybrid. The F1 seed harvested from the A line rows is the hybrid seed sold to farmers.

1 Q3. Give two advantages and one disadvantage of SRI.

- Advantages: it saves seed and water (about 25-40 per cent of water) and increases tillering and yield. Disadvantage: it needs more frequent weeding and careful water management (alternate wetting and drying), which is labour-intensive.

Long-answer questions with model-answer outlines

1 Q1. Explain SRI in detail with its principles, layout, advantages and limitations.

- Outline: (i) origin (Madagascar) and the concept of giving each plant more space, air and light; (ii) six principles with the science behind each; (iii) layout diagram at 25 x 25 cm; (iv) comparison table against conventional rice (seed, water, spacing, weeding, yield); (v) worked example of plant population (1.6 lakh hills/ha) and seed (8 kg/ha); (vi) advantages - seed, water, yield; (vii) limitations - weeding labour, need for discipline, smaller gains on heavy soils with abundant water; (viii) conclusion on where SRI is best suited.

1 Q2. Describe hybrid rice cultivation and hybrid seed production.

- Outline: (i) concept of heterosis and the 15-20 per cent yield advantage; (ii) the three-line system - A, B and R lines and their roles; (iii) the two-line system and its advantage; (iv) steps in seed production - isolation distance, planting ratio, staggered sowing, GA3 spray, supplementary pollination, roguing; (v) cultivation of the hybrid crop - low seed rate, higher nitrogen, careful pest management; (vi) a note on cost and the need for good management; (vii) conclusion.

Lecture 4 - Maize: Origin, Distribution, Economic Importance, Soil and Climatic Requirement, Varieties, Cultural Practices and Yield

4.1 Origin and distribution

Botanical identity

- Common name: Maize (English), Makka cholam (Tamil), Corn. Botanical name: Zea mays L. Family: Poaceae. Chromosome number 2n = 20.
- Maize is a monoecious annual grass - the male inflorescence (tassel) is at the top of the plant and the female inflorescence (ear/silk) is borne in the middle. It is cross-pollinated by wind.

Origin

- Maize originated in Central America, in the region of southern Mexico and Guatemala. Its wild ancestor is teosinte (Zea mexicana). It was domesticated about 7000-9000 years ago.
- The crop then spread to the Americas, and after the voyages of Columbus it spread to Europe, Africa and Asia. Maize is therefore a New World crop.

Geographic distribution

- World area is about 200 million hectares. The United States (the Corn Belt) is the largest producer, followed by China, Brazil, the European Union, Argentina, Mexico and India.
- In India the main states are Karnataka, Madhya Pradesh, Bihar, Maharashtra, Andhra Pradesh, Telangana, Tamil Nadu, Rajasthan and Uttar Pradesh.
- In Tamil Nadu maize is grown in Perambalur, Salem, Namakkal, Erode, Coimbatore, Tiruppur, Dindigul, Karur, Villupuram and other districts, largely as an irrigated crop, both for grain and for fodder.

4.2 Economic importance and nutritional value

- Maize is called the 'queen of cereals' because of its high yield potential and its many uses. It is used as human food (in the form of flour, grits, corn flakes and popcorn), as livestock and poultry feed (about 60 per cent of Indian maize goes to feed), as green and dry fodder, and as an industrial raw material.
- Industrial products from maize include starch, dextrose, glucose, corn syrup, corn oil, corn flakes, alcohol and ethanol (fuel and potable), and corn gluten feed.
- Maize is the third most important cereal in India after rice and wheat, and the most important feed grain.
- Nutritional value (per 100 g): about 70-72 per cent carbohydrate (starch), 9-10 per cent protein (low in lysine and tryptophan, which is why quality-protein maize, QPM, was developed), 4-5 per cent fat, about 342 kcal energy, and useful amounts of vitamin A (carotene in yellow maize), thiamine and iron.

- Green maize fodder is succulent and palatable and is a good fodder for cattle; it can be used for silage.

Nutrient (per 100 g maize grain)	Approximate content
Energy	about 342 kcal
Carbohydrate	70-72 g
Protein	9-10 g
Fat	4-5 g
Crude fibre	2-3 g
Vitamin A (yellow maize)	good source (carotene)
Limiting amino acids	lysine, tryptophan (improved in QPM)

4.3 Soil and climatic requirement

- Soil: maize grows best on deep, fertile, well-drained loam and sandy loam soils rich in organic matter. It does not tolerate waterlogging or salinity. Optimum pH is 5.5-7.5.
- Climate: maize is a warm-weather crop of the tropics and subtropics. Optimum temperature is about 21-27 degrees C. Germination needs a minimum of about 10 degrees C. It is very sensitive to frost and to waterlogging.
- Water: the crop needs about 500-800 mm of water. It is sensitive to moisture stress at tasselling, silking and grain filling - the critical stages - and to excess water at any stage.
- Maize is a short-day plant in many types, but the modern hybrids grown in Tamil Nadu are largely photo-insensitive, allowing them to be grown in several seasons.

Requirement	Optimum for maize
Soil	Deep, fertile, well-drained loam/sandy loam
Soil pH	5.5-7.5
Temperature	21-27 C
Water need	500-800 mm
Critical moisture stages	Tasselling, silking, grain filling
Sensitive to	Waterlogging, frost, salinity
Day length	Mostly photo-insensitive (modern hybrids)

4.4 Varieties

Maize varieties for Tamil Nadu include TNAU composites, TNAU hybrids and public/private single-cross hybrids. Hybrids out-yield composites and are preferred for grain and fodder.

- CO 1 - TNAU composite, medium duration (about 95-100 days), yellow grain, moderate yield.
- CO 6 - TNAU composite, about 95-100 days, orange-yellow flint grain; also used for fodder.
- COH(M) 5 - TNAU single-cross hybrid, short duration (about 85-90 days), high yield (about 6-7 t/ha), the standard short-duration hybrid.
- COH(M) 6, COH(M) 8, COH(M) 9 - TNAU single-cross hybrids of short to medium duration with high yield and good grain quality.
- NK 6240 - popular private single-cross hybrid of medium duration with high yield.

- Other public hybrids and composites (for example, from ICAR-IIMR) and quality-protein maize (QPM) hybrids are also available. Sweet corn and baby corn types are grown for specialty markets.

Choose hybrids for high yield and composites for stability and low seed cost. For fodder, tall leafy types and the fodder maize hybrids (see Lecture 16) are used.

4.5 Land preparation and sowing

- Plough the field once or twice with an iron/disc plough, then harrow to a fine tilth. Maize needs a well-drained, deeply prepared seed-bed.
- Form ridges and furrows at the time of sowing, or sow on the flat and form ridges later, to allow irrigation and drainage.
- Sow on the side of the ridge at about 5 cm depth. Treat seed with a fungicide and with Azospirillum and phosphobacteria.
- Sowing time: as an irrigated crop, maize is sown throughout the year in Tamil Nadu; the main seasons are June-July (kharif/Adi), September-October (rabi) and January-February (summer). As a rain-fed crop it is sown with the onset of the south-west monsoon in June-July.
- Maize can be raised as a sole crop or intercropped with pulses (cowpea, black gram) and with vegetables.

4.6 Seed rate and spacing

- Seed rate: about 20 kg/ha for hybrids and about 25 kg/ha for composites (larger, variable seed).
- Spacing: 60 x 20 cm for grain maize, giving about 83,000 plants per hectare. For fodder maize a closer spacing of 30 x 15 cm or 45 x 15 cm is used. Sweet corn and baby corn use closer spacing.
- Plant population: for grain, aim at 80,000-85,000 plants per hectare at harvest (sow a little more to allow for losses).

4.7 Nutrient management

- Standard fertiliser dose for irrigated hybrid maize: 250 : 75 : 75 kg N : P2O5 : K2O per hectare. (For some hybrids the recommended nitrogen is 200-250 kg/ha depending on the target yield.)
- Nitrogen is applied in three splits: half basal, one-quarter at knee-high stage (about 25-30 days), and one-quarter at tasselling/pre-tasselling (about 45-50 days).
- Apply 10-12.5 t/ha of farmyard manure or compost before sowing. Apply zinc sulphate 25 kg/ha where deficient.
- For fodder maize a higher nitrogen dose (up to 250-300 kg N/ha) with split application gives more green fodder.
- Bio-fertilisers (Azospirillum, phosphobacteria) and organic manures improve soil health and reduce the inorganic requirement.

Fertiliser for irrigated hybrid maize	Dose
FYM/compost	10-12.5 t/ha
N	250 kg/ha (3 splits)
P2O5	75 kg/ha (basal)
K2O	75 kg/ha (basal)
ZnSO4 (deficient soils)	25 kg/ha
N split timing	half basal; 1/4 at 25-30 d; 1/4 at 45-50 d

4.8 Water management

- Maize is sensitive to both drought and waterlogging. Irrigate at the critical stages - seedling establishment, knee-high, tasselling, silking and grain filling.
- As a guide, irrigate once every 10-12 days on light soils and once every 12-15 days on heavy soils in summer; the crop needs about 8-10 irrigations in the summer season and fewer in the rainy season.
- Drain excess water immediately after heavy rain; maize cannot tolerate standing water.
- Ridge planting and furrow irrigation allow both irrigation and drainage in the same field.

4.9 Weed management

- The first 30-45 days are the critical period for crop-weed competition. Weeds reduce yield heavily if not controlled early.
- Pre-emergence herbicide: apply atrazine (or a recommended alternate) immediately after sowing on moist soil.
- Hand weeding: one or two hand weedings, at about 20-25 days and 40-45 days, or a mechanical weeder between rows.
- Earthing-up: after the second weeding, earth up soil to the base of the plant to support the stem and cover weeds.

4.10 Pest and disease management

Major insect pests

- Fall Armyworm (Spodoptera frugiperda) - the most serious recent pest; the larva feeds in the whorl and gives ragged holes and moist sawdust-like frass. Management: scouting, hand-picking of larvae and egg masses, whorl application of sand/soil, biological control (Trichogramma, Bacillus thuringiensis, NPV), and need-based insecticide in the whorl. Early sowing and intercropping help.
- Stem borer (Chilo partellus, and the pink stem borer Sesamia inferens) - causes 'dead heart' in young plants and stem tunnelling later. Management: destroy stubbles, use Trichogramma and pheromone traps, and need-based insecticide.
- Shoot fly (Atherigona spp.) in the seedling stage; corn earworm and aphids are minor pests.

Major diseases

- Turcicum leaf blight (Exserohilum turcicum) - long, cigar-shaped grey-green lesions; severe in cool humid weather.
- Maydis leaf blight (Bipolaris maydis) and banded leaf and sheath blight (Rhizoctonia solani).
- Downy mildew (Peronosclerospora) - chlorotic streaks and 'crazy top'; serious in some tracts.
- Management: grow resistant hybrids, treat seed, rotate crops, destroy stubbles, and use fungicide only when needed.

4.11 Harvest and yield

- Harvest the grain crop when the husk (leaf sheath) turns dry and brown, the grains are hard and glazed, and a black layer forms at the base of the kernel - about 95-110 days for most hybrids and composites.
- For fodder, harvest at the milk to early-dough stage for the best balance of yield and quality; harvest green fodder at about 60-75 days.
- Dry the grain to 12-14 per cent moisture before storage. Shell the cobs and store in dry, aerated bins.

- Yield: about 6-7 t/ha for short-duration hybrids (COH(M) 5) and 7-9 t/ha for well-managed medium-duration hybrids under irrigation; composites yield about 4-5 t/ha. Rain-fed yields are lower (about 3-4 t/ha). Fodder maize gives 40-60 t/ha of green fodder.

4.12 Worked example - fertiliser for a given area

Problem: A farmer has 0.8 hectare of hybrid maize and wants to apply the standard 250 : 75 : 75 kg N : P2O5 : K2O per hectare. Calculate the quantity of urea (46 per cent N), single super phosphate (SSP, 16 per cent P2O5) and muriate of potash (MOP, 60 per cent K2O) required.

- For 0.8 ha, N needed = 250 x 0.8 = 200 kg; P2O5 = 75 x 0.8 = 60 kg; K2O = 75 x 0.8 = 60 kg.
- Urea = N needed / 0.46 = 200 / 0.46 = about 435 kg.
- SSP = P2O5 needed / 0.16 = 60 / 0.16 = 375 kg.
- MOP = K2O needed / 0.60 = 60 / 0.60 = 100 kg.
- Apply all SSP and MOP and half the urea (about 218 kg) as basal; apply one-quarter urea (about 109 kg) at the knee-high stage and one-quarter (about 109 kg) at tasselling.
- Check: 435 kg urea x 0.46 = 200 kg N; 375 kg SSP x 0.16 = 60 kg P2O5; 100 kg MOP x 0.60 = 60 kg K2O. The calculation matches the target dose for 0.8 hectare.

4.13 Maize growth stages - text diagram

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Maize growth stages (about 90-110 days):
  Emergence -> 6-leaf (V6) -> Knee-high (V8-V10) -> Tasselling/Silking
     (0-10 d)    (20-25 d)      (30-40 d)            (50-60 d)
                                                      |
  Critical stages for water and N: tasselling and silking (50-60 d)
     |
  Grain filling (milky -> dough -> dent) -> Maturity / black layer (95-110 d)
```

4.15 Cropping systems, intercropping and value addition

Maize in cropping systems

- Maize-pulse: maize followed by a short-duration pulse (black gram, green gram) or intercropped with a pulse.
- Maize-maize: two maize crops a year under irrigation (kharif and rabi), common in the poultry-feed belts.
- Maize-vegetables: maize followed by vegetables in garden land.
- Maize as an intercrop: maize + cowpea, maize + black gram and maize + vegetables are common; the legume fixes nitrogen and adds income.

Intercropping

- Maize is an excellent intercrop component because of its upright growth. Maize + cowpea (or black gram) in a 1:1 or 2:1 row ratio adds protein and nitrogen; maize + groundnut is also grown. Intercropping uses light, water and nutrients more efficiently and reduces the risk of a single-crop failure.

Value addition and special maize types

- Sweet corn: harvested at the milky stage for the table and the processing industry; commands a high price. Baby corn: young cobs harvested for the canning and restaurant trade; a short-duration, high-value crop.
- Popcorn and quality protein maize (QPM): QPM has about twice the lysine and tryptophan of normal maize and is promoted for nutrition.

- Industrial uses: starch, dextrose, glucose, corn syrup, corn oil and ethanol; maize silage for dairy cattle.

4.16 Comparison of important maize varieties

Variety	Duration	Type	Salient feature
CO 1	95-100 d	Composite	Yellow grain, moderate yield
CO 6	95-100 d	Composite	Orange-yellow flint; dual purpose
COH(M) 5	85-90 d	Single cross hybrid	Short duration, high yield (6-7 t/ha)
COH(M) 8	95-100 d	Single cross hybrid	High yield, good grain quality
COH(M) 9	90-95 d	Single cross hybrid	High yield, stress tolerance
NK 6240	100-110 d	Private hybrid	Popular high-yielding hybrid

Field calendar and common problems

Crop calendar (operations schedule)

Operation	Time (days after sowing)	Remarks
Land preparation	Before sowing	Deep plough; ridges and furrows; FYM
Sowing	0	20 kg/ha hybrid seed at 60 x 20 cm
Basal fertiliser	0	Half N, full P and K
Thinning and first weeding	15-20	Thin to one plant/hill; atrazine pre-em
Knee-high top-dress N	25-30	One-quarter N
Tasselling top-dress N	45-50	One-quarter N
Irrigation	Critical at tasselling/silking	8-10 irrigations in summer
Harvest	85-110	At black layer; dry to 12-14 per cent

Common field problems, causes and remedies

Problem	Likely cause	Remedy
Ragged whorls with frass	Fall armyworm	Scout; Bt/NPV; whorl insecticide
Dead heart in seedlings	Stem borer / shoot fly	Destroy stubble; Trichogramma; insecticide
Poor grain filling	Moisture stress at silking	Irrigate at tasselling and silking
Lodging	Excess N; wind	Reduce N; grow sturdy hybrids
Cigar-shaped leaf lesions	Turcicum leaf blight	Resistant hybrid; fungicide if severe

Objective-type practice questions

Choose the correct answer. Answers are given at the end of the set.

1 Maize belongs to the family?

2 The origin of maize is?

3 The wild ancestor of maize is?
4 Maize is called the?
5 The standard NPK dose for irrigated hybrid maize is?
6 The seed rate for hybrid maize is?
7 The critical moisture stages in maize are?
8 The most serious recent pest of maize is?
9 Maize grain contains about this much protein?
10 Quality protein maize is rich in?

Answers: 1-Poaceae; 2-Central America; 3-teosinte; 4-queen of cereals; 5-250:75:75; 6-20 kg/ha; 7-tasselling and silking; 8-fall armyworm; 9-9-10 per cent; 10-lysine and tryptophan

4.14 Exam focus

Key points to remember

- Zea mays, family Poaceae, 2n = 20; monoecious and wind-pollinated; origin Central America (Mexico/Guatemala), ancestor teosinte.
- Maize is the 'queen of cereals' - food, feed, fodder and industrial raw material; the third most important cereal of India.
- Grain: about 70-72 per cent carbohydrate, 9-10 per cent protein, 4-5 per cent fat, about 342 kcal per 100 g; low in lysine and tryptophan (improved in QPM).
- Soil: deep, well-drained loam, pH 5.5-7.5; sensitive to waterlogging and frost. Optimum temperature 21-27 C; water need 500-800 mm.
- TNAU varieties: CO 6 (composite), COH(M) 5 (short hybrid, 85-90 days), COH(M) 6/8/9; also NK 6240 and other hybrids.
- Seed rate 20 kg/ha (hybrid), 25 kg/ha (composite); spacing 60 x 20 cm; NPK 250 : 75 : 75 kg/ha.
- Major pests: fall armyworm, stem borer. Major diseases: turcicum leaf blight, downy mildew, banded leaf and sheath blight.
- Yield 6-9 t/ha (irrigated hybrids); fodder maize 40-60 t/ha green fodder.

Short-answer questions with model answers

1 Q1. Why is maize called the queen of cereals?

- Maize is called the queen of cereals because of its very high yield potential among cereals and its wide range of uses - human food, livestock and poultry feed, green and dry fodder, and many industrial products such as starch, oil, syrup, flakes and ethanol. No other cereal has such diverse use.

1 Q2. State the critical stages of moisture stress in maize.

- The critical stages are tasselling, silking and grain filling. Moisture stress at these stages causes poor pollination, incomplete grain filling and heavy yield loss. Maize is also intolerant of waterlogging at all stages.

1 Q3. Give the seed rate, spacing and fertiliser dose for irrigated hybrid maize.

- Seed rate 20 kg/ha; spacing 60 x 20 cm (about 83,000 plants/ha); fertiliser 250 : 75 : 75 kg N : P2O5 : K2O per hectare with nitrogen in three splits.

Long-answer questions with model-answer outlines

1 Q1. Write the complete package of practices for irrigated hybrid maize in Tamil Nadu.

- Outline: (i) origin and importance in one line; (ii) soil and climate; (iii) variety choice - COH(M) 5, COH(M) 8, NK 6240; (iv) land preparation - deep ploughing, ridges and furrows, FYM; (v) seed rate 20 kg/ha and spacing 60 x 20 cm with seed treatment; (vi) fertiliser 250:75:75 with split nitrogen and zinc; (vii) irrigation at critical stages; (viii) weed management with atrazine and hand weeding plus earthing-up; (ix) pest and disease management with emphasis on fall armyworm and leaf blight; (x) harvest at black layer and yield; (xi) economics in one line.

1 Q2. Describe the origin, distribution and economic importance of maize.

- Outline: (i) botanical identity and monoecious nature; (ii) origin in Central America and teosinte as ancestor; (iii) spread to the world and India; (iv) world and Indian distribution with leading states; (v) economic importance - food, feed, fodder and industry, with the feed share; (vi) nutritional value with figures; (vii) conclusion on why maize area is expanding in India.

Lecture 5 - Sorghum and Pearl Millet: Origin, Distribution, Economic Importance, Soil and Climatic Requirement, Varieties, Cultural Practices and Yield

Sorghum and pearl millet are the two most important millets of the semi-arid tropics. Both are drought-tolerant C4 cereals that secure food and fodder in rain-fed Tamil Nadu. This lecture treats each crop in turn, then compares them.

PART A - SORGHUM (Great Millet / Cholam)

5.1 Origin and distribution

- Common name: Sorghum, Jowar (Hindi), Cholam (Tamil), Great millet. Botanical name: Sorghum bicolor (L.) Moench. Family: Poaceae. Chromosome number 2n = 20.
- Origin: sorghum originated in Africa, in the Ethiopia-Sudan region (the Abyssinian centre of Vavilov), which is the primary centre of diversity; India is a secondary centre of diversity.
- Distribution: sorghum is the fifth most important cereal of the world. The main producers are the United States, Nigeria, India, Mexico, Sudan, Ethiopia, Argentina, China and Brazil. In India the main states are Maharashtra, Karnataka, Rajasthan, Madhya Pradesh, Andhra Pradesh, Telangana, Tamil Nadu and Gujarat.
- In Tamil Nadu sorghum is grown mainly as a rain-fed crop in Coimbatore, Erode, Salem, Namakkal, Dharmapuri, Krishnagiri, Virudhunagar, Madurai, Tirunelveli, Thoothukudi and Ramanathapuram, and as an irrigated crop for grain and fodder.

5.2 Economic importance and nutritional value

- Sorghum is a dual-purpose crop - grain for food and stover (dry fodder) for cattle. In the semi-arid tropics the stover is often as valuable as the grain.
- Grain uses: food (rotis, porridge), poultry and cattle feed, and industrial uses - starch, syrup, alcohol, and biofuel. Sweet sorghum is grown for juice and ethanol. Sorghum is gluten-free and is promoted for people with gluten intolerance.
- Nutritional value (per 100 g): about 70-72 per cent carbohydrate, 10-12 per cent protein, about 2-3 per cent fat, 1.6-2 per cent crude fibre, and about 349 kcal energy. It is a good source of iron and B vitamins, but the protein is limiting in lysine.
- Stover is a valuable dry fodder; some types are grown purely for fodder (see Lecture 15).

Nutrient (per 100 g sorghum grain)	Approximate content
Energy	about 349 kcal
Carbohydrate	70-72 g
Protein	10-12 g
Fat	2-3 g
Crude fibre	1.6-2 g
Notable minerals	iron, calcium, B vitamins

5.3 Soil and climatic requirement

- Soil: sorghum grows on a wide range of soils, from sandy loam to heavy clay. It performs best on deep, well-drained loams with pH 6.0-8.0. It is more tolerant of salinity and alkalinity than maize, and is fairly tolerant of poor soils.
- Climate: a warm-weather crop of the semi-arid tropics. Optimum temperature is about 25-30 degrees C. Sorghum is drought-tolerant because of its deep root system, waxy leaves and the ability to roll its leaves. It needs about 400-600 mm of water. It is sensitive to low temperature at flowering and to waterlogging.
- Sorghum is a short-day plant, but the modern hybrids and varieties grown in Tamil Nadu are largely photo-insensitive and can be grown in kharif, rabi and summer.

5.4 Varieties

- CO 30 - TNAU variety, medium duration (about 105-110 days), high-yielding dual purpose (grain and fodder), widely grown.
- CO 26, CO 28 - TNAU varieties of medium duration with good grain and fodder.
- CO 25 - TNAU variety of medium duration, dual purpose.
- K 12 - popular variety of medium duration grown for grain and fodder.
- COH 4 - TNAU hybrid with high yield potential (about 4-5 t/ha under irrigation).
- CSV (CSV 15, CSV 20 etc.) and CSH (CSH 16 etc.) are ICAR-bred varieties and hybrids of national importance.
- Multi-cut and single-cut fodder sorghums are separate (see Lecture 15).

5.5 Cultural practices

- Land preparation: plough once or twice and harrow to a fine tilth; form ridges and furrows. Sorghum needs a well-drained seed-bed.
- Sowing time: rain-fed kharif sowing with the onset of the south-west monsoon (June-July); irrigated crops in June-July, September-October and January-February.
- Seed rate: about 10 kg/ha for grain sorghum (line sowing); 12-15 kg/ha for fodder types. Treat seed with a fungicide and with Azospirillum.
- Spacing: 45 x 15 cm for grain sorghum (about 1.5 lakh plants/ha). For fodder, closer spacing of 30 x 10 cm or 45 x 10 cm is used.
- Thinning: thin the seedlings at 10-15 days to one plant per hill at the recommended spacing, to avoid overcrowding.
- Nutrient management: irrigated sorghum 100 : 50 : 50 kg N : P2O5 : K2O per hectare; rain-fed sorghum 40 : 20 : 0 to 50 : 25 : 0 kg/ha. Apply nitrogen in two splits (basal and 30-35 days). Apply FYM 10 t/ha.

- Water management: sorghum is largely rain-fed. Where irrigated, irrigate at the critical stages - boot leaf, flowering and grain filling. Avoid waterlogging.
- Weed management: two hand weedings or a pre-emergence herbicide (atrazine) followed by one hand weeding and earthing-up. The first 30 days are critical.
- Harvest and yield: harvest when the grain is hard and the peduncle turns yellow, about 100-115 days for medium varieties. Grain yield about 3-4 t/ha under irrigation and 1.5-2.5 t/ha rain-fed; stover 4-6 t/ha.

PART B - PEARL MILLET (Cumbu / Bajra)

5.6 Origin and distribution

- Common name: Pearl millet, Bajra (Hindi), Cumbu (Tamil). Botanical name: Pennisetum glaucum (L.) R. Br. (syn. P. typhoides, P. americanum). Family: Poaceae. Chromosome number 2n = 14.
- Origin: pearl millet originated in Africa, in the Sahel region of West Africa (the centre of diversity is the Sahel and India is a secondary centre).
- Distribution: pearl millet is the most widely grown millet of the world by area. The main producers are India, Nigeria, Niger, Mali, Burkina Faso, Sudan, Senegal and Pakistan. In India the main states are Rajasthan, Maharashtra, Gujarat, Uttar Pradesh, Haryana, Karnataka, Andhra Pradesh and Tamil Nadu.
- In Tamil Nadu pearl millet is grown in Coimbatore, Erode, Salem, Namakkal, Dharmapuri, Krishnagiri, Virudhunagar, Tirunelveli, Thoothukudi and Ramanathapuram, mostly as a rain-fed crop.

5.7 Economic importance and nutritional value

- Pearl millet is the hardiest cereal - the most drought-tolerant of all cereals - and is the food of the driest, most marginal lands. It is a dual-purpose crop (grain and fodder).
- Uses: food (bajra roti, porridge), poultry and cattle feed, and fodder (green and dry). It is also used for malting and for biofuel in some countries.
- Nutritional value (per 100 g): about 67-70 per cent carbohydrate, 11-12 per cent protein (higher than rice and wheat), about 5 per cent fat, 1.2-2 per cent crude fibre, and about 361 kcal energy. It is rich in iron, and pearl millet is relatively high in protein among cereals.

5.8 Soil and climatic requirement

- Soil: pearl millet grows on light, sandy, well-drained soils and tolerates poor fertility, salinity and alkalinity better than most cereals. It performs best on sandy loam with pH 6.5-8.0.
- Climate: the crop of the hottest, driest areas. Optimum temperature about 25-35 degrees C. It needs only about 300-500 mm of water and tolerates high temperature and drought because of its deep, dense root system and rapid growth. It is very sensitive to waterlogging.

5.9 Varieties

- CO 9 - TNAU variety, short duration (about 85-90 days), high-yielding, dual purpose; the standard short-duration cumbu.
- CO 10 - TNAU variety, short duration, good grain and fodder.
- CO 8 - TNAU variety of short duration.
- X 7 - improved variety of short duration.
- WCC 75 - a popular high-yielding variety.
- ICMV 221 - an ICRISAT-bred variety of short duration with downy mildew resistance, widely grown.

- Hybrids such as the ICAR/private bajra hybrids give higher yields where inputs are assured.

5.10 Cultural practices

- Land preparation: a shallow ploughing and harrowing are enough; pearl millet needs a fine, well-drained seed-bed and does not need deep tillage.
- Sowing time: rain-fed kharif with the onset of the monsoon (June-July); it can also be grown in summer under irrigation (January-February).
- Seed rate: about 4 kg/ha for line sowing and 6-8 kg/ha for broadcasting. Seed is small, so the rate is low.
- Spacing: 45 x 15 cm for grain (about 1.5 lakh plants/ha); closer spacing for fodder.
- Nutrient management: rain-fed crop 40 : 20 : 0 kg N : P2O5 : K2O per hectare; irrigated crop about 80 : 40 : 20 kg/ha (some recommendations use up to 100 : 50 : 50 with high management). Apply FYM 5-10 t/ha.
- Water management: mostly rain-fed; where irrigated, irrigate at tillering, flowering and grain filling. Drain excess water.
- Weed management: one or two hand weedings or a pre-emergence herbicide; the crop is a poor competitor early and needs a weed-free first 30 days.
- Harvest and yield: harvest when the ear heads are mature and grain is hard, about 85-95 days for short-duration varieties. Grain yield about 2-2.5 t/ha under irrigation and 1-1.5 t/ha rain-fed; fodder 4-5 t/ha.

5.11 Comparison of sorghum and pearl millet

Feature	Sorghum (Cholam)	Pearl millet (Cumbu)
Botanical name	Sorghum bicolor	Pennisetum glaucum
Chromosome number	2n = 20	2n = 14
Origin	Africa (Ethiopia-Sudan)	Africa (Sahel, West Africa)
Duration (TNAU)	about 105-110 days (CO 30)	about 85-90 days (CO 9)
Protein (per 100 g)	10-12 g	11-12 g
Energy (per 100 g)	about 349 kcal	about 361 kcal
Drought tolerance	High	Very high (hardiest cereal)
Seed rate	about 10 kg/ha	about 4 kg/ha
Spacing	45 x 15 cm	45 x 15 cm
NPK (irrigated)	100 : 50 : 50	80 : 40 : 20
NPK (rain-fed)	40 : 20 : 0	40 : 20 : 0
Irrigated yield	3-4 t/ha	2-2.5 t/ha

5.12 Worked example - seed rate from desired plant population

Problem: A farmer wants a final plant population of 1,50,000 plants per hectare of pearl millet at a spacing of 45 x 15 cm. The seed has 85 per cent germination and 90 per cent field survival, and the 1000-grain weight is 8 g. Calculate the seed rate.

- Target plants per hectare = 1,50,000.

- Allow for germination and field survival: seed to be placed = 1,50,000 / (0.85 x 0.90) = 1,50,000 / 0.765 = 1,96,078 seeds per hectare.
- Seed weight = 1,96,078 x 8 g / 1000 = 1,569 g = about 1.57 kg per hectare of pure, viable seed.
- The standard recommendation is 4 kg/ha because it includes a margin for bird damage, uneven establishment and a small amount of seed that is not placed at the correct spacing. The calculation shows the biological minimum; the recommendation adds a practical safety margin.

5.14 Cropping systems, intercropping and value addition

Sorghum and pearl millet in cropping systems

- Sorghum-pulse: sorghum followed by a short-duration pulse; sorghum + pigeon pea or sorghum + cowpea as intercrops.
- Pearl millet-pulse: pearl millet followed by a pulse or intercropped with cowpea, green gram or cluster bean.
- Millet-groundnut and millet-cotton rotations are common in the dry tracts.
- These millets are the main crops of rain-fed dryland agriculture, where a mixed or intercropped stand spreads the risk of a monsoon failure.

Intercropping

- Sorghum + pigeon pea (2:1 or 4:2 rows), sorghum + cowpea, pearl millet + cowpea and pearl millet + cluster bean are standard dryland intercrops. The legume fixes nitrogen, provides protein and improves fodder quality.

Value addition and special uses

- Sorghum: roti flour, malt (for beverages and health drinks), poha and flakes; sweet sorghum for juice, syrup and ethanol; the stover for dry fodder.
- Pearl millet: flour and porridge (bajra), malt, ready-to-eat foods, and the green fodder for cattle. Both are gluten-free and are promoted as nutri-cereals with a price premium.
- Both crops also support the livestock economy through stover and green fodder, which is often as valuable as the grain in dryland areas.

Field calendar and common problems

Crop calendar (operations schedule)

Operation	Time (days)	Remarks
Land preparation	Before sowing	Shallow plough; ridges and furrows
Sowing	0	Sorghum 10 kg/ha; pearl millet 4 kg/ha
Thinning	10-15	One plant/hill at 45 x 15 cm
Basal fertiliser	0	Full P; part N
Top-dress N	30-35	Remaining N
Weeding	20 and 35	One or two weedings; earthing-up
Harvest	85-115	Sorghum 105-110 d; pearl millet 85-90 d

Common field problems, causes and remedies

Problem	Likely cause	Remedy
Dead heart in seedlings	Shoot fly	Timely sowing; seed treatment; insecticide
Ear head / panicle not filling	Sorghum midge; drought	Manage midge; irrigate at flowering
Downy mildew (chlorotic streaks)	Peronosclerospora	Resistant varieties; seed treatment
Grain mould in wet harvest	Fungal complex	Timely harvest; dry quickly
Low rain-fed yield	Moisture stress	Drought-tolerant varieties; moisture conservation

Objective-type practice questions

Choose the correct answer. Answers are given at the end of the set.

1 Sorghum belongs to the family?
2 The origin of sorghum is?
3 The chromosome number of pearl millet is?
4 The hardiest, most drought-tolerant cereal is?
5 The TNAU short-duration pearl millet variety is?
6 The standard NPK for irrigated sorghum is?
7 The seed rate for pearl millet is?
8 Shoot fly and stem borer are pests of?
9 Downy mildew is an important disease of?
10 Pearl millet grain contains about this much protein?

Answers: 1-Poaceae; 2-Africa (Ethiopia-Sudan); 3-2n=14; 4-pearl millet; 5-CO 9; 6-100:50:50; 7-4 kg/ha; 8-sorghum; 9-sorghum and pearl millet; 10-11-12 per cent

5.13 Exam focus

Key points to remember

- Sorghum: Sorghum bicolor, 2n = 20, origin Africa (Ethiopia-Sudan); dual purpose (grain + stover); TNAU variety CO 30 (105-110 days); NPK 100:50:50 (irrigated) and 40:20:0 (rain-fed); seed 10 kg/ha; spacing 45 x 15 cm.
- Pearl millet: Pennisetum glaucum, 2n = 14, origin Africa (Sahel); the hardiest, most drought-tolerant cereal; TNAU variety CO 9 (85-90 days); NPK 80:40:20 (irrigated) and 40:20:0 (rain-fed); seed 4 kg/ha; spacing 45 x 15 cm.
- Both are C4, short-day, largely photo-insensitive cereals tolerant of poor soils, salinity and drought; both are sensitive to waterlogging.
- Sorghum grain about 10-12 per cent protein and 349 kcal; pearl millet about 11-12 per cent protein and 361 kcal per 100 g.
- Major pests: shoot fly, stem borer, shoot bug, sorghum midge. Major diseases: downy mildew, grain mould, charcoal rot, ergot.
- Irrigated yields: sorghum 3-4 t/ha, pearl millet 2-2.5 t/ha; rain-fed yields are lower.

Short-answer questions with model answers

1 Q1. Why is pearl millet considered the hardiest cereal?

- Pearl millet is the most drought-tolerant cereal because of its deep, dense and extensive root system, its rapid growth, its ability to reduce transpiration and its tolerance of high temperature, poor soils, salinity and low rainfall (300-500 mm). It can produce a crop where other cereals fail.

1 Q2. Give the botanical name and chromosome number of sorghum and pearl millet.

- Sorghum: Sorghum bicolor, 2n = 20. Pearl millet: Pennisetum glaucum, 2n = 14. Both belong to the family Poaceae.

1 Q3. State the fertiliser dose for irrigated and rain-fed sorghum.

- Irrigated sorghum: 100 : 50 : 50 kg N : P2O5 : K2O per hectare. Rain-fed sorghum: 40 : 20 : 0 kg/ha. Nitrogen is applied in two splits.

Long-answer questions with model-answer outlines

1 Q1. Describe the origin, distribution, economic importance and cultural practices of pearl millet.

- Outline: (i) botanical identity (Pennisetum glaucum, 2n = 14) and origin in the West African Sahel; (ii) world and Indian distribution with leading states; (iii) economic importance - hardiest cereal, food, feed and fodder, dual purpose; (iv) nutritional value with figures; (v) soil and climate (sandy, pH 6.5-8.0, 25-35 C, 300-500 mm); (vi) varieties CO 9, CO 10, WCC 75, ICMV 221; (vii) cultural practices - land preparation, seed 4 kg/ha, spacing 45 x 15 cm, NPK 80:40:20 (irrigated), weed and water management; (viii) harvest and yield; (ix) a one-line comparison with sorghum.

1 Q2. Compare sorghum and pearl millet under the headings of origin, botany, nutritional value, agronomy and yield.

- Outline: a structured comparison using the comparison table - botanical names and chromosome numbers, centres of origin, durations, protein and energy contents, drought tolerance, seed rates, spacing, fertiliser doses, and irrigated/rain-fed yields. Conclude with the point that pearl millet is the more drought-hardy and short-duration crop, while sorghum is the higher-yielding dual-purpose crop with better stover.

Lecture 6 - Finger Millet and Minor Millets: Origin, Distribution, Economic Importance, Soil and Climatic Requirement, Varieties, Cultural Practices and Yield

PART A - FINGER MILLET (Ragi / Kezhvaragu)

6.1 Origin and distribution

- Common name: Finger millet, Ragi (Kannada/Hindi), Kezhvaragu (Tamil). Botanical name: Eleusine coracana (L.) Gaertn. Family: Poaceae. Chromosome number 2n = 36.
- Origin: finger millet originated in Africa, in the Ethiopian and Ugandan highlands, and India is a secondary centre of diversity. It is one of the oldest cultivated millets of Africa and India.
- Distribution: the crop is important in India and in East and Southern Africa (Uganda, Kenya, Tanzania, Ethiopia, Zimbabwe). In India the main states are Karnataka, Tamil Nadu, Uttarakhand, Maharashtra, Odisha, Andhra Pradesh and Jharkhand. In Tamil Nadu finger millet is grown in Salem, Namakkal, Dharmapuri, Krishnagiri, Erode, Coimbatore, Vellore, Tiruvannamalai, Madurai, Dindigul and the Nilgiris, largely as a rain-fed crop but also under irrigation.

6.2 Economic importance and nutritional value

- Finger millet is the most important small millet of India after sorghum and pearl millet. It is a staple in the hilly and dry tracts and a crop of the poor and the marginal lands.

- Uses: food (ragi mudde, kali, porridge, malt, health foods, biscuits), and its straw is a valued dry fodder. Ragi malt and sprouted ragi are popular health foods, especially for children.
- Finger millet is nutritionally the richest of the common millets. It is exceptionally rich in calcium, which makes it valuable for growing children, pregnant women and the elderly.
- Nutritional value (per 100 g): about 72-73 per cent carbohydrate, 6-7 per cent protein, about 1.3 per cent fat, 2-3 per cent crude fibre, about 336 kcal energy, and about 0.3-0.4 per cent calcium (one of the richest cereal sources of calcium). It also contains iron and B vitamins.

Finger millet is the richest cereal source of calcium (roughly 300-350 mg per 100 g), far higher than rice, wheat or maize. This single fact is a common examination point.

6.3 Soil and climatic requirement

- Soil: finger millet grows on a wide range of soils, from red sandy loam to black and clay loams, and on poor, gravelly and lateritic soils. It is fairly tolerant of acidity (pH 5.0-8.0) and does well on well-drained soils rich in organic matter.
- Climate: it is grown from the plains to high altitudes (up to about 2000 m). Optimum temperature about 20-30 degrees C. It needs about 700-1000 mm of water and tolerates moderate drought. It is not tolerant of waterlogging.

6.4 Varieties

- CO(Ra) 14 - TNAU variety of medium duration with high yield and good quality.
- CO 12 - TNAU variety of short duration (about 100-105 days), suited to rain-fed and irrigated conditions.
- CO 13, CO 15 - TNAU varieties of medium duration with high yield and good grain quality.
- GPU 28 - a popular high-yielding variety (from GKVK, Bengaluru) with blast tolerance, widely grown in South India.
- ML 365 - a variety with good yield and quality.
- Indaf 9 and VL 149 (hills) are other important varieties. Blast-resistant varieties are preferred in high-rainfall and hilly areas.

6.5 Cultural practices

- Land preparation: two ploughings and harrowing to a fine tilth; finger millet seed is small, so a fine, firm, well-drained seed-bed is needed. Form beds and channels where irrigated.
- Sowing time: rain-fed kharif with the onset of the monsoon (June-July); irrigated crops in June-July and September-October; transplanted ragi (for high yields) is raised in a nursery and transplanted at 25-30 days.
- Seed rate: about 10 kg/ha for line sowing and 20 kg/ha for broadcasting (or about 5-7 kg/ha for transplanted ragi).
- Spacing: 22.5 x 10 cm for line sowing; wider spacing in fertile soils.
- Nutrient management: irrigated finger millet about 60 : 30 : 30 kg N : P2O5 : K2O per hectare; rain-fed crop about 50 : 25 : 25 or 40 : 20 : 20 kg/ha. Apply FYM 10 t/ha. Nitrogen is applied in two splits.
- Water management: mostly rain-fed; where irrigated, irrigate at establishment, tillering, ear-head emergence and grain filling. Avoid waterlogging.
- Weed management: two hand weedings or a pre-emergence herbicide; the crop is a weak competitor early and needs a weed-free first 30-35 days. Earthing-up after weeding supports the plant.
- Harvest and yield: harvest when the ear heads turn brown and the grain is hard, about 100-115 days. Grain yield about 2.5-3 t/ha under irrigation and 1.5-2 t/ha rain-fed; straw 3-4 t/ha.

PART B - MINOR MILLETS

6.6 The minor millets and their identity

The minor millets are a group of small-grained cereals grown on marginal lands. They are highly nutritious, drought-tolerant and climate-resilient, and are promoted as 'nutri-cereals'. The main ones are kodo, little, foxtail, proso and barnyard millet.

Minor millet	Botanical name	Chromosome (2n)	Tamil name
Kodo millet	Paspalum scrobiculatum	40	Varagu
Little millet	Panicum sumatrense	36	Samai
Foxtail millet	Setaria italica	18	Tenai
Proso millet	Panicum miliaceum	36	Panivaragu
Barnyard millet	Echinochloa frumentacea	54	Kuthiraivali

6.7 Origin and distribution of minor millets

- Kodo millet: origin in India (peninsular India is a centre of diversity); grown in Madhya Pradesh, Maharashtra, Karnataka, Tamil Nadu and Odisha.
- Little millet: origin in India (the Indian sub-continent is a centre of origin); grown in Odisha, Madhya Pradesh, Tamil Nadu, Karnataka, Andhra Pradesh and the North-East.
- Foxtail millet: one of the oldest cultivated millets, origin in East Asia (China) with a secondary centre in India; grown in Karnataka, Andhra Pradesh, Telangana, Tamil Nadu, Maharashtra and the North-East.
- Proso millet: origin in Central and East Asia; grown in small areas in India (Karnataka, Tamil Nadu, the North-East).
- Barnyard millet: origin in India and East Asia; grown in Uttarakhand, Tamil Nadu, Andhra Pradesh and the North-East.

6.8 Economic importance and nutritional value

- Minor millets are grown on poor, dry, marginal soils where other crops fail, and are important for food and nutritional security of tribal and hill communities. They are now promoted as nutri-cereals for their high fibre, minerals and antioxidants, and for being gluten-free with a low glycemic index.
- Fodder: the straw of these millets is a valued dry fodder.
- Nutritional value: minor millets generally contain about 65-72 per cent carbohydrate, 7-12 per cent protein, 2-5 per cent fat, and useful fibre, iron, calcium and B vitamins. Foxtail and kodo millets are notably high in protein and minerals; barnyard millet is rich in fibre.

Millet	Protein (g/100g)	Fat (g/100g)	Calcium (mg/100g)	Energy (kcal)
Finger millet	6-7	1.3	300-350	336
Kodo millet	8-9	2-4	27-35	350-353
Little millet	7-8	4-5	15-20	340-346
Foxtail millet	11-12	4	30-40	350-360
Proso millet	11-12	3-4	15-20	350-360
Barnyard millet	6-11	2-4	20-30	300-350

The figures above are standard textbook ranges; exact values vary with variety and locality. Finger millet is outstanding for calcium; foxtail and proso are high in protein.

6.9 Soil and climatic requirement of minor millets

- Soil: minor millets grow on the poorest soils - light, sandy, gravelly, lateritic and shallow soils - and tolerate low fertility and moderate acidity. They are not suited to heavy, waterlogged soils.
- Climate: short-duration, warm-weather crops of the dry tropics. They need only about 300-500 mm of water, tolerate high temperature and drought, and are the most climate-resilient of cereals. They are very sensitive to waterlogging.

6.10 Varieties of minor millets

- TNAU has released improved varieties of the minor millets, for example CO 4 and CO 5 in little millet, CO 5 and CO 6 in foxtail millet, and CO 3 in kodo millet. The exact current list should be checked against the latest TNAU Crop Production Guide.
- ICAR-IIMR and the All India Coordinated Research Project on Small Millets have released varieties (for example, in kodo, little, foxtail and barnyard millet) that are high-yielding and disease-resistant.
- Duration of minor millets is short, generally about 70-110 days depending on the crop and season.

6.11 Cultural practices of minor millets

- Land preparation: one or two shallow ploughings and harrowing to a fine tilth; these crops need a fine, well-drained seed-bed and no deep tillage.
- Sowing time: rain-fed kharif with the monsoon (June-July); some are also grown in rabi and summer under irrigation.
- Seed rate: about 4-8 kg/ha for line sowing (the seed is very small); broadcasting uses a higher rate.
- Spacing: 22.5 x 10 cm to 30 x 10 cm depending on the crop.
- Nutrient management: mostly rain-fed, about 40 : 20 : 0 kg N : P2O5 : K2O per hectare; irrigated crops receive about 40-60 : 20-30 : 0-20 kg/ha. FYM 5 t/ha. Nitrogen in two splits.
- Water management: rain-fed; where irrigated, light irrigations at establishment and grain filling. Drain excess water.
- Weed management: one or two hand weedings; the crops are poor competitors early.
- Harvest and yield: harvest when the ear heads/panicles mature and grain hardens, about 70-110 days. Grain yield about 1-1.5 t/ha rain-fed and up to 1.5-2 t/ha under good management; straw 2-3 t/ha.

6.12 Comparison of finger millet and the minor millets

Feature	Finger millet	Minor millets (general)
Botanical name	Eleusine coracana	Paspalum, Panicum, Setaria, Echinochloa
Chromosome number	2n = 36	2n = 18 to 54 (varies)
Origin	Africa (Ethiopian highlands)	India and East/Central Asia (varies)
Duration	about 100-115 days	about 70-110 days
Special nutrition	Highest calcium (300-350 mg/100 g)	High protein (foxtail, kodo) and fibre
Seed rate	10-20 kg/ha	4-8 kg/ha (line)
Spacing	22.5 x 10 cm	22.5-30 x 10 cm
NPK (irrigated)	60 : 30 : 30	40-60 : 20-30 : 0-20

Feature	Finger millet	Minor millets (general)
Yield (rain-fed)	1.5-2 t/ha	1-1.5 t/ha

6.13 Worked example - cost of cultivation and return for a minor millet

Problem: A farmer grows 1 hectare of rain-fed little millet. Costs: land preparation Rs. 3,000; seed Rs. 500; FYM Rs. 2,500; fertiliser Rs. 1,500; sowing and weeding labour Rs. 4,000; harvesting Rs. 3,000; total Rs. 14,500. Yield is 1.2 t/ha and the grain sells at Rs. 3,500 per quintal. Calculate gross and net return.

- Gross return = 1.2 t/ha = 12 quintals x Rs. 3,500 = Rs. 42,000 per hectare.
- Net return = Rs. 42,000 - Rs. 14,500 = Rs. 27,500 per hectare.
- Benefit-cost ratio = 42,000 / 14,500 = about 2.9.
- Minor millets give a lower yield than major cereals but a higher price (nutri-cereal premium) and a very low cost of cultivation, which is why the B:C ratio is attractive on marginal land.

6.15 Processing, value addition and cropping systems

Processing and value addition

- Finger millet: sprouting (malting) improves the availability of minerals and vitamins; ragi malt, ragi flour, ragi mudde and health mixes are high-value products. Decortication (removing the seed coat) improves cooking quality.
- Minor millets: dehulling, milling into flour, popping (for kodo and little millet), and ready-to-cook/ready-to-eat mixes. Millet flakes, puffed millet and millet-based snacks are growing markets because of the nutri-cereal promotion.
- Millet value chains are supported by State and central schemes that provide a minimum support price for millets and encourage millet-based products.

Cropping systems

- Finger millet-pulse: ragi followed by a short-duration pulse, or ragi + field bean/avare (a common mixed crop in Karnataka and Tamil Nadu).
- Millet-groundnut and millet-millet rotations in the dry tracts.
- Minor millets are often grown as mixed crops with other millets, pulses and oilseeds on marginal land; the mixture spreads risk and gives a range of foods.
- In the hilly tracts (the Nilgiris and the Western Ghats), finger millet is grown with potato, beans and other crops in terraced and mixed systems.

Field calendar and common problems

Crop calendar (operations schedule)

Operation	Time (days)	Remarks
Land preparation	Before sowing	Fine seed-bed; beds and channels
Sowing / nursery	0	Ragi 10 kg/ha line (5-7 kg for transplanting)
Transplanting (ragi)	25-30 day seedlings	For high yields
Basal fertiliser	0	Full P and K; part N
Top-dress N	25-30	Remaining N
Weeding and earthing-up	20 and 40	Keep weed-free first month

Operation	Time (days)	Remarks
Harvest	70-115	Finger millet 100-115 d; minor millets 70-110 d

Common field problems, causes and remedies

Problem	Likely cause	Remedy
Finger / ear head blast	Blast fungus	Resistant variety (GPU 28); balanced N
Poor establishment	Small seed; crusting; drought	Fine seed-bed; light irrigation; thin
Low yield on poor soil	Low fertility	FYM; small fertiliser dose; better variety
Lodging in ragi	Excess N	Reduce N; grow sturdy varieties
Waterlogging damage	Poor drainage	Beds and channels; drain

Objective-type practice questions

Choose the correct answer. Answers are given at the end of the set.

1 Finger millet belongs to the family?
2 The chromosome number of finger millet is?
3 The cereal richest in calcium is?
4 Finger millet is called?
5 Kodo millet is botanically?
6 Little millet is botanically?
7 Foxtail millet is botanically?
8 Barnyard millet is botanically?
9 The chromosome number of foxtail millet is?
10 Minor millets are now called?

Answers: 1-Poaceae; 2-2n=36; 3-finger millet; 4-ragi / kezhvaragu; 5-Paspalum scrobiculatum; 6-Panicum sumatrense; 7-Setaria italica; 8-Echinochloa frumentacea; 9-2n=18; 10-nutri-cereals

6.14 Exam focus

Key points to remember

- Finger millet: Eleusine coracana, 2n = 36, origin Africa (Ethiopian highlands); richest cereal in calcium (300-350 mg/100 g); TNAU varieties CO(Ra) 14, CO 12, CO 13, CO 15; seed 10-20 kg/ha; spacing 22.5 x 10 cm; NPK 60:30:30 (irrigated).
- Minor millets: kodo (Varagu), little (Samai), foxtail (Tenai), proso (Panivaragu), barnyard (Kuthiraivali); chromosome numbers vary from 2n = 18 (foxtail) to 2n = 54 (barnyard).
- Minor millets are short-duration (70-110 days), drought-tolerant, climate-resilient nutri-cereals grown on marginal soils.
- Foxtail and proso millets are high in protein (about 11-12 per cent); barnyard is high in fibre; kodo is high in protein and minerals.
- Seed rates for minor millets are low (4-8 kg/ha) because the seed is very small; they need only 300-500 mm of water.
- All millets are sensitive to waterlogging and respond to a fine seed-bed and a weed-free first month.

Short-answer questions with model answers

1 Q1. Why is finger millet important in the diet of children and pregnant women?

- Finger millet is the richest cereal source of calcium (about 300-350 mg per 100 g), and also contains iron and B vitamins. Calcium and iron are especially needed by growing children and by pregnant and lactating women, which makes finger millet a valuable protective food.

1 Q2. Give the botanical names of any four minor millets.

- Kodo millet - Paspalum scrobiculatum; little millet - Panicum sumatrense; foxtail millet - Setaria italica; barnyard millet - Echinochloa frumentacea. (Proso millet is Panicum miliaceum.)

1 Q3. State the seed rate and spacing for finger millet.

- Seed rate about 10 kg/ha for line sowing and 20 kg/ha for broadcasting (about 5-7 kg/ha for transplanted ragi); spacing about 22.5 x 10 cm for line sowing.

Long-answer questions with model-answer outlines

1 Q1. Describe the origin, importance and cultural practices of finger millet.

- Outline: (i) botanical identity (Eleusine coracana, 2n = 36) and African origin with India as secondary centre; (ii) distribution in India and Tamil Nadu; (iii) economic importance and the exceptional calcium content; (iv) nutritional value with figures; (v) soil and climate; (vi) varieties CO(Ra) 14, CO 12, CO 13, GPU 28; (vii) cultural practices - land preparation, seed rate, spacing, NPK 60:30:30, water and weed management; (viii) harvest and yield; (ix) a line on its role as a nutri-cereal.

1 Q2. Write short notes on the minor millets of Tamil Nadu.

- Outline: a crop-by-crop note on kodo, little, foxtail, proso and barnyard millet - botanical name, Tamil name, chromosome number, origin, duration, special nutritional feature, seed rate, spacing and yield; then a common paragraph on their agronomy (marginal soils, low water need, short duration, weed-free first month) and their importance as climate-resilient nutri-cereals.

Lecture 7 - Pigeon Pea: Origin, Distribution, Economic Importance, Soil and Climatic Requirement, Varieties, Cultural Practices and Yield

7.1 Origin and distribution

- Common name: Pigeon pea, Red gram, Tur/Arhar (Hindi), Thuvarai (Tamil). Botanical name: Cajanus cajan (L.) Millsp. Family: Fabaceae (Leguminosae), sub-family Papilionaceae. Chromosome number 2n = 22.
- Origin: pigeon pea originated in India, in peninsular India (the Indo-Burma region), which is the primary centre of diversity. Africa is a secondary centre.
- Distribution: India is the largest producer and consumer of pigeon pea in the world. Other producers are Myanmar, Malawi, Tanzania, Kenya, Uganda, Nigeria and the Dominican Republic. In India the main states are Maharashtra, Karnataka, Madhya Pradesh, Gujarat, Uttar Pradesh, Andhra Pradesh, Telangana, Jharkhand and Tamil Nadu.
- In Tamil Nadu pigeon pea is grown as a rain-fed crop and as an irrigated crop, and it is a very important component of intercropping systems (with sorghum, groundnut, cotton and maize). Main districts include Coimbatore, Erode, Salem, Namakkal, Dharmapuri, Krishnagiri, Vellore, Tiruvannamalai, Virudhunagar, Madurai, Tirunelveli, Ramanathapuram and Cuddalore.

7.2 Economic importance and nutritional value

- Pigeon pea is the second most important pulse of India (after chickpea). It is a major source of dietary protein for a largely vegetarian population, and it is a dual-purpose crop - grain for food and stalks for fuel and thatch.
- It is a deep-rooted legume that fixes atmospheric nitrogen through Rhizobium, improves soil fertility, and is an excellent component of intercropping and crop-rotation systems. It is a classic 'restorative' crop of dryland agriculture.
- Uses: dhal (split seed), whole seed, green pods as a vegetable, and the stalks as fuel; the leaves are used as fodder.
- Nutritional value (per 100 g): about 20-22 per cent protein (among the highest among pulses), 57-60 per cent carbohydrate, 1.5-2 per cent fat, and about 343 kcal energy. It is rich in lysine (complementing the lysine-poor cereals) but low in methionine. It is a good source of iron, calcium and B vitamins.

Nutrient (per 100 g pigeon pea, dry)	Approximate content
Energy	about 343 kcal
Protein	20-22 g
Carbohydrate	57-60 g
Fat	1.5-2 g
Crude fibre	1.5-2 g
Limiting amino acid	methionine (rich in lysine)
Notable minerals	iron, calcium, B vitamins

7.3 Soil and climatic requirement

- Soil: pigeon pea grows on a wide range of soils, from sandy loam to heavy black cotton soils. It performs best on deep, well-drained loams with pH 6.5-7.5, and tolerates a wider range of pH 5.5-8.5 and poor soils. It is sensitive to waterlogging and to salinity.
- Climate: a warm-weather, semi-arid crop. Optimum temperature about 20-30 degrees C. It needs about 600-1000 mm of water and is fairly drought-tolerant because of its deep tap root. It is a short-day plant, and the traditional long-duration types are strongly photoperiod-sensitive.

7.4 Varieties

Pigeon pea varieties are grouped by duration into short (about 120-140 days), medium (about 150-180 days) and long (above 180 days). Short and medium duration varieties suit Tamil Nadu and fit into intercropping and multiple cropping.

- CO 5 - TNAU variety, medium duration, high yield.
- CO 6 - TNAU variety, short-medium duration (about 120-140 days), suited to rain-fed and intercropping situations.
- CO 7 - TNAU variety, medium duration, with good yield and wilt tolerance.
- CO(Rg) 7 - a TNAU release of medium duration.
- VBN 1, VBN 2, VBN 3 - TNAU varieties of short-medium duration with good yield and disease tolerance.
- APK 1 - a variety suited to rain-fed conditions.
- SA 1 - a variety with good performance in the southern districts.
- BRG 2 and LRG 41 - varieties of medium duration grown in the pulse tracts.

- Wilt-resistant and sterility-mosaic-resistant varieties are preferred, because wilt and sterility mosaic are the two most destructive diseases of pigeon pea.

7.5 Land preparation and sowing

- Land preparation: one or two ploughings and harrowing to a fine tilth; form ridges and furrows. On heavy soils, ensure good drainage. Pigeon pea, being deep-rooted, benefits from a deep initial ploughing.
- Sowing time: rain-fed kharif with the onset of the south-west monsoon (June-July); it can also be sown in September-October as a rabi crop and in some areas in February-March under irrigation.
- Seed treatment: treat seed with Rhizobium and phosphobacteria (for nitrogen fixation and phosphorus solubilisation) and with a fungicide (for wilt and other seed-borne diseases).
- Seed rate: about 15 kg/ha for a sole crop (short/medium duration); 12-15 kg/ha where intercropped.
- Spacing: 60 x 30 cm for medium duration; 45 x 30 cm for short duration; wider spacing (90 x 20 cm or 150 x 30 cm) for long-duration types and for intercropping.

7.6 Nutrient management

- Standard fertiliser dose: 25 : 50 : 25 kg N : P2O5 : K2O per hectare. Being a legume, pigeon pea needs little nitrogen, but a starter dose of 25 kg N/ha helps early growth.
- Apply FYM or compost 5-10 t/ha. Apply zinc sulphate and sulphur where deficient. Apply gypsum or lime as per soil test.
- Rhizobium seed inoculation can meet much of the nitrogen requirement and improves soil nitrogen for the succeeding crop.

7.7 Water, weed, pest and disease management

Water management

- Pigeon pea is largely rain-fed and drought-tolerant, but it responds to irrigation at the flowering and pod-filling stages. Avoid waterlogging, which causes wilt and root rot.

Weed management

- Keep the crop weed-free for the first 30-40 days. One or two hand weedings, or a pre-emergence herbicide followed by one hand weeding. Earthing-up after weeding supports the plant.

Major insect pests

- Pod borer (Helicoverpa armigera) - the most serious pest; the larva bores into the pod and destroys the seed. Management: pheromone traps, NPV (HaNPV), biological control, timely sowing, and need-based insecticide.
- Pod fly (Melanagromyza obtusa), blister beetle, jassids, aphids, pod-sucking bugs and the plume moth are other pests. IPM is essential because pigeon pea is a long-duration crop exposed to many pest generations.

Major diseases

- Wilt (Fusarium udum) - the most destructive disease; causes sudden wilting and death. Management: resistant varieties, crop rotation, seed treatment, and use of Trichoderma.
- Sterility mosaic virus (spread by the eriophyid mite) - causes bushy, pale, sterile plants. Management: resistant varieties, mite control, and removal of infected plants.
- Phytophthora blight, root rot and powdery mildew are other diseases.

7.8 Harvest and yield

- Harvest when the pods turn brown and the seed rattles inside, about 120-140 days (short duration), 150-180 days (medium) and 180-200 days (long duration). Do not delay, or the pods shatter.
- Harvest in two or three pickings for the long-duration types, or as a single harvest for short-duration types. Dry, thresh and winnow the seed and store at 8-10 per cent moisture.
- Yield: about 1.0-1.5 t/ha for medium-duration varieties under good management; short-duration varieties yield about 0.8-1.2 t/ha; rain-fed yields are lower (about 0.6-1.0 t/ha).

7.9 Worked example - plant population and seed rate

Problem: A farmer grows pigeon pea (medium duration, CO 7) as a sole crop at 60 x 30 cm on 1 hectare. The 1000-grain weight is 80 g, germination 90 per cent and field survival 90 per cent. Calculate the number of plants per hectare and the seed rate.

- Plants per hectare = 10000 / (0.60 x 0.30) = 10000 / 0.18 = 55,556 plants per hectare (about 55,500).
- Seed to be placed = 55,556 / (0.90 x 0.90) = 55,556 / 0.81 = 68,588 seeds per hectare.
- Seed weight = 68,588 x 80 g / 1000 = 5,487 g = about 5.5 kg per hectare of pure viable seed.
- The standard recommendation is about 15 kg/ha because pigeon pea seed is larger, establishment is uneven, and a margin is kept for bird damage and gap-filling. The biological minimum of about 5.5 kg shows the effect of wide spacing and large seed; the recommendation adds a practical margin.

7.10 Package of practices at a glance - pigeon pea

Item	Recommendation
Season	Kharif (June-July); also rabi (Sep-Oct)
Varieties	CO 5, CO 6, CO 7, VBN 1-3, APK 1, SA 1
Duration	Short 120-140 d; medium 150-180 d
Seed rate	about 15 kg/ha (sole crop)
Spacing	60 x 30 cm (medium); 45 x 30 cm (short)
Seed treatment	Rhizobium + phosphobacteria + fungicide
FYM	5-10 t/ha
NPK (kg/ha)	25 : 50 : 25
Water	Mostly rain-fed; irrigate at flowering and pod filling
Major pests	Pod borer (Helicoverpa), pod fly
Major diseases	Wilt (Fusarium), sterility mosaic
Yield	1.0-1.5 t/ha (medium duration)

7.12 Cropping systems and intercropping

Pigeon pea in cropping systems

- Pigeon pea is the classic intercrop component of dryland agriculture. Its deep root system, slow early growth and upright habit make it an ideal companion for shorter, faster crops.
- Common intercrops: pigeon pea + sorghum (2:1 or 1:1), pigeon pea + groundnut (1:3 or 1:2), pigeon pea + cotton, pigeon pea + maize, pigeon pea + black gram and pigeon pea + castor.

- Row ratios are chosen so that the total plant population of the two crops is close to the sole-crop population of the main crop. For example, in a 2:1 pigeon pea-sorghum system, the sorghum rows are thinned to about two-thirds of their sole-crop population.
- Benefits: better use of light, water and nutrients (different rooting depths and growth rhythms); nitrogen fixation by pigeon pea; and a spread of risk - if one crop fails, the other survives.

Crop rotation

- Pigeon pea is a restorative crop and is rotated with cereals (sorghum, maize, rice) to break pest and disease cycles (especially wilt) and to restore soil nitrogen. A pigeon pea-cereal rotation is a standard recommendation for dryland and for wilt-prone fields.

Field calendar and common problems

Crop calendar (operations schedule)

Operation	Time (days)	Remarks
Land preparation	Before sowing	Deep plough; ridges and furrows
Sowing	0	15 kg/ha at 60 x 30 cm; Rhizobium seed treatment
Basal fertiliser	0	Full P and K; starter N
First weeding	30-40	Keep weed-free early
Flowering irrigation	90-110	If irrigated; also pod filling
Pest management	Flowering to podding	Pod borer IPM (NPV, pheromone traps)
Harvest	120-180	By duration group; avoid shattering

Common field problems, causes and remedies

Problem	Likely cause	Remedy
Sudden wilting and death	Fusarium wilt	Resistant variety; rotation; Trichoderma
Bushy, pale, sterile plants	Sterility mosaic (mite)	Resistant variety; mite control; rogue out
Pods with holes / damaged seed	Pod borer, pod fly	IPM; timely sowing; need-based insecticide
Poor pod set	Drought or excess N	Irrigate at flowering; balance fertiliser
Slow early growth	Weed competition	Early weeding; correct spacing

Objective-type practice questions

Choose the correct answer. Answers are given at the end of the set.

1 Pigeon pea belongs to the family?
2 The chromosome number of pigeon pea is?
3 The origin of pigeon pea is?
4 The most serious pest of pigeon pea is?
5 Wilt in pigeon pea is caused by?
6 Sterility mosaic is transmitted by?
7 The standard NPK for pigeon pea is?
8 The seed rate for pigeon pea is?
9 Pigeon pea is a?

10 The duration of medium-duration pigeon pea is?

Answers: 1-Fabaceae; 2-2n=22; 3-India (peninsular); 4-pod borer (Helicoverpa); 5-Fusarium udum; 6-the eriophyid mite; 7-25:50:25; 8-about 15 kg/ha; 9-deep-rooted legume (restorative crop); 10-150-180 days

7.11 Exam focus

Key points to remember

- Cajanus cajan, family Fabaceae, 2n = 22; origin peninsular India (primary centre), Africa secondary.
- India is the largest producer and consumer of pigeon pea; the second most important pulse of India.
- Protein 20-22 per cent (rich in lysine, low in methionine), about 343 kcal per 100 g; a dual-purpose crop (grain + stalks).
- Deep-rooted legume that fixes nitrogen and improves soil; an excellent intercrop and restorative crop.
- Soil pH 6.5-7.5 (tolerant 5.5-8.5), sensitive to waterlogging; temperature 20-30 C; water need 600-1000 mm.
- TNAU varieties: CO 5, CO 6, CO 7, VBN 1-3, APK 1, SA 1; seed rate about 15 kg/ha; spacing 60 x 30 cm; NPK 25:50:25.
- Major pest: pod borer (Helicoverpa). Major diseases: wilt (Fusarium) and sterility mosaic.
- Yield about 1.0-1.5 t/ha (medium duration).

Short-answer questions with model answers

1 Q1. Why is pigeon pea called a restorative crop?

- Pigeon pea is a deep-rooted legume that fixes atmospheric nitrogen in root nodules through Rhizobium, adds organic matter through leaf fall and root residues, and draws nutrients from deeper layers to the surface. It thus improves soil fertility and is called a restorative crop in dryland agriculture.

1 Q2. Name the two most destructive diseases of pigeon pea and their causal agents.

- Wilt, caused by Fusarium udum; and sterility mosaic, caused by the sterility mosaic virus transmitted by the eriophyid mite. Both are managed mainly through resistant varieties.

1 Q3. Give the seed rate, spacing and fertiliser dose for pigeon pea.

- Seed rate about 15 kg/ha; spacing 60 x 30 cm for medium duration (45 x 30 cm for short duration); fertiliser 25 : 50 : 25 kg N : P2O5 : K2O per hectare.

Long-answer questions with model-answer outlines

1 Q1. Describe the origin, importance, varieties and cultural practices of pigeon pea.

- Outline: (i) botanical identity and origin in peninsular India; (ii) distribution and Indian states; (iii) economic importance - protein source, dual purpose, nitrogen fixation, intercropping; (iv) nutritional value with figures; (v) soil and climate; (vi) varieties by duration group with emphasis on TNAU varieties and disease resistance; (vii) cultural practices - land preparation, seed treatment with Rhizobium, seed rate, spacing, NPK 25:50:25, water and weed management; (viii) pest and disease management with emphasis on pod borer and wilt; (ix) harvest and yield.

1 Q2. Explain the role of pigeon pea in intercropping and dryland farming.

- Outline: (i) deep root system and different rooting depth from cereals; (ii) nitrogen fixation and soil improvement; (iii) suitable intercrops - sorghum, groundnut, cotton, maize - and the row ratios; (iv) reduction of risk (two crops in one season); (v) efficient use of resources - water, light, nutrients; (vi) the role of short-duration varieties in multiple cropping; (vii) a note on the effect on the succeeding crop's yield.

Lecture 8 - Green Gram, Black Gram and Cowpea; Agronomy of Rice Fallow Pulses

The three short-duration kharif pulses - green gram, black gram and cowpea - are the main pulse crops of Tamil Nadu. They fix nitrogen, need little water, mature in about two months and fit into the rice-fallow and dryland windows. This lecture treats the three crops in turn, compares them, and then explains the special agronomy of rice fallow pulses.

PART A - GREEN GRAM (Mung / Pachaippayaru)

8.1 Origin, distribution and importance

- Common name: Green gram, Mung bean, Moong (Hindi), Pachaippayaru (Tamil). Botanical name: Vigna radiata (L.) Wilczek. Family: Fabaceae. Chromosome number 2n = 22.
- Origin: India, in the Indo-Burma region (the primary centre of diversity). It spread early to South-East Asia and Africa.
- Distribution: India, Myanmar, China, Thailand, Indonesia, Bangladesh and Pakistan are the main producers; India is the largest producer. In Tamil Nadu it is grown in almost all districts, especially Coimbatore, Erode, Salem, Namakkal, Dharmapuri, Vellore, Tiruvannamalai, Madurai, Virudhunagar, Tirunelveli, Cuddalore and the delta districts.
- Importance: a short-duration (60-75 day) pulse that is grown as a sole crop, as an intercrop and as a rice-fallow crop. It is used as dhal (whole and split), as a vegetable (green pods), and as green manure; it fixes nitrogen and improves soil.
- Nutritional value (per 100 g): about 24 per cent protein, 56-60 per cent carbohydrate, 1-1.5 per cent fat, about 334 kcal energy, and useful iron, calcium and B vitamins. Like other pulses it is rich in lysine and low in methionine.

8.2 Soil, climate, varieties and agronomy

- Soil: grows on a wide range of soils, from sandy loam to black soils; best on well-drained loams with pH 6.5-7.5; sensitive to waterlogging and salinity.
- Climate: a warm-weather, short-day crop; optimum temperature 25-30 degrees C; needs about 350-500 mm of water; sensitive to heavy rain and waterlogging.
- Varieties: CO 6, CO 7, CO 8 (TNAU, short duration, high yield); VBN 2, VBN 3, VBN 4 (TNAU, resistant to mung bean yellow mosaic virus, MYMV); KM 2; and other State/ICAR varieties. MYMV-resistant varieties are preferred because yellow mosaic is the most serious disease.
- Sowing time: kharif (June-July), rabi (Sep-Oct) and summer/rice-fallow (Dec-Feb).
- Seed rate: about 15-20 kg/ha (20-25 kg/ha for broadcasting and rice-fallow).
- Spacing: 30 x 10 cm for a sole crop; wider rows where intercropped.
- Nutrient management: 25 : 50 : 25 kg N : P2O5 : K2O per hectare; Rhizobium and phosphobacteria seed treatment; FYM 5 t/ha. Being a legume, green gram needs little nitrogen.
- Water management: mostly rain-fed; where irrigated, light irrigations at flowering and pod filling. Drain excess water.
- Weed management: keep weed-free for the first 25-30 days; one or two hand weedings or a pre-emergence herbicide.
- Pest and disease management: pests - pod borer, aphids, jassids, whitefly (vector of MYMV), thrips. Diseases - mung bean yellow mosaic virus (MYMV, spread by whitefly), powdery mildew, root rot and leaf spot. Management: resistant varieties, whitefly control, seed treatment, IPM.

- Harvest and yield: harvest when pods turn black/brown and dry, about 60-75 days. Yield about 0.8-1.2 t/ha (irrigated/good management), 0.4-0.7 t/ha rain-fed.

PART B - BLACK GRAM (Urad / Ulundu)

8.3 Origin, distribution and importance

- Common name: Black gram, Urad (Hindi), Ulundu (Tamil). Botanical name: Vigna mungo (L.) Hepper. Family: Fabaceae. Chromosome number 2n = 22.
- Origin: India (the Indo-Burma region is the primary centre of diversity).
- Distribution: India, Myanmar, Pakistan, Bangladesh, Sri Lanka and Thailand; India is the largest producer. In Tamil Nadu it is grown in the delta districts (as a rice-fallow crop), and in Coimbatore, Erode, Salem, Madurai, Virudhunagar, Tirunelveli and other districts.
- Importance: black gram is the main pulse of the Cauvery delta rice-fallow system and a key ingredient of idli/dosa batter (with rice), which gives it steady demand. It fixes nitrogen and is used as a green-manure and cover crop.
- Nutritional value (per 100 g): about 24 per cent protein, 56-60 per cent carbohydrate, 1-1.5 per cent fat, about 341 kcal energy, and useful iron, calcium and B vitamins. Rich in lysine.

8.4 Soil, climate, varieties and agronomy

- Soil: grows on a wide range of soils; best on heavy and black soils that hold moisture; pH 6.5-7.5; sensitive to waterlogging.
- Climate: warm-weather, short-day crop; optimum temperature 25-30 degrees C; needs about 350-500 mm of water; very sensitive to waterlogging, which makes it ideal for the moisture-retentive delta soils.
- Varieties: CO 5, CO 6 (TNAU); VBN 4, VBN 5, VBN 6, VBN 8 (TNAU, with MYMV and disease tolerance); ADT 3, ADT 5 (short duration, suited to rice fallow); TMV 1; T 9. Rice-fallow and short-duration types are preferred for the delta.
- Sowing time: kharif (June-July), rabi (Sep-Oct) and rice-fallow (Nov-Feb).
- Seed rate: about 15-20 kg/ha (25-30 kg/ha for broadcasting in rice fallow).
- Spacing: 30 x 10 cm for a sole crop.
- Nutrient management: 25 : 50 : 25 kg N : P2O5 : K2O per hectare; Rhizobium and phosphobacteria seed treatment; FYM 5 t/ha.
- Water management: mostly rain-fed or on residual moisture; light irrigation at flowering and pod filling where available. Drain excess water.
- Weed management: keep weed-free for the first 25-30 days; one or two hand weedings or a pre-emergence herbicide.
- Pest and disease management: pests - whitefly (vector of MYMV), aphids, jassids, pod borer, thrips. Diseases - mung bean yellow mosaic virus, powdery mildew, leaf spot, root rot. Management: resistant varieties, whitefly control, seed treatment, IPM.
- Harvest and yield: harvest when pods turn black and dry, about 60-70 days. Yield about 0.7-1.0 t/ha (good management), 0.3-0.6 t/ha rain-fed/rice-fallow.

PART C - COWPEA (Lobia / Karamani)

8.5 Origin, distribution and importance

- Common name: Cowpea, Lobia (Hindi), Karamani (Tamil), Black-eyed pea. Botanical name: Vigna unguiculata (L.) Walp. Family: Fabaceae. Chromosome number 2n = 22.

- Origin: Africa, in the West African savanna (the primary centre of diversity); India is a secondary centre.
- Distribution: widely grown in Africa, India, Brazil and the southern United States. In Tamil Nadu it is grown for grain, for green pods (vegetable), and as a fodder and green-manure crop.
- Importance: cowpea is the most versatile of the kharif pulses - used as grain (dhal), as a vegetable (tender pods and green seed), as fodder (green and hay) and as green manure/cover crop. It is drought-tolerant, fixes nitrogen and smothers weeds.
- Nutritional value (per 100 g): about 22-24 per cent protein, 55-60 per cent carbohydrate, 1-2 per cent fat, about 340 kcal energy, and useful iron, calcium and B vitamins. Rich in lysine.

8.6 Soil, climate, varieties and agronomy

- Soil: grows on a wide range of soils, from sandy to clay; best on well-drained sandy loam and loam; pH 5.5-7.5; tolerates poor soils and moderate drought but not waterlogging.
- Climate: warm-weather crop; optimum temperature 25-30 degrees C; needs about 400-600 mm of water; drought-tolerant.
- Varieties: CO 2, CO 3, CO 4 (TNAU, grain and dual-purpose); VBN 1, VBN 2 (TNAU); Pusa Komal and other State/ICAR varieties; fodder cowpea types are separate (see Lecture 15).
- Sowing time: kharif (June-July) and summer (Jan-Feb); also as a rice-fallow and intercrop.
- Seed rate: about 20-25 kg/ha for grain; closer and higher for fodder and green manure.
- Spacing: 45 x 15 cm for grain (or 30 x 10 cm for vegetable types and fodder).
- Nutrient management: 25 : 50 : 25 kg N : P2O5 : K2O per hectare; Rhizobium and phosphobacteria seed treatment; FYM 5 t/ha.
- Water management: mostly rain-fed; irrigate at flowering and pod filling where available.
- Weed management: keep weed-free for the first 30 days; the crop then covers the ground and suppresses weeds.
- Pest and disease management: pests - aphids, jassids, pod borer, thrips. Diseases - cowpea mosaic, bacterial blight, powdery mildew, root rot. Management: resistant varieties, seed treatment, IPM.
- Harvest and yield: for grain, harvest when pods dry and turn brown, about 70-90 days; for vegetable, pick tender pods at 50-60 days. Grain yield about 0.8-1.2 t/ha; green fodder 20-30 t/ha.

8.7 Comparison of the three kharif pulses

Feature	Green gram	Black gram	Cowpea
Botanical name	Vigna radiata	Vigna mungo	Vigna unguiculata
Chromosome (2n)	22	22	22
Origin	India (Indo-Burma)	India (Indo-Burma)	Africa (W. Africa)
Duration	60-75 days	60-70 days	70-90 days
Protein (per 100 g)	about 24 g	about 24 g	22-24 g
Seed rate	15-20 kg/ha	15-20 kg/ha	20-25 kg/ha
Spacing	30 x 10 cm	30 x 10 cm	45 x 15 cm
NPK	25 : 50 : 25	25 : 50 : 25	25 : 50 : 25
Key disease	MYMV (whitefly)	MYMV (whitefly)	Cowpea mosaic
Yield	0.8-1.2 t/ha	0.7-1.0 t/ha	0.8-1.2 t/ha

PART D - AGRONOMY OF RICE FALLOW PULSES

8.8 Concept of rice fallow pulses

- Rice fallow pulses are short-duration pulse crops (mainly black gram and green gram) grown in a rice field after the rice harvest, using the residual soil moisture stored in the soil profile, with little or no irrigation.
- The system is widespread in the Cauvery delta (Thanjavur, Thiruvarur, Nagapattinam, Mayiladuthurai) and in other rice tracts, where the Thaladi or Samba rice is harvested in December-January and the land would otherwise lie fallow until the next rice crop.
- The pulse crop is sown into the standing moisture, usually without ploughing (zero or minimum tillage), either by broadcasting and covering the seed lightly or by sowing in the rice stubble.

8.9 Agronomy of rice fallow pulses

- Land preparation: none, or only a light harrowing; the seed is broadcast or dibbled into the moist soil immediately after the rice harvest, before the soil dries. Zero tillage conserves moisture and saves cost.
- Sowing time: immediately after rice harvest, generally late November to mid-January in the delta; late sowing means less residual moisture and lower yield.
- Varieties: short-duration (60-65 day) rice-fallow types - black gram ADT 3, ADT 5 and VBN 4; green gram CO 6, CO 7 and VBN 2.
- Seed rate: higher than a normal sole crop - about 25-30 kg/ha - because broadcasting needs a thicker stand and establishment is less certain.
- Spacing: broadcast or line-sown at about 25-30 x 10 cm.
- Nutrient management: a small starter dose of 25 : 50 : 25 kg N : P2O5 : K2O per hectare, or a foliar spray of 2 per cent DAP/urea if the crop is pale; Rhizobium seed treatment. On residual moisture, fertiliser response is modest.
- Water management: the crop depends on residual moisture; if the soil dries and the crop shows stress at flowering/pod filling, one protective irrigation gives a large benefit.
- Weed management: the pulse crop usually follows a puddled, weed-free rice field, so weed pressure is low; one hand weeding may be enough.
- Pest and disease management: the main problems are whitefly-transmitted MYMV, aphids, and pod borers. Follow IPM and use resistant varieties.
- Harvest and yield: harvest at 60-65 days. Yield is lower than an irrigated sole crop - about 300-500 kg/ha for black gram and green gram - but the cost is very low, so the net return per hectare is attractive.

8.10 Benefits of rice fallow pulses

- Additional income from land that would otherwise be fallow, with very low cost.
- Nitrogen fixation and addition of organic matter, which improve the fertility of the following rice crop (a saving of about 20-25 kg N/ha for the next rice).
- Weed suppression, better soil structure and reduced soil erosion.
- Improved household nutrition and the use of a short window before the next rice season.

8.11 Worked example - net return from a rice fallow black gram crop

Problem: A delta farmer broadcasts black gram (ADT 5) in 1 hectare of rice fallow. Costs: seed 28 kg at Rs. 100/kg = Rs. 2,800; seed treatment and fertiliser Rs. 1,500; sowing labour Rs. 1,000; one hand weeding Rs. 1,500; harvesting and threshing Rs. 3,000; total Rs. 9,800. Yield is 450 kg/ha (4.5 quintals) and the grain sells at Rs. 7,000 per quintal. Calculate the net return and the B:C ratio.

- Gross return = 4.5 quintals x Rs. 7,000 = Rs. 31,500 per hectare.
- Net return = Rs. 31,500 - Rs. 9,800 = Rs. 21,700 per hectare.
- Benefit-cost ratio = 31,500 / 9,800 = about 3.2.
- Add the value of nitrogen fixed (about 20-25 kg N/ha, worth roughly Rs. 1,200-1,500) as a benefit to the next rice crop. The very low cost of cultivation is the key to the high B:C ratio.

8.13 Cropping systems, seed production and value addition

Pulses in cropping systems

- Pulses are grown as sole crops, as intercrops and as rice-fallow crops. Their nitrogen fixation and short duration make them ideal for fitting into gaps between two cereal crops.
- Common intercrops: green gram + sorghum, black gram + maize, cowpea + sorghum or maize, and groundnut + pigeon pea. Cereal + pulse intercropping gives a balanced output and improves soil fertility.
- Crop rotation: cereal-pulse rotations (rice-pulse, sorghum-pulse, maize-pulse) are standard, and pulses are also grown after a cereal to use residual moisture and nutrients.

Seed production of pulses

- Pulses are largely self-pollinated, so seed production is easier than for cross-pollinated crops. Maintain an isolation distance (about 3-5 m from other varieties of the same crop).
- Rogue out off-types and diseased plants (especially those with yellow mosaic) throughout the season. Harvest the seed crop from the middle of the field, dry to about 8-10 per cent moisture and store with a seed treatment.
- Seed replacement and certified seed are important because the on-farm seed of pulses is often of low quality and carries seed-borne viruses.

Value addition

- Black gram and rice are the base of idli/dosa batter, a large and growing industry in Tamil Nadu. Dhal milling, papad and vadam making, and the preparation of pulse flour and protein isolates add value.
- The pulses are also used as green manure (cowpea, sunhemp) and as fodder, so the whole plant is used.

Field calendar and common problems

Crop calendar (operations schedule)

Operation	Time (days)	Remarks
Land preparation	Before sowing	Fine tilth; or zero tillage in rice fallow
Sowing	0	15-20 kg/ha (25-30 in rice fallow)
Seed treatment	Before sowing	Rhizobium + phosphobacteria + fungicide
Basal fertiliser	0	25:50:25 (or starter in rice fallow)
Weeding	20-30	One or two weedings

Operation	Time (days)	Remarks
Protective irrigation	Flowering/pod filling (if dry)	One irrigation gives a large benefit
Harvest	60-90	Pods turn black/brown and dry

Common field problems, causes and remedies

Problem	Likely cause	Remedy
Yellow mosaic (yellow patches)	MYMV via whitefly	Resistant variety; control whitefly
Poor stand in rice fallow	Sowing delayed; soil too dry	Sow immediately after harvest; higher seed rate
Powdery mildew	Fungus in humid weather	Resistant variety; need-based fungicide
Flower drop and low pod set	Moisture stress	One protective irrigation; balance fertiliser
Seed-borne virus in next crop	Infected seed	Use certified seed; roguing

Objective-type practice questions

Choose the correct answer. Answers are given at the end of the set.

1 Green gram is botanically?
2 Black gram is botanically?
3 Cowpea is botanically?
4 The origin of cowpea is?
5 The most serious disease of green gram and black gram is?
6 MYMV is transmitted by?
7 Rice-fallow pulses are grown on?
8 The seed rate for a rice-fallow pulse is?
9 The standard NPK for pulses is?
10 The protein content of pulses is about?

Answers: 1-Vigna radiata; 2-Vigna mungo; 3-Vigna unguiculata; 4-West Africa; 5-MYMV (yellow mosaic); 6-whitefly; 7-residual soil moisture; 8-25-30 kg/ha; 9-25:50:25; 10-22-24 per cent

8.12 Exam focus

Key points to remember

- Green gram: Vigna radiata, 2n = 22, origin India; 60-75 days; seed 15-20 kg/ha; spacing 30 x 10 cm; NPK 25:50:25; yield 0.8-1.2 t/ha; key disease MYMV.
- Black gram: Vigna mungo, 2n = 22, origin India; 60-70 days; seed 15-20 kg/ha; spacing 30 x 10 cm; NPK 25:50:25; key disease MYMV; the main rice-fallow pulse of the delta.
- Cowpea: Vigna unguiculata, 2n = 22, origin West Africa; 70-90 days; seed 20-25 kg/ha; spacing 45 x 15 cm; the most versatile pulse (grain, vegetable, fodder, green manure).
- All three fix nitrogen through Rhizobium, are rich in lysine, and contain about 22-24 per cent protein.
- Rice fallow pulses are short-duration (60-65 day) black gram and green gram grown on residual moisture after rice, mostly without tillage and with little or no irrigation.
- Rice-fallow sowing must be immediate (Nov-Jan); seed rate is higher (25-30 kg/ha); yield is 300-500 kg/ha but the B:C ratio is high (about 3 or more) because the cost is very low.

Short-answer questions with model answers

1 Q1. What are rice fallow pulses? Name two suitable crops.

- Rice fallow pulses are short-duration pulse crops grown after the rice harvest on residual soil moisture, with little or no irrigation and usually without tillage. Black gram and green gram are the two main rice-fallow pulses of the Cauvery delta.

1 Q2. Give the botanical names and chromosome numbers of green gram, black gram and cowpea.

- Green gram - Vigna radiata; black gram - Vigna mungo; cowpea - Vigna unguiculata. All three belong to the family Fabaceae with chromosome number 2n = 22.

1 Q3. Why is the seed rate of a rice-fallow pulse higher than that of a normal sole crop?

- A rice-fallow pulse is broadcast on residual moisture without tillage, so establishment is less certain and the stand must be thick enough to cover the ground. The seed rate is therefore raised to about 25-30 kg/ha against 15-20 kg/ha for a sole crop.

Long-answer questions with model-answer outlines

1 Q1. Write the agronomy of rice fallow pulses in the Cauvery delta.

- Outline: (i) definition and the delta context; (ii) why land lies fallow after Thaladi/Samba and how the pulse uses the window; (iii) crops and varieties - black gram ADT 3, ADT 5, VBN 4; green gram CO 6, VBN 2; (iv) zero/minimum tillage and immediate sowing to conserve moisture; (v) higher seed rate (25-30 kg/ha) and broadcast/line sowing; (vi) starter fertiliser and one protective irrigation if needed; (vii) low weed pressure and pest/disease management (MYMV, aphids, pod borers); (viii) harvest at 60-65 days and yield of 300-500 kg/ha; (ix) worked net return and the nitrogen benefit to the next rice crop; (x) conclusion on the system's profitability and sustainability.

1 Q2. Compare green gram, black gram and cowpea under the headings of origin, botany, duration, seed rate, spacing and yield.

- Outline: a structured comparison using the table - botanical names, chromosome number, centres of origin (India for green and black gram, West Africa for cowpea), durations, protein content, seed rates, spacing, fertiliser dose (common 25:50:25), key diseases and yields. Conclude with the point that all three are short-duration, nitrogen-fixing pulses of the kharif season, differing mainly in origin, duration and their place in the cropping system.

Lecture 9 - Mid-Semester Examination (Revision and Consolidation of Lectures 1-8)

The syllabus lists a mid-semester examination at this point. This chapter is a consolidation of Lectures 1 to 8 - the two rice lectures, the special methods of rice cultivation, and the cereals, millets and pulses covered so far. It gathers the standard figures into revision tables and provides practice questions in the usual examination pattern.

9.1 Scope of the mid-semester examination

- The mid-semester examination covers the first half of the syllabus: rice (origin, distribution, importance, varieties, soil and climate; cultural practices, yield, economics), SRI and hybrid rice, maize, sorghum and pearl millet, finger millet and minor millets, pigeon pea, and green gram, black gram and cowpea including rice fallow pulses.
- Typical pattern: short-answer questions (2-5 marks), short notes (5 marks), and one or two long-answer questions (10-15 marks), with a compulsory numerical question on seed rate, plant population or fertiliser.

9.2 Master revision table - cereals and millets

Crop	Botanical name (2n)	Origin	Duration	Seed rate	Spacing	NPK (irrigated)
Rice	Oryza sativa (24)	Indo-Burma/ China	105-140 d	40-60 kg/ha	20 x 10 cm	150:50:50
Maize	Zea mays (20)	C. America	85-110 d	20-25 kg/ha	60 x 20 cm	250:75:75
Sorghum	Sorghum bicolor (20)	Africa	105-110 d	10 kg/ha	45 x 15 cm	100:50:50
Pearl millet	Pennisetum glaucum (14)	Africa (Sahel)	85-90 d	4 kg/ha	45 x 15 cm	80:40:20
Finger millet	Eleusine coracana (36)	Africa	100-115 d	10-20 kg/ha	22.5 x 10 cm	60:30:30
Little millet	Panicum sumatrense (36)	India	70-100 d	4-8 kg/ha	22.5-30 x 10 cm	40-60:20-30:0-20

9.3 Master revision table - pulses

Pulse	Botanical name (2n)	Origin	Duration	Seed rate	Spacing	NPK
Pigeon pea	Cajanus cajan (22)	India	120-180 d	15 kg/ha	60 x 30 cm	25:50:25
Green gram	Vigna radiata (22)	India	60-75 d	15-20 kg/ha	30 x 10 cm	25:50:25
Black gram	Vigna mungo (22)	India	60-70 d	15-20 kg/ha	30 x 10 cm	25:50:25
Cowpea	Vigna unguiculata (22)	W. Africa	70-90 d	20-25 kg/ha	45 x 15 cm	25:50:25
Rice-fallow pulse	Vigna spp. (22)	India	60-65 d	25-30 kg/ha	broadcast	25:50:25 (starter)

9.4 Standard figures to memorise

- Standard fertiliser doses (kg N : P2O5 : K2O per ha): rice 150:50:50; maize 250:75:75; sorghum 100:50:50 (irrigated) and 40:20:0 (rain-fed); pearl millet 80:40:20 (irrigated) and 40:20:0 (rain-fed); finger millet 60:30:30; pulses 25:50:25.
- Standard seed rates: rice 40-60 kg/ha; maize 20-25 kg/ha; sorghum 10 kg/ha; pearl millet 4 kg/ha; finger millet 10-20 kg/ha; pigeon pea 15 kg/ha; green gram and black gram 15-20 kg/ha; cowpea 20-25 kg/ha; rice-fallow pulses 25-30 kg/ha.
- Standard spacings: rice 20 x 10 cm; maize 60 x 20 cm; sorghum and pearl millet 45 x 15 cm; finger millet 22.5 x 10 cm; pigeon pea 60 x 30 cm; green and black gram 30 x 10 cm; cowpea 45 x 15 cm.
- SRI: 8-12 day old seedlings, 1 seedling per hill, 25 x 25 cm spacing, seed 5-8 kg/ha, alternate wetting and drying, rotary weeder.
- Hybrid rice: F1 seed, 15-20 per cent heterosis, A/B/R lines, seed rate 15-20 kg/ha.

9.5 Consolidated worked-example formulas

- Plant population per hectare = 10000 (m2) / (row spacing in m x plant spacing in m).

- Seed rate (kg/ha) = (plant population x 1000-grain weight in g) / (germination fraction x survival fraction x 1000000), then convert to kg. Or use the biological minimum and add a practical margin.
- Fertiliser quantity = nutrient dose (kg/ha) x area (ha) / nutrient fraction in the fertiliser (urea 0.46 N; SSP 0.16 P2O5; MOP 0.60 K2O; DAP 0.46 N and 0.46 P2O5).
- Gross return = yield x price; Net return = gross return - cost of cultivation; B:C ratio = gross return / cost.
- Yield gap = potential yield - actual yield; Yield gap index (per cent) = (yield gap / potential yield) x 100.

9.6 Practice short-answer questions

1 Q1. Give the botanical name, family and chromosome number of any four cereals covered so far.

- Rice - Oryza sativa (Poaceae, 2n = 24); maize - Zea mays (Poaceae, 2n = 20); sorghum - Sorghum bicolor (Poaceae, 2n = 20); pearl millet - Pennisetum glaucum (Poaceae, 2n = 14).

1 Q2. State the six principles of SRI.

- Young seedlings (8-12 days), one seedling per hill, wide square spacing (25 x 25 cm), careful shallow transplanting, alternate wetting and drying, and mechanical weeding with organic matter.

1 Q3. Why is finger millet nutritionally important?

- It is the richest cereal source of calcium (about 300-350 mg per 100 g) and also supplies iron and B vitamins; it is valuable for children and pregnant women.

1 Q4. Define rice fallow pulses and name the two main crops.

- Short-duration pulses grown after rice on residual moisture without tillage and with little irrigation; the two main crops are black gram and green gram.

1 Q5. Give the standard fertiliser dose for irrigated maize and irrigated rice.

- Maize 250 : 75 : 75 kg N : P2O5 : K2O per hectare; rice 150 : 50 : 50 kg/ha.

9.7 Practice long-answer questions

1 Q1. Write the package of practices for irrigated hybrid maize, including a numerical on fertiliser for 0.5 hectare.

- Outline: variety, land preparation, seed rate and spacing, NPK 250:75:75 with splits, irrigation at critical stages, weed and pest management (fall armyworm), harvest and yield; then the numerical (0.5 ha: urea 272 kg, SSP 234 kg, MOP 63 kg approximately).

1 Q2. Compare the agronomy of any three kharif pulses.

- Outline: use the pulse comparison table - origin, duration, seed rate, spacing, NPK, key disease and yield; add one paragraph on their common features (nitrogen fixation, short duration, low water need, lysine-rich protein).

1 Q3. Explain SRI with principles, layout and a worked example of plant population.

- Outline: origin and concept; six principles; 25 x 25 cm layout diagram; worked example (1.6 lakh hills/ha, 8 kg seed/ha); advantages and limitations.

9.9 Model question paper (mid-semester pattern)

Practice with the following paper, which follows the usual mid-semester pattern for AGR 201.

Section A - Short answers (2 marks each, answer any ten)

1 Define kharif season and name three kharif crops.

2 Give the botanical name and chromosome number of rice.
3 What is khaira in rice?
4 State the six principles of SRI.
5 Define heterosis and name the three lines used in hybrid rice.
6 Give the seed rate and spacing for irrigated hybrid maize.
7 Which millet is the richest source of calcium?
8 Define rice-fallow pulses.
9 Name two diseases of pigeon pea.
10 What is ginning outturn?
11 Give the NPK dose for irrigated rice.
12 What is the difference between weather and climate?

Section B - Short notes (5 marks each, answer any four)

1 System of Rice Intensification.
2 Hybrid rice seed production.
3 Millets as nutri-cereals.
4 Rice fallow pulse agronomy.
5 Integrated Pest Management in rice.
6 Cropping systems of the Cauvery delta.

Section C - Long answers (10 marks each, answer any two)

1 Describe the origin, importance and cultural practices of rice.
2 Write the package of practices for irrigated hybrid maize with a numerical on fertiliser.
3 Compare sorghum and pearl millet under origin, botany, agronomy and yield.
4 Explain the agronomy of the kharif pulses with a comparison table.
5 Explain the SRI method with its principles, layout and a worked example.

Section D - Numerical (compulsory)

1 A farmer grows pigeon pea at 60 x 30 cm. Given a 1000-grain weight of 80 g, 90 per cent germination and 90 per cent survival, calculate the plant population and the seed rate. Then compute the urea, SSP and MOP needed to apply 25:50:25 kg NPK per hectare on 1 hectare.

9.8 Exam focus

Key points to remember

- The mid-semester examination covers Lectures 1-8; numericals on seed rate, plant population and fertiliser are almost certain.
- Keep the master tables of cereals/millets and pulses in memory - botanical name, chromosome number, origin, duration, seed rate, spacing and NPK.
- Standard doses: rice 150:50:50, maize 250:75:75, sorghum 100:50:50, pearl millet 80:40:20, finger millet 60:30:30, pulses 25:50:25.
- SRI principles and hybrid rice (A/B/R lines) are frequently asked short notes.
- Rice fallow pulses (black gram, green gram) and their agronomy are an important delta-specific topic.
- Always state the botanical name, family and chromosome number in any crop answer.

Short-answer questions with model answers

1 Q1. List the formulas for plant population, seed rate and fertiliser quantity.

- Plant population/ha = 10000 / (row spacing x plant spacing in m). Seed rate = (population x 1000-grain weight) / (germination x survival). Fertiliser quantity = nutrient dose x area / nutrient fraction in the fertiliser.

1 Q2. Which crops in Lectures 1-8 are C4 plants?

- Maize, sorghum, pearl millet, finger millet and the other millets are C4 plants; rice is a C3 plant. The C4 pathway gives the millets and maize higher water-use efficiency and heat tolerance.

Long-answer questions with model-answer outlines

1 Q1. Write short notes on any four of the following: SRI; hybrid rice; rice-fallow pulses; fall armyworm; zinc deficiency in rice; the Cauvery delta cropping system.

- Outline: one paragraph each with definition, key features, an important figure and a management point; for example, SRI (principles, 25 x 25 cm, seed 5-8 kg), hybrid rice (heterosis 15-20 per cent, A/B/R lines), rice-fallow pulses (residual moisture, 60-65 days, 300-500 kg/ha), fall armyworm (whorl damage, IPM), zinc deficiency (khaira, ZnSO4 25 kg/ha), the delta system (Kuruvai-Thaladi-pulse).

Lecture 10 - Groundnut: Origin, Distribution, Economic Importance, Soil and Climatic Requirements, Varieties, Cultural Practices, Yield and Economics

10.1 Origin and distribution

- Common name: Groundnut, Peanut, Monkey nut, Kadalai (Tamil). Botanical name: Arachis hypogaea L. Family: Fabaceae (Leguminosae). Chromosome number 2n = 40 (tetraploid).
- Origin: groundnut originated in South America, in the region of southern Brazil and Peru (the Brazilian centre of origin). It spread to Africa, and from Africa to India and other parts of Asia in the post-Columbian period. It is a New World crop.
- Distribution: groundnut is grown in over 100 countries. The largest producers by area are India and China; China leads in production because of higher yields. Other producers are Nigeria, the United States, Sudan, Myanmar, Argentina, Indonesia and Senegal.
- In India the main states are Gujarat, Rajasthan, Tamil Nadu, Andhra Pradesh, Karnataka, Telangana and Maharashtra. In Tamil Nadu groundnut is the most important oilseed, grown in Vellore, Tiruvannamalai, Dharmapuri, Krishnagiri, Salem, Namakkal, Erode, Virudhunagar, Madurai, Tirunelveli, Ramanathapuram, Cuddalore, Villupuram and the delta and coastal tracts.

10.2 Types of groundnut

- Bunch type (Spanish and Valencia groups): erect, compact, short duration (about 95-110 days), with pods clustered near the base. Varieties: TMV 2, TMV 7, CO 7, ALR 3.
- Spreading type (Virginia group): semi-erect to spreading, longer duration (about 120-140 days), with pods spread over the root zone. Varieties: TMV 10, some VRI types.
- The bunch type suits shorter seasons, closer spacing and rice-fallow/rice-bund situations; the spreading type gives higher yield over a longer season.

10.3 Economic importance and nutritional value

- Groundnut is the most important oilseed of India by area. It is a dual-purpose crop - the kernel yields edible oil (about 44-50 per cent), and the haulm (leaf and stem) is a valuable fodder. The cake after oil extraction is a protein-rich cattle feed.

- Uses: edible oil, roasted and salted kernels, peanut butter, confectionery, and industrial products (soap, cosmetics, lubricants). The oil cake and haulm support the dairy industry.
- Nutritional value (per 100 g kernel): about 44-50 per cent oil, 25-30 per cent protein, 15-20 per cent carbohydrate, about 567 kcal energy, and useful amounts of vitamin E, B vitamins and minerals. Groundnut protein is rich in lysine but low in methionine. The kernels are a major source of dietary energy and protein.

Nutrient (per 100 g groundnut kernel)	Approximate content
Energy	about 567 kcal
Oil	44-50 g
Protein	25-30 g
Carbohydrate	15-20 g
Crude fibre	2-3 g
Notable vitamins	vitamin E, B vitamins (niacin)

10.4 Soil and climatic requirements

- Soil: groundnut performs best on well-drained, light-coloured sandy loam, sandy and red loam soils that allow easy penetration of pegs and easy lifting of pods. Optimum pH is 6.0-7.5. Groundnut is sensitive to waterlogging, salinity and heavy clay soils (which cause pod rot and poor pegging).
- Calcium: groundnut has a high calcium requirement, particularly at the pegging and pod-filling stage, because the pod absorbs calcium directly from the soil solution around it. This is why gypsum is applied at flowering.
- Climate: a warm-weather crop. Optimum temperature is about 25-30 degrees C. It needs about 500-700 mm of water. It is drought-tolerant to a degree because of its deep root system, but moisture stress at flowering and pegging causes heavy yield loss. It is a short-day plant, but most improved varieties are photo-insensitive.

Requirement	Optimum for groundnut
Soil	Well-drained sandy loam, sandy, red loam
Soil pH	6.0-7.5
Temperature	25-30 C
Water need	500-700 mm
Critical stages	Flowering, pegging, pod development
Special need	Calcium (gypsum) at pegging
Sensitive to	Waterlogging, salinity, heavy clay

10.5 Varieties

- TMV 2 - the classic bunch-type variety of Tamil Nadu, short duration (about 100-105 days), small kernel, moderate yield; still widely grown.
- TMV 7 - bunch type, short duration, high yield.
- TMV 10 - spreading type, longer duration, high yield.
- TMV 13, TMV 14 - improved bunch-type varieties of short duration with good yield and oil content.
- VRI 2, VRI 6, VRI 7, VRI 8 - varieties from the Regional Research Station, Vridhachalam, of short duration with good yield, oil content and disease tolerance.

- CO 7 - TNAU bunch variety, short duration.
- CO(Gn) 5 - TNAU variety of short duration with good yield.
- ALR 3 - a variety suited to certain tracts; TG 37A and TMV(Gn) 3 are other released varieties.
- Varieties with tolerance to tikka leaf spot, rust and bud necrosis are preferred in disease-prone areas.

10.6 Land preparation and sowing

- Land preparation: plough once or twice and harrow to a fine, deep, well-drained tilth. Groundnut needs a loose, friable soil so that pegs can enter and pods can be lifted without loss. Form beds and channels where irrigated.
- Sowing time: rain-fed kharif with the onset of the monsoon (June-July); irrigated crops in April-May (Chithirai pattam), June-July, September-October and December-January. The delta and rice-fallow situations use December-February.
- Seed treatment: treat seed with a fungicide (for seed-borne and soil-borne diseases) and with Rhizobium and phosphobacteria. Shell the pods just before sowing for good germination; use kernels for sowing (about 100 kg kernels from about 140 kg pods).
- Seed rate: about 100 kg of kernels per hectare for bunch types and about 80 kg/ha for spreading types (spreading types have larger kernels and wider spacing).
- Spacing: 30 x 10 cm for bunch types; 45 x 15 cm for spreading types. The number of plants per hectare is about 3.3 lakh for bunch types.

10.7 Nutrient management

- Standard fertiliser dose: 25 : 50 : 75 kg N : P2O5 : K2O per hectare for irrigated groundnut (some recommendations use 25:50:75 or 17:34:54; the important point is a low N, moderate P and higher K dose).
- Apply FYM or compost 10 t/ha before sowing. Apply gypsum 400 kg/ha at the flowering/pegging stage (about 40-45 days) to supply calcium and sulphur for pod filling.
- Apply zinc sulphate 25 kg/ha where deficient. Rhizobium seed inoculation improves nitrogen fixation.
- For rain-fed groundnut a lower dose (about 10-25 : 25-50 : 0-25 kg/ha) is used.

Fertiliser for irrigated groundnut	Dose
FYM/compost	10 t/ha
N	25 kg/ha
P2O5	50 kg/ha
K2O	75 kg/ha
Gypsum	400 kg/ha at flowering/pegging
ZnSO4 (deficient soils)	25 kg/ha
Seed treatment	Rhizobium + phosphobacteria + fungicide

10.8 Water management

- Groundnut is sensitive to moisture stress at flowering, pegging and pod development - the critical stages. Stress at pegging reduces the number of pods and at pod filling reduces kernel size and oil content.
- Irrigate at about 7-10 day intervals in summer; provide about 8-10 irrigations for a bunch-type crop and more for spreading types. Stop irrigation about 10-15 days before harvest.

- Avoid waterlogging, which causes pod rot, leaf yellowing and yield loss. Beds and channels and good drainage are essential in the rainy season.

10.9 Weed management

- Keep the crop weed-free for the first 30-40 days. Weeds compete for water, nutrients and light and reduce yield.
- One or two hand weedings or a pre-emergence herbicide (for example, pendimethalin) followed by one hand weeding. Mechanical weeding must be shallow to avoid damaging pegs and pods.
- Earth up the soil lightly around the base at pegging to help peg entry.

10.10 Pest and disease management

Major insect pests

- Leaf miner (Aproaerema modicella) - the larva mines the leaflets and webs them; serious in the kharif. Management: IPM, need-based insecticide, conservation of natural enemies.
- White grub (Holotrichia spp.) - the grub feeds on roots and causes wilting; serious in light soils.
- Thrips (vector of bud necrosis virus), jassids (leaf hopper) and aphids - sucking pests; jassid causes 'hopper burn' (yellowing at the leaf margins).
- Red hairy caterpillar (Amsacta albistriga) and other caterpillars in the kharif; leaf-eating caterpillars and termites.

Major diseases

- Tikka leaf spot (Cercospora arachidicola and C. personata) - the most important disease; dark brown/black spots on leaves cause heavy defoliation. Management: resistant varieties, seed treatment, crop rotation and fungicide.
- Rust (Puccinia arachidis) - orange-brown pustules on the lower leaf surface.
- Stem rot/root rot (Sclerotium rolfsii, Aspergillus niger) - causes wilting and collar rot, favoured by high moisture and high temperature.
- Bud necrosis virus (spread by thrips) - a serious viral disease; management through resistant varieties and thrips control.
- Management: integrated - resistant varieties, seed treatment, crop rotation, clean cultivation, balanced fertiliser and need-based fungicide.

10.11 Harvest and yield

- Harvest when the crop matures - the leaves turn yellow and fall, the inner shell of the pod shows a dark brown/black layer and the kernels are fully developed. Bunch types mature at about 100-110 days and spreading types at about 120-140 days.
- Lift the plants with a spade or groundnut lifter, shake off soil, and dry the pods in the sun for a few days to reduce moisture to about 8-10 per cent for safe storage.
- Yield: about 2-3 t/ha of pods for irrigated bunch types and 1-1.5 t/ha for rain-fed crops; spreading types may yield up to 2.5-3 t/ha under irrigation. Haulm (dry fodder) yield is about 3-5 t/ha. The shelling percentage is about 70-75 per cent.

10.12 Economics - worked example

Problem: A farmer grows 1 hectare of irrigated bunch-type groundnut (TMV 7) and obtains 2.5 t/ha of pods. Pods sell at Rs. 6,000 per quintal. Calculate gross return, net return and B:C ratio using the cost items below.

Cost item (per hectare)	Approximate amount (Rs.)
Land preparation	5,000
Seed (100 kg kernels)	9,000
FYM 10 t and gypsum	7,000
Fertiliser (25:50:75)	4,500
Sowing, weeding and earthing	6,000
Plant protection	4,000
Irrigation	3,000
Harvesting, drying, shelling	8,000
Total cost of cultivation	46,500

- Gross return = 2.5 t/ha = 25 quintals x Rs. 6,000 = Rs. 1,50,000 per hectare.
- Net return = Rs. 1,50,000 - Rs. 46,500 = Rs. 1,03,500 per hectare.
- Benefit-cost ratio = 1,50,000 / 46,500 = about 3.2.
- Add the value of haulm (about 4 t at Rs. 2,000/t = Rs. 8,000) as an extra benefit; the net return then exceeds Rs. 1,10,000. Groundnut is one of the most profitable kharif crops of Tamil Nadu when prices are good.

10.13 Groundnut pegging - text diagram

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Groundnut growth and pegging (bunch type, about 100-105 days):
  Vegetative (0-30 d) -> Flowering (30-45 d) -> Pegging (45-60 d)
     |                      |                      |
  Leaf development     Yellow flowers       Peg (gynophore) grows down
                                            into soil; pod develops underground
     |
  Pod filling (60-90 d) -> Maturity (100-105 d)
     |                        |
  Kernels fill, oil forms    Leaves yellow; inner shell darkens -> harvest
  Note: gypsum is applied at flowering/pegging for calcium;
  moisture stress at pegging is the most damaging.
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10.15 Cropping systems, intercropping and value addition

Groundnut in cropping systems

- Groundnut-pulse: groundnut followed by a short-duration pulse; groundnut-rice and groundnut-millet rotations in the tank-fed and dry tracts.
- Groundnut + pigeon pea (3:1 or 4:2 rows), groundnut + castor (border rows), groundnut + sesame and groundnut + red gram are standard rain-fed intercrops.
- In the delta and garden land, groundnut fits as a short-duration crop between two rice crops or after a rice crop.

Value addition

- Oil extraction: groundnut oil is a premium edible oil. The oil cake (groundnut cake) is a protein-rich cattle feed and fertiliser. Filtered and cold-pressed oils command a higher price.
- Food products: roasted and salted kernels, peanut butter, chikki, and confectionery. Graded and branded kernels get a price premium.
- The haulm (leaf and stem) is a valuable fodder, and the shells can be used as fuel or as a soil amendment after composting.

10.16 Comparison of important groundnut varieties

Variety	Type	Duration	Salient feature
TMV 2	Bunch	100-105 d	Classic TN variety, small kernel
TMV 7	Bunch	100-105 d	High yield, good oil
TMV 10	Spreading	120-140 d	High yield, longer season
CO 7	Bunch	100-105 d	TNAU high-yielding short-duration
CO(Gn) 5	Bunch	100-105 d	Good yield and quality
VRI 7, VRI 8	Bunch	100-110 d	High yield, disease tolerance
ALR 3	Bunch	100-105 d	Suited to certain tracts

Field calendar and common problems

Crop calendar (operations schedule)

Operation	Time (days)	Remarks
Land preparation	Before sowing	Deep, friable tilth; beds and channels
Sowing	0	100 kg kernels/ha (bunch) at 30 x 10 cm
Basal fertiliser	0	25:50:75; FYM 10 t/ha
Flowering/pegging	40-45	Apply gypsum 400 kg/ha
Weeding and earthing	20-40	Shallow intercultivation; earth up at pegging
Irrigation	Critical at flowering and pegging	8-10 irrigations in summer
Harvest	100-140	Leaves yellow; inner shell darkens

Common field problems, causes and remedies

Problem	Likely cause	Remedy
Dark leaf spots, defoliation	Tikka leaf spot	Resistant variety; fungicide; rotation
Orange pustules on leaves	Rust	Resistant variety; fungicide
Empty pods / 'pops'	Calcium deficiency	Apply gypsum at pegging
Wilting, collar rot	Stem/root rot	Rotation; seed treatment; drainage
Webbed, mined leaflets	Leaf miner	IPM; need-based insecticide

Objective-type practice questions

Choose the correct answer. Answers are given at the end of the set.

1 Groundnut belongs to the family?
2 The chromosome number of groundnut is?
3 The origin of groundnut is?
4 The oil content of groundnut kernel is?
5 Gypsum is applied to groundnut at?

6 The seed rate for bunch groundnut is?

7 The most important leaf disease of groundnut is?

8 Bud necrosis virus is transmitted by?

9 The pegging organ of groundnut is the?

10 The standard NPK for irrigated groundnut is?

Answers: 1-Fabaceae; 2-2n=40; 3-South America; 4-44-50 per cent; 5-flowering/pegging; 6-100 kg kernels/ha; 7-tikka leaf spot; 8-thrips; 9-gynophore; 10-25:50:75

10.14 Exam focus

Key points to remember

- Arachis hypogaea, family Fabaceae, 2n = 40; origin South America (Brazil/Peru); a New World crop.
- Groundnut is the most important oilseed of India; dual purpose (oil + haulm fodder); kernel has 44-50 per cent oil, 25-30 per cent protein, about 567 kcal per 100 g.
- Soil: well-drained sandy loam/red loam, pH 6.0-7.5; calcium (gypsum) is critical at pegging; sensitive to waterlogging and heavy clay.
- Climate: 25-30 C, water need 500-700 mm; critical stages are flowering, pegging and pod development.
- Types: bunch (Spanish/Valencia, short, erect) and spreading (Virginia, long). TNAU varieties: TMV 2, TMV 7, CO 7, CO(Gn) 5, VRI 2/6/7/8, ALR 3.
- Seed rate about 100 kg kernels/ha (bunch) and 80 kg/ha (spreading); spacing 30 x 10 cm (bunch) and 45 x 15 cm (spreading); NPK 25:50:75 with gypsum 400 kg/ha.
- Major pests: leaf miner, white grub, thrips, jassids. Major diseases: tikka leaf spot, rust, stem rot, bud necrosis.
- Yield 2-3 t/ha (irrigated bunch) and 1-1.5 t/ha (rain-fed); net return over Rs. 1,00,000/ha at good prices.

Short-answer questions with model answers

1 Q1. Why is gypsum applied to groundnut, and when?

- Gypsum (calcium sulphate) is applied at the flowering/pegging stage (about 40-45 days) to supply calcium and sulphur. Groundnut pods absorb calcium directly from the soil solution around them, and calcium is essential for pod and kernel development; a deficiency causes poor pod filling ('pops'). Gypsum is applied at about 400 kg/ha.

1 Q2. Distinguish bunch and spreading types of groundnut.

- Bunch (Spanish/Valencia) types are erect, compact, short duration (95-110 days), with pods clustered at the base, and are grown at closer spacing. Spreading (Virginia) types are semi-erect to spreading, longer duration (120-140 days), with pods spread over the root zone, and are grown at wider spacing.

1 Q3. Name two major diseases of groundnut and their pathogens.

- Tikka leaf spot, caused by Cercospora arachidicola and C. personata; and stem rot/root rot, caused by Sclerotium rolfsii. Rust (Puccinia arachidis) and bud necrosis virus are also important.

Long-answer questions with model-answer outlines

1 Q1. Write the complete package of practices for irrigated groundnut in Tamil Nadu.

- Outline: (i) origin and importance; (ii) soil and climate with the calcium requirement; (iii) variety (TMV 7, CO 7, VRI 7); (iv) land preparation for easy pegging and lifting; (v) seed rate 100 kg kernels/ha, shelling just before sowing, seed treatment with Rhizobium and fungicide; (vi) spacing 30 x 10 cm; (vii)

NPK 25:50:75 plus FYM 10 t and gypsum 400 kg/ha at pegging; (viii) irrigation at flowering, pegging and pod filling; (ix) weed management with shallow intercultivation; (x) pest and disease management (leaf miner, tikka, rust, bud necrosis); (xi) harvest at maturity and yield; (xii) a one-line economics.

1 Q2. Explain the economics of groundnut cultivation with a worked example.

- Outline: (i) cost items (seed is the biggest single cost, then harvesting and plant protection); (ii) gross return from pods and the extra return from haulm; (iii) worked example with 2.5 t/ha at Rs. 6,000/quintal giving a net return of about Rs. 1,03,500 and a B:C ratio of about 3.2; (iv) sensitivity to price and yield; (v) the role of value addition (oil, cake) in raising returns.

Lecture 11 - Sesame and Soybean: Origin, Distribution, Economic Importance, Soil and Climatic Requirements, Varieties, Cultural Practices and Yield

Sesame and soybean are the two important kharif oilseeds of this lecture. Sesame is an ancient Indian oilseed grown for its high-quality edible oil; soybean is a dual-purpose pulse-oilseed grown for oil and protein. Each is treated in turn and then compared.

PART A - SESAME (Gingelly / Ellu)

11.1 Origin and distribution

- Common name: Sesame, Gingelly, Til (Hindi), Ellu (Tamil). Botanical name: Sesamum indicum L. Family: Pedaliaceae. Chromosome number 2n = 26.
- Origin: sesame is one of the oldest oilseed crops. It originated in India (some authorities place the primary centre in Africa with India as a secondary centre of diversity). It has been cultivated in the Indian sub-continent for several thousand years.
- Distribution: India, China, Myanmar, Sudan, Nigeria, Ethiopia, Tanzania and Uganda are the main producers; India is one of the largest producers and exporters. In India the main states are Gujarat, Rajasthan, Madhya Pradesh, Uttar Pradesh, Maharashtra, Karnataka, Andhra Pradesh, Telangana, Tamil Nadu and West Bengal.
- In Tamil Nadu sesame is grown in Virudhunagar, Ramanathapuram, Sivaganga, Madurai, Tirunelveli, Thoothukudi, Salem, Erode, Villupuram, Cuddalore and the delta and rain-fed tracts, both as a rain-fed kharif crop and as a rice-fallow/summer crop.

11.2 Economic importance and nutritional value

- Sesame is called the 'queen of oilseeds' for the high quality of its oil. The oil is used for edible purposes (as a cooking oil and for confectionery) and for non-edible uses (cosmetics, pharmaceuticals, soaps, and as a carrier for medicines). The seed and oil are used in many traditional preparations.
- Uses: oil (about 45-50 per cent of the seed), the seed in bakery and confectionery (til chikki, gajak), and the oil cake as a protein-rich cattle feed. Sesame oil is rich in antioxidants (sesamol, sesamin), which give it a long shelf life.
- Nutritional value (per 100 g seed): about 45-50 per cent oil, 18-20 per cent protein, 20-25 per cent carbohydrate, about 563 kcal energy, and useful calcium, iron and B vitamins. Sesame seed is a rich source of calcium (especially the unhulled seed) and of the amino acid methionine.

11.3 Soil and climatic requirements

- Soil: sesame grows on a wide range of soils, from sandy loam to light black soils; it performs best on well-drained, light to medium loams with pH 6.0-8.0. It is sensitive to waterlogging and to salinity, and does poorly on heavy clay and on waterlogged soils.
- Climate: a warm-weather crop of the tropics and subtropics. Optimum temperature is about 25-30 degrees C. It needs about 400-600 mm of water. It is fairly drought-tolerant but sensitive to waterlogging and to heavy rain, especially at flowering. It is a short-day plant, and many traditional types are photoperiod-sensitive.

11.4 Varieties

- TMV 3, TMV 4, TMV 5, TMV 6 - TNAU varieties of short duration (about 75-85 days) with good yield and oil content; several are white-seeded and suited to the rain-fed tracts.
- TMV(Sv) 7 - TNAU variety of short duration with good yield.
- CO 1 - TNAU variety of short duration.
- VRI 1, VRI 2 - varieties from the Regional Research Station, Vridhachalam, of short duration.
- SVPR 1 - a variety suited to the southern districts, of short duration.
- Varieties with tolerance to phyllody, powdery mildew and leaf spot are preferred in disease-prone areas.

11.5 Cultural practices

- Land preparation: one or two shallow ploughings and harrowing to a fine, well-drained tilth; form beds and channels where irrigated. Sesame seed is very small, so a fine, firm seed-bed is needed.
- Sowing time: rain-fed kharif with the onset of the monsoon (June-July); also sown in September-October and as a summer/rice-fallow crop in January-February.
- Seed rate: about 4-5 kg/ha (the seed is very small); mix the seed with sand for even sowing.
- Spacing: 30 x 15 cm or 30 x 10 cm for line sowing; broadcasting uses a higher seed rate.
- Nutrient management: about 35 : 23 : 23 kg N : P2O5 : K2O per hectare (some recommendations use 40 : 20 : 20 or 25 : 50 : 25; the exact dose depends on the soil test). Apply FYM 5 t/ha. Nitrogen in two splits.
- Water management: mostly rain-fed; where irrigated, irrigate at establishment, flowering and capsule filling. Avoid waterlogging.
- Weed management: keep weed-free for the first 25-30 days; the seedling is a weak competitor. One or two hand weedings or a pre-emergence herbicide. Thinning to the correct spacing at 10-15 days is important.
- Pest and disease management: pests - leaf webber and capsule borer, gall fly, sphinx caterpillar, aphids, thrips, and the sesame leaf caterpillar. Diseases - phyllody (a mycoplasma-like disease spread by leaf hoppers, which causes the flowers to turn into leafy structures and no seed to form), powdery mildew, leaf spot (Cercospora) and root rot. Management: resistant varieties, vector control, crop rotation, seed treatment and need-based fungicide.
- Harvest and yield: sesame is harvested when the lower capsules turn yellow-brown and the seed is mature, about 80-90 days; the plants are cut, stacked and dried, then threshed. Yield about 400-600 kg/ha (good management) and 200-350 kg/ha rain-fed.

PART B - SOYBEAN

11.6 Origin and distribution

- Common name: Soybean, Soya bean, Bhat (Hindi), Soya (Tamil). Botanical name: Glycine max (L.) Merrill. Family: Fabaceae. Chromosome number 2n = 40.

- Origin: soybean originated in China, in the region of the Yellow River valley, and is one of the oldest cultivated crops of East Asia. It spread to Japan, Korea and South-East Asia, and in the twentieth century to the Americas.
- Distribution: the United States, Brazil, Argentina, China and India are the largest producers; the United States and Brazil dominate world trade. In India the main states are Madhya Pradesh, Maharashtra, Rajasthan, Karnataka, Telangana, Andhra Pradesh, Chhattisgarh and Gujarat. In Tamil Nadu soybean is grown in Coimbatore, Erode, Salem, Namakkal, Tiruppur, Dindigul, Vellore and other districts, mainly as an irrigated crop.

11.7 Economic importance and nutritional value

- Soybean is the world's most important oilseed and protein crop. It is unique among kharif crops in supplying both oil (about 18-20 per cent) and high-quality protein (about 40 per cent) at the same time.
- Uses: edible oil; soy protein (soy flour, soya milk, tofu, textured vegetable protein, soy nuggets); animal and poultry feed (soybean meal is the standard protein supplement); industrial uses (soy lecithin, paints, plastics, biodiesel). It is a nitrogen-fixing legume that improves soil.
- Nutritional value (per 100 g seed): about 40 per cent protein (the highest among common food crops), 18-20 per cent oil, 20-25 per cent carbohydrate, about 432 kcal energy, and useful calcium, iron and B vitamins. Soy protein is of good quality and contains all essential amino acids except methionine, which is low.

11.8 Soil and climatic requirements

- Soil: soybean grows on a wide range of soils but performs best on well-drained loams and clay loams with pH 6.0-7.5; it is sensitive to waterlogging and salinity, and to highly acidic soils. It responds to lime on acid soils.
- Climate: a warm-weather crop of the tropics and subtropics. Optimum temperature is about 20-30 degrees C. It needs about 450-700 mm of water. It is a short-day plant, and the modern varieties are largely photo-insensitive. Moisture stress at flowering and pod filling reduces yield.

11.9 Varieties

- CO 1, CO 2, CO 3 - TNAU varieties of short to medium duration (about 85-100 days) suited to Tamil Nadu conditions.
- JS 335 - a popular high-yielding variety of medium duration, widely grown in India.
- JS 9560 - a variety with good yield and quality.
- Hardee - a short-duration variety (about 85-90 days) that fits into multiple cropping.
- PK 472 and other ICAR/State varieties are grown where suited. Varieties with tolerance to yellow mosaic and rust are preferred.

11.10 Cultural practices

- Land preparation: one or two ploughings and harrowing to a fine, well-drained tilth; form ridges and furrows. Soybean needs a fine seed-bed and good drainage.
- Sowing time: rain-fed kharif with the monsoon (June-July); irrigated crops in June-July, September-October and January-February.
- Seed treatment: treat seed with a fungicide and with Rhizobium and phosphobacteria. Do not treat with nitrogen fertiliser near the seed, as it harms the Rhizobium.
- Seed rate: about 70-75 kg/ha (the seed is large); some recommendations use a lower rate (about 40-45 kg/ha) with closer spacing and good management.

- Spacing: 45 x 5 cm or 30 x 10 cm for line sowing; about 3-4 lakh plants per hectare.
- Nutrient management: about 30 : 60 : 40 kg N : P2O5 : K2O per hectare. Soybean needs a small starter dose of nitrogen; the rest comes from Rhizobium fixation. Apply FYM 5-10 t/ha, and sulphur (soybean responds to sulphur). Apply lime on acid soils.
- Water management: mostly rain-fed; where irrigated, irrigate at flowering and pod filling. Avoid waterlogging.
- Weed management: keep weed-free for the first 30-40 days; one or two hand weedings or a pre-emergence herbicide followed by one hand weeding.
- Pest and disease management: pests - stem fly, girdle beetle, semilooper, tobacco caterpillar (Spodoptera litura), pod borer, and sucking pests. Diseases - yellow mosaic virus (spread by whitefly), soybean rust, bacterial pustule, Rhizoctonia blight and root rot. Management: resistant varieties, vector control, seed treatment, crop rotation and IPM.
- Harvest and yield: harvest when the pods turn brown and the leaves have fallen, about 90-100 days. Yield about 2-2.5 t/ha under irrigation and good management, and 1-1.5 t/ha rain-fed.

11.11 Comparison of sesame and soybean

Feature	Sesame	Soybean
Botanical name	Sesamum indicum	Glycine max
Family	Pedaliaceae	Fabaceae
Chromosome (2n)	26	40
Origin	India (Africa secondary)	China
Duration	80-90 days	85-100 days
Oil content	45-50 per cent	18-20 per cent
Protein content	18-20 per cent	about 40 per cent
Seed rate	4-5 kg/ha	70-75 kg/ha
Spacing	30 x 15 cm	45 x 5 cm / 30 x 10 cm
NPK	about 35:23:23	about 30:60:40
Key disease	Phyllody (mycoplasma)	Yellow mosaic (whitefly)
Yield	0.4-0.6 t/ha	2-2.5 t/ha (irrigated)

11.12 Worked example - fertiliser for sesame on 0.5 hectare

Problem: A farmer wants to apply 35 : 23 : 23 kg N : P2O5 : K2O per hectare to 0.5 hectare of sesame. Calculate the urea, SSP and MOP required.

- For 0.5 ha: N = 17.5 kg; P2O5 = 11.5 kg; K2O = 11.5 kg.
- Urea = 17.5 / 0.46 = about 38 kg.
- SSP = 11.5 / 0.16 = about 72 kg.
- MOP = 11.5 / 0.60 = about 19 kg.
- Apply half the urea with all the SSP and MOP as basal; apply the rest of the urea at flowering.
- Check: 38 x 0.46 = 17.5 kg N; 72 x 0.16 = 11.5 kg P2O5; 19 x 0.60 = 11.4 kg K2O. The calculation matches the target for 0.5 hectare.

11.14 Cropping systems and value addition

Sesame and soybean in cropping systems

- Sesame-pulse, sesame-millet and sesame-groundnut rotations are common in the rain-fed tracts. Sesame is also grown as a rice-fallow and summer crop and on field bunds.
- Soybean-maize, soybean-sorghum and soybean-wheat rotations are common in central India; in Tamil Nadu soybean fits as an irrigated crop in rotation with cereals and pulses.
- Both are also grown as intercrops: sesame + green gram, soybean + pigeon pea and soybean + maize are practised.

Value addition

- Sesame: oil extraction (cold-pressed oil is a premium product), the seed in bakery and confectionery (chikki, gajak), tahini and sesame paste, and the oil cake as feed. Sesame oil's antioxidants give it a long shelf life.
- Soybean: oil extraction, and the many soy foods - soya milk, tofu, soy flour, textured vegetable protein (soy chunks), soy nuggets and soy-fortified foods. Soybean meal is the standard protein supplement in poultry and cattle feed.
- Both crops are also promoted for their health value (sesame for calcium and antioxidants; soybean for high-quality protein and isoflavones).

Field calendar and common problems

Crop calendar (operations schedule)

Operation	Time (days)	Remarks
Land preparation	Before sowing	Fine seed-bed; beds and channels
Sowing (sesame)	0	4-5 kg/ha at 30 x 15 cm
Sowing (soybean)	0	70-75 kg/ha at 45 x 5 cm; Rhizobium
Thinning	10-15	Sesame needs thinning to stand
Basal fertiliser	0	Sesame 35:23:23; soybean 30:60:40
Weeding	25-30	Keep weed-free early
Harvest	80-100	Sesame 80-90 d; soybean 90-100 d

Common field problems, causes and remedies

Problem	Likely cause	Remedy
Floral parts become leafy (sesame)	Phyllody (mycoplasma)	Resistant variety; control leaf hopper
Yellow mosaic (soybean)	Virus via whitefly	Resistant variety; control whitefly
Poor sesame stand	Very small seed; crusting	Fine seed-bed; mix seed with sand; thin
Pods damaged (soybean)	Girdle beetle, semilooper, caterpillars	IPM; need-based insecticide
Low oil content	Late harvest; poor nutrition	Timely harvest; sulphur as per soil test

Objective-type practice questions

Choose the correct answer. Answers are given at the end of the set.

1 Sesame belongs to the family?
2 The chromosome number of sesame is?
3 Sesame is called the?
4 The disease in which sesame flowers become leafy is?
5 Soybean belongs to the family?
6 The chromosome number of soybean is?
7 The origin of soybean is?
8 The protein content of soybean is about?
9 Yellow mosaic of soybean is transmitted by?
10 The seed rate for soybean is about?

Answers: 1-Pedaliaceae; 2-2n=26; 3-queen of oilseeds; 4-phyllody; 5-Fabaceae; 6-2n=40; 7-China; 8-40 per cent; 9-whitefly; 10-70-75 kg/ha

11.13 Exam focus

Key points to remember

- Sesame: Sesamum indicum, family Pedaliaceae, 2n = 26; origin India; 'queen of oilseeds'; 45-50 per cent oil, 18-20 per cent protein; short duration 80-90 days; seed 4-5 kg/ha; spacing 30 x 15 cm; key disease phyllody.
- Soybean: Glycine max, family Fabaceae, 2n = 40; origin China; the world's most important oilseed and protein crop; 40 per cent protein, 18-20 per cent oil; 85-100 days; seed 70-75 kg/ha; key disease yellow mosaic.
- Both are warm-weather, short-day crops needing well-drained soils and about 400-700 mm of water; both are sensitive to waterlogging.
- Sesame is very small-seeded and needs a fine seed-bed and thinning; soybean is a legume and fixes nitrogen, needing a starter dose of N.
- Sesame NPK about 35:23:23; soybean NPK about 30:60:40.
- Yields: sesame 0.4-0.6 t/ha; soybean 2-2.5 t/ha (irrigated).

Short-answer questions with model answers

1 Q1. Why is sesame called the queen of oilseeds?

- Sesame is called the queen of oilseeds because of the high quality of its oil - it has a high oil content (45-50 per cent), a pleasant taste, a long shelf life because of its natural antioxidants (sesamol, sesamin), and it is used both as an edible oil and in medicine and cosmetics.

1 Q2. What is phyllody in sesame and how is it managed?

- Phyllody is a disease caused by a mycoplasma-like organism transmitted by leaf hoppers. The floral parts are transformed into green leafy structures, so no capsules or seed are formed. It is managed by growing resistant varieties, controlling the leaf hopper vector, and removing infected plants.

1 Q3. Give the nutritional importance of soybean.

- Soybean contains about 40 per cent protein (the highest among common food crops) and 18-20 per cent oil, and is rich in calcium, iron and B vitamins. It is used for edible oil, soy protein foods and animal feed, and is the world's most important protein crop.

Long-answer questions with model-answer outlines

1 Q1. Describe the origin, importance, soil and climate, and cultural practices of sesame.

- Outline: (i) botanical identity and ancient Indian origin; (ii) distribution and Indian/Tamil Nadu areas; (iii) economic importance - queen of oilseeds, edible and industrial oil, cake; (iv) nutritional value; (v) soil and climate; (vi) varieties TMV 3-6, VRI 1/2, SVPR 1; (vii) cultural practices - land preparation, seed 4-5 kg/ha, spacing, NPK 35:23:23, thinning, weed and water management; (viii) pest and disease management with phyllody; (ix) harvest and yield.

1 Q2. Compare sesame and soybean as kharif oilseeds.

- Outline: a structured comparison using the table - botanical name, family, chromosome number, origin, duration, oil and protein content, seed rate, spacing, NPK, key diseases and yield. Conclude with the point that sesame is an oil-first crop while soybean is a dual-purpose protein-and-oil crop, and that soybean, being a legume, enriches the soil while sesame does not.

Lecture 12 - Castor: Origin, Distribution, Economic Importance, Soil and Climatic Requirements, Varieties, Cultural Practices, Yield and Economics

12.1 Origin and distribution

- Common name: Castor, Castor bean, Arandi (Hindi), Amanakku (Tamil). Botanical name: Ricinus communis L. Family: Euphorbiaceae. Chromosome number 2n = 20.
- Origin: castor originated in Africa, in the Ethiopian (Abyssinian) region, which is the primary centre of diversity; India and the Mediterranean region are secondary centres. Castor has been cultivated in India for a very long time and India is now the world's largest producer.
- Distribution: India, China, Brazil, Mozambique, Myanmar, Ethiopia and Thailand are the main producers. India is the largest producer and exporter of castor seed and castor oil. In India the main states are Gujarat (which alone accounts for the largest share), Rajasthan, Andhra Pradesh, Telangana, Karnataka and Tamil Nadu.
- In Tamil Nadu castor is grown in Namakkal, Salem, Erode, Coimbatore, Tiruppur, Karur, Dindigul, Madurai, Virudhunagar, Tirunelveli, Thoothukudi, Ramanathapuram and the rain-fed tracts, and it is often grown on field bunds and as a border crop.

12.2 Economic importance and nutritional value

- Castor is grown mainly for its oil, which is unique because it contains ricinoleic acid (about 85-90 per cent of the fatty acids). This gives castor oil its high viscosity and its many industrial uses.
- Uses of castor oil: lubricants (especially for high-speed engines and jet engines), hydraulic brake fluid, paints and varnishes, soaps, cosmetics, pharmaceuticals (as a purgative in medicine), plastics, nylon-11, and bio-based lubricants. Castor oil is a strategic industrial oil, and castor is an important export crop for India.
- Uses of the seed and cake: the oil cake is used as a fertiliser and, after detoxification, as a feed. The seed contains the toxin ricin, so raw cake and seed are poisonous and must be handled carefully. The plant is also used to rear the eri silkworm (castor is the host plant of the eri silkworm in some areas).
- Nutritional value of the seed (per 100 g): about 45-55 per cent oil, 18-20 per cent protein, and a high energy value (about 565 kcal). The protein is rich in lysine but the presence of ricin makes the seed unfit for direct food use.

Nutrient (per 100 g castor seed)	Approximate content
Oil	45-55 g
Protein	18-20 g

Nutrient (per 100 g castor seed)	Approximate content
Carbohydrate	20-25 g
Energy	about 565 kcal
Special feature	Ricinoleic acid (85-90 per cent of oil)
Toxin	Ricin (in the seed)

12.3 Soil and climatic requirements

- Soil: castor grows on a wide range of soils, from sandy loam to heavy clay, and tolerates poor soils. It performs best on deep, well-drained loams and on red and black soils with pH 6.0-8.0. It is fairly tolerant of salinity and alkalinity but sensitive to waterlogging.
- Climate: a warm-weather crop of the tropics and subtropics. Optimum temperature is about 20-30 degrees C. It needs about 500-700 mm of water and is fairly drought-tolerant because of its deep root system. It is a long-duration crop (about 150-180 days) and is grown mainly as a rain-fed kharif crop; the seed does not set well in very high temperature and dry conditions, and flowering is delayed by long days.

12.4 Varieties

- TMV 5 - TNAU variety, medium duration, high-yielding, with good oil content; the standard variety of Tamil Nadu.
- TMV 6 - TNAU variety of medium duration with good yield.
- CO 1 - TNAU variety of medium duration.
- CO(Rc) 1 - a TNAU release suited to Tamil Nadu conditions.
- GCH 4 - a high-yielding hybrid (from Gujarat) of medium duration.
- DCH 177, DCH 519 - popular hybrids with high yield and oil content.
- 48-1, Aruna and YTP 1 - other varieties/hybrids grown in different states.
- Hybrids give a much higher yield than varieties where inputs and management are assured, but they need good management and, in some cases, a higher plant population.

12.5 Land preparation and sowing

- Land preparation: one or two deep ploughings and harrowing to a fine tilth; castor is deep-rooted and benefits from deep initial tillage. Form ridges and furrows, and provide drainage.
- Sowing time: rain-fed kharif with the onset of the monsoon (June-July); where irrigated, it can be sown a little earlier or as a summer crop.
- Seed treatment: treat seed with a fungicide (for seedling blight and wilt) and with Trichoderma. Because the seed is large, place it carefully.
- Seed rate: about 10-12 kg/ha for varieties and about 8-10 kg/ha for hybrids (the seed is large and spacing is wide).
- Spacing: 90 x 60 cm for the spreading/tall types and hybrids; 60 x 45 cm or 60 x 60 cm for compact, short-statured types; the recommended plant population is about 18,000-25,000 plants per hectare for wide spacing.

12.6 Nutrient management

- Standard fertiliser dose: about 40 : 50 : 30 kg N : P2O5 : K2O per hectare for varieties; some recommendations use 60 : 40 : 20 or, for high-yielding hybrids with irrigation, up to 100 : 50 : 50 kg/ha.

- Apply FYM or compost 10 t/ha before sowing. Apply nitrogen in two splits (basal and at the time of flowering/first capsule formation).
- Castor responds to sulphur (for oil formation) and to zinc where deficient. Apply gypsum or elemental sulphur as per soil test.

Fertiliser for castor	Dose
FYM/compost	10 t/ha
N	40-60 kg/ha (up to 100 for hybrids)
P2O5	40-50 kg/ha
K2O	20-30 kg/ha
Sulphur	as per soil test (for oil)
N split timing	basal + at flowering/first capsule

12.7 Water management

- Castor is largely rain-fed and drought-tolerant, but it responds to irrigation. The critical stages are flowering and capsule development; moisture stress at these stages reduces seed set and oil content.
- Where irrigated, irrigate at about 12-15 day intervals; avoid waterlogging, which causes wilt and root rot. Provide drainage in the rainy season.

12.8 Weed management

- Keep the crop weed-free for the first 40-45 days. The wide spacing means weeds can establish easily; one or two hand weedings or a pre-emergence herbicide followed by one hand weeding.
- Intercropping with short-duration pulses (for example, black gram) can suppress weeds and add income.

12.9 Pest and disease management

Major insect pests

- Semilooper (Achaea janata) - the most serious pest; the caterpillar defoliates the plant, sometimes completely. Management: IPM, hand collection of larvae, biological control (Trichogramma, NPV) and need-based insecticide.
- Castor capsule borer (Conogethes punctiferalis) - the larva bores into the capsules and destroys the seed. Management: timely harvest, destruction of infested capsules and need-based insecticide.
- Shoot fly (in the seedling stage), leaf hopper (a vector of some diseases), thrips, jassids and the whitefly are other pests.

Major diseases

- Wilt (Fusarium oxysporum) - the most important disease; causes sudden wilting. Management: resistant varieties, crop rotation, seed treatment and Trichoderma.
- Seedling blight (Phytophthora) - causes collar rot and damping off in wet conditions.
- Rust, leaf spot (Alternaria, Cercospora) and powdery mildew are other diseases; manage with resistant varieties and need-based fungicide.

12.10 Harvest and yield

- Castor is a long-duration crop. The capsules mature in stages; harvest in two or three pickings when the capsules turn brown and the seed is hard, starting at about 150-160 days and continuing to about 180-190 days.
- Dry the capsules, thresh and clean the seed. Store at low moisture (below about 8 per cent) to avoid aflatoxin and quality loss.
- Yield: about 1.0-1.5 t/ha for varieties and up to 1.5-2.0 t/ha for hybrids under good management; rain-fed yields are lower (about 0.5-1.0 t/ha).

12.11 Economics - worked example

Problem: A farmer grows 1 hectare of castor (TMV 5) and obtains 1.2 t/ha of seed. Castor seed sells at Rs. 5,500 per quintal. Calculate gross return, net return and B:C ratio using the cost items below.

Cost item (per hectare)	Approximate amount (Rs.)
Land preparation	4,000
Seed (11 kg)	1,500
FYM 10 t	5,000
Fertiliser and sulphur	4,000
Sowing and thinning labour	2,500
Weeding and intercultivation	4,000
Plant protection	3,000
Harvesting (2-3 pickings) and threshing	6,000
Total cost of cultivation	30,000

- Gross return = 1.2 t/ha = 12 quintals x Rs. 5,500 = Rs. 66,000 per hectare.
- Net return = Rs. 66,000 - Rs. 30,000 = Rs. 36,000 per hectare.
- Benefit-cost ratio = 66,000 / 30,000 = 2.2.
- Castor has a modest net return per hectare but a low cost of cultivation and good drought tolerance, which makes it a useful crop for rain-fed and marginal land. Hybrids with a higher yield can raise the net return substantially.

12.13 Intercropping, cropping systems and the value chain

Intercropping and cropping systems

- Castor is grown as a sole crop and on field bunds and borders. Because of its wide spacing and slow early growth, it is well suited to intercropping with short-duration pulses.
- Common intercrops: castor + black gram, castor + green gram, castor + cowpea and castor + groundnut. The intercrop uses the space between the wide-spaced castor rows, adds nitrogen and income, and suppresses weeds.
- Rotations: castor-pulse and castor-cereal rotations are used in the dry tracts; castor is also grown after a cereal in the rain-fed areas.

The castor value chain

- Seed is crushed for oil; the oil is refined for industrial use (lubricants, hydraulic fluid, paints, cosmetics, medicine) and the cake is used as fertiliser or, after detoxification, as feed.
- India dominates world castor trade, exporting seed and oil to the United States, the European Union, China and Japan. The industrial demand gives castor a relatively stable and remunerative price.

- Because the seed contains ricin, safe handling and processing are important; the oil is safe, but raw cake must be detoxified before use as feed.

Field calendar and common problems

Crop calendar (operations schedule)

Operation	Time (days)	Remarks
Land preparation	Before sowing	Deep ploughing; ridges and furrows
Sowing	0	10-12 kg/ha at 90 x 60 cm
Thinning	15-20	One plant/hill at the wide spacing
Basal fertiliser	0	40:50:30; FYM 10 t/ha
Weeding	40-45	Wide spacing allows weeds; keep clean
Pest management	Vegetative	Semilooper and capsule borer control
Harvest	150-190	Two or three pickings of mature capsules

Common field problems, causes and remedies

Problem	Likely cause	Remedy
Complete defoliation	Semilooper (Achaea janata)	Hand collection; NPV; need-based insecticide
Capsules with holes	Capsule borer	Timely harvest; destroy infested capsules
Sudden wilting	Fusarium wilt	Resistant variety; rotation; Trichoderma
Poor capsule set	High temperature/drought	Irrigate at flowering; choose right season
Low oil content	Late harvest; immature seed	Harvest mature capsules; dry properly

Objective-type practice questions

Choose the correct answer. Answers are given at the end of the set.

1 Castor belongs to the family?
2 The chromosome number of castor is?
3 The origin of castor is?
4 The unique fatty acid of castor oil is?
5 The toxin present in castor seed is?
6 The most serious pest of castor is?
7 Wilt in castor is caused by?
8 The seed rate for castor varieties is?
9 The spacing for castor is commonly?
10 Castor oil is used mainly for?

Answers: 1-Euphorbiaceae; 2-2n=20; 3-Africa (Ethiopian region); 4-ricinoleic acid; 5-ricin; 6-semilooper (Achaea janata); 7-Fusarium; 8-10-12 kg/ha; 9-90 x 60 cm; 10-industrial purposes

12.12 Exam focus

Key points to remember

- Ricinus communis, family Euphorbiaceae, 2n = 20; origin Africa (Ethiopian region), India a secondary centre; India is the largest producer and exporter.
- Castor oil contains 45-55 per cent oil rich in ricinoleic acid (85-90 per cent of the oil), which gives it unique industrial uses (lubricants, hydraulic fluid, paints, medicine). The seed contains the toxin ricin.
- Soil: wide range, best on well-drained loam/red soil, pH 6.0-8.0; tolerant of salinity but not waterlogging. Temperature 20-30 C; water need 500-700 mm; long duration 150-180 days.
- TNAU varieties TMV 5, TMV 6, CO 1, CO(Rc) 1; hybrids GCH 4, DCH 177, DCH 519.
- Seed rate 10-12 kg/ha; spacing 90 x 60 cm (or 60 x 45 cm for compact types); NPK about 40:50:30 with sulphur.
- Major pest: semilooper (Achaea janata) and capsule borer. Major disease: wilt (Fusarium).
- Yield 1.0-1.5 t/ha (varieties) and up to 1.5-2.0 t/ha (hybrids); net return about Rs. 36,000/ha at Rs. 5,500/quintal.

Short-answer questions with model answers

1 Q1. What makes castor oil industrially important?

- Castor oil contains ricinoleic acid (about 85-90 per cent of its fatty acids), which gives the oil high viscosity, a stable structure and resistance to high and low temperatures. This makes it valuable for lubricants (including jet engines), hydraulic brake fluid, paints, plastics and medicines, and makes India a leading exporter of castor oil.

1 Q2. Why is castor seed cake dangerous as a feed?

- Castor seed contains the toxic protein ricin (and the allergen ricinine). Raw castor cake therefore cannot be fed to livestock; it must be detoxified (for example, by heat or chemical treatment) before use. The seed is otherwise rich in protein and oil.

1 Q3. Give the seed rate, spacing and fertiliser dose for castor.

- Seed rate about 10-12 kg/ha (8-10 kg/ha for hybrids); spacing 90 x 60 cm (or 60 x 45 cm for compact types); fertiliser about 40 : 50 : 30 kg N : P2O5 : K2O per hectare, with sulphur as per soil test.

Long-answer questions with model-answer outlines

1 Q1. Describe the origin, importance, varieties and cultural practices of castor.

- Outline: (i) botanical identity and African origin with India as a secondary centre; (ii) distribution and India's position as the largest producer/exporter; (iii) economic importance - ricinoleic acid and industrial uses, export value, cake and eri silkworm; (iv) nutritional/chemical composition; (v) soil and climate; (vi) varieties TMV 5, TMV 6, CO 1 and hybrids GCH 4, DCH 177; (vii) cultural practices - deep land preparation, seed rate, spacing, NPK 40:50:30 with sulphur, water and weed management; (viii) pest and disease management (semilooper, capsule borer, wilt); (ix) harvest in pickings and yield.

1 Q2. Explain the economics of castor cultivation and why it suits rain-fed areas.

- Outline: (i) the cost items with harvesting as a major cost because of multiple pickings; (ii) gross return from 1.2 t/ha at Rs. 5,500/quintal; (iii) net return of about Rs. 36,000 and a B:C ratio of about 2.2; (iv) the effect of using a hybrid (higher yield, higher return); (v) why castor suits rain-fed and marginal land - deep root system, drought tolerance, low cost, tolerance of poor soils; (vi) a note on price stability because of industrial demand and export.

Lecture 13 - Cotton: Origin, Distribution, Economic Importance, Soil and Climatic Requirements, Varieties, Cultural Practices and Yield

13.1 Origin and distribution

- Common name: Cotton, Paruthi (Tamil), Kapas (Hindi, for the unginned seed cotton). Botanical name: Gossypium spp. Family: Malvaceae.
- The four cultivated species and their chromosome numbers:
- Gossypium hirsutum - American upland cotton, 2n = 52 (New World, tetraploid); the most widely grown species in the world and in India.
- Gossypium barbadense - Egyptian and Sea Island cotton, 2n = 52 (New World, tetraploid); the longest and finest fibre, but a smaller area.
- Gossypium arboreum - Asiatic/desi cotton, 2n = 26 (Old World, diploid); native to India.
- Gossypium herbaceum - Asiatic/desi cotton, 2n = 26 (Old World, diploid); native to India and Africa.
- Origin: the two Asiatic species (G. arboreum and G. herbaceum) originated in the Indian sub-continent and Africa, and the two American species (G. hirsutum and G. barbadense) originated in Central and South America (Mexico, Peru). Cotton was thus domesticated independently in both the Old and the New World. India is the original home of the Asiatic cottons.
- Distribution: the leading producers are China, India, the United States, Brazil, Pakistan, Turkey, Australia and Uzbekistan. India is the largest producer and the second largest consumer of cotton in the world. In India the main states are Gujarat, Maharashtra, Telangana, Andhra Pradesh, Karnataka, Tamil Nadu, Punjab, Haryana, Rajasthan and Madhya Pradesh.
- In Tamil Nadu cotton is grown in Coimbatore, Tiruppur, Erode, Salem, Namakkal, Karur, Dindigul, Madurai, Virudhunagar, Tirunelveli, Thoothukudi, Ramanathapuram, Vellore, Tiruvannamalai and the southern districts.

13.2 Economic importance and nutritional value

- Cotton is the most important natural fibre crop of the world and the 'king of fibres'. It is grown for its lint (fibre), which is spun into yarn and woven into cloth; the seed yields edible oil (cotton seed oil) and the seed cake/meal is a protein-rich animal feed.
- Uses: lint for textiles, yarn, hosiery and non-woven fabrics; short lint (linters) for cellulose, rayon, paper and absorbent cotton; cotton seed oil for cooking and for soap; cotton seed cake/meal as cattle feed (after removing gossypol); the stalks for fuel. Cotton is a major source of farm income and employment in the textile industry.
- Nutritional value: cotton seed contains about 20-25 per cent oil and about 20-25 per cent protein. Cotton seed oil is an edible oil after refining (it is rich in polyunsaturated fatty acids, including vitamin E). The seed contains the pigment gland gossypol, which is toxic, so the meal is fed only after detoxification or by using glandless varieties. Cotton lint itself is not a food.
- Cotton is also an important commercial (cash) crop and supports a very large value chain - ginning, spinning, weaving, garmenting and export.

Product (per 100 g)	Approximate content
Cotton seed oil	20-25 g
Cotton seed protein	20-25 g
Cotton seed cake/meal	protein-rich animal feed (after detoxification)
Lint (fibre)	cellulose, used for textiles

Product (per 100 g)	Approximate content
Linters	cellulose for rayon, paper, absorbent cotton
Toxic principle	gossypol (in the seed glands)

13.3 Soil and climatic requirements

- Soil: cotton performs best on deep, well-drained black soils (regur) and on fertile loams and alluvial soils. Optimum pH is 6.0-8.0; cotton tolerates a fairly wide range and is fairly tolerant of salinity and alkalinity, but it is sensitive to waterlogging. Deep soils that hold moisture suit the rain-fed crop.
- Climate: a warm-weather crop of the tropics and subtropics. Optimum temperature is about 21-30 degrees C. It needs about 500-1000 mm of water over a long season. Cotton needs bright sunshine during boll development; cloudy weather and heavy late rain reduce yield and quality and cause boll rot. It is sensitive to frost.
- Cotton is a short-day plant in many types, but the modern varieties grown in Tamil Nadu are largely day-neutral.

Requirement	Optimum for cotton
Soil	Deep well-drained black soil, loam, alluvium
Soil pH	6.0-8.0 (wide tolerance)
Temperature	21-30 C
Water need	500-1000 mm
Critical stages	Flowering, boll development
Sensitive to	Waterlogging, frost, cloudy weather
Day length	Largely day-neutral (modern types)

13.4 Varieties

- MCU 5, MCU 7 - TNAU hirsutum varieties of medium duration with good yield; long-staple.
- MCU 12, MCU 13 - improved TNAU hirsutum varieties with high yield and good fibre quality.
- SVPR 2, SVPR 3, SVPR 4 - varieties from the Regional Research Station, Srivilliputhur, suited to the southern cotton tracts.
- CO 14, CO 17 - TNAU varieties of medium duration.
- K 10, K 11 - varieties of medium duration with good fibre quality.
- TCHB 213, TCHB 451 - TNAU hirsutum varieties/hybrids with good yield and fibre quality.
- Suraj, Kanchana, LRA 5166 - other popular varieties grown in the country.
- Desi (arboreum/herbaceum) varieties have a shorter staple but are hardy and suited to rain-fed and marginal areas; the hirsutum types and Bt hybrids dominate the irrigated areas. Bt cotton hybrids (with resistance to the American bollworm) cover a large part of the area and need refugia and resistance management.

13.5 Land preparation and sowing

- Land preparation: one or two deep ploughings and harrowing to a fine, well-drained tilth. On black soils, summer ploughing helps moisture conservation and reduces pests. Form ridges and furrows and provide drainage.

- Sowing time: rain-fed kharif with the onset of the monsoon (June-July); irrigated crops in April-May (summer/Chithirai) and August-September. The main Tamil Nadu season is the winter/rain-fed crop sown in June-July or September-October.
- Seed treatment: treat delinted seed with a fungicide and with an insecticide (for sucking pests and early bollworm) as per the recommendation. Acid-delint the seed to remove fuzz and improve sowing.
- Seed rate: about 2.5-3 kg/ha of delinted hirsutum seed (about 5-7 kg/ha of fuzzy seed); about 15-20 kg/ha for desi (arboreum) cotton because the seed is small and the spacing is closer.
- Spacing: 60 x 30 cm or 75 x 30 cm for hirsutum; 45 x 30 cm for desi cotton; the plant population is about 55,000-75,000 plants per hectare for hirsutum and higher for desi.

13.6 Nutrient management

- Standard fertiliser dose: 80 : 40 : 40 kg N : P2O5 : K2O per hectare for irrigated cotton (some recommendations use 100 : 50 : 50 for high-yielding types and 40 : 20 : 0 for rain-fed cotton).
- Apply FYM or compost 10-12.5 t/ha before sowing. Apply nitrogen in two or three splits - half basal and the rest at squaring and at flowering.
- Apply zinc sulphate and magnesium sulphate where deficient; cotton responds to foliar sprays of nutrients at boll development. Apply gypsum on alkaline soils.

Fertiliser for cotton	Dose
FYM/compost	10-12.5 t/ha
N	80 kg/ha (irrigated); 40 (rain-fed)
P2O5	40 kg/ha (irrigated); 20 (rain-fed)
K2O	40 kg/ha (irrigated); 0 (rain-fed)
N split timing	half basal; rest at squaring and flowering
Micronutrients	ZnSO4, MgSO4 where deficient

13.7 Water management

- Cotton is fairly drought-tolerant but responds well to irrigation. The critical stages are flowering and boll development; moisture stress at these stages causes square and boll shedding and reduces yield.
- Where irrigated, irrigate at about 10-15 day intervals in summer, and provide about 8-10 irrigations. Avoid excess water, which causes vegetative growth at the cost of bolls, and waterlogging, which causes boll rot and root rot.
- On rain-fed black soils, moisture conservation through summer ploughing, ridges and furrows and mulching is important.

13.8 Weed management

- Keep the crop weed-free for the first 45-60 days; cotton is a slow starter and a poor competitor early. One or two hand weedings or a pre-emergence herbicide followed by one hand weeding.
- Intercropping with short-duration pulses (black gram, green gram, cowpea) suppresses weeds and gives extra income, and is common in the rain-fed tracts.

13.9 Pest and disease management

Major insect pests

- Pink bollworm (Pectinophora gossypiella) - now the most serious pest of cotton; the larva bores into the square, flower and boll and damages the lint and seed. Management: timely sowing, destruction of infested bolls and crop residue, pheromone traps, monitoring, and need-based insecticide; rotate Bt and non-Bt crops to slow resistance.
- American bollworm (Helicoverpa armigera) - bores into the square and boll; controlled by Bt cotton and IPM.
- Spotted bollworm (Earias spp.) - bores into the shoot tip (in young plants) and the boll.
- Sucking pests - whitefly, jassids (leaf hopper), aphids and thrips - cause yellowing, curling and 'hopper burn', and reduce yield and quality; whitefly also transmits cotton leaf curl virus.
- Other pests: mealybug, red spider mite, leaf roller and the cotton stem weevil.

Major diseases

- Fusarium wilt and Verticillium wilt - cause wilting and internal browning; managed by resistant varieties, crop rotation and soil health.
- Bacterial blight (Xanthomonas) - angular water-soaked leaf spots and boll rot; managed by resistant varieties and seed treatment.
- Grey mildew, root rot, boll rot (in wet weather) and the cotton leaf curl virus are other important diseases.
- Management: integrated - resistant varieties and Bt hybrids with refugia, clean cultivation, destruction of crop residue, balanced fertiliser, monitoring with pheromone traps, conservation of natural enemies, and need-based pesticide.

13.10 Harvest and yield

- Cotton is harvested by picking the seed cotton from the opened bolls. Picking starts at about 150-180 days and continues in several pickings; the crop may extend to 200-240 days for long-duration types.
- Pick dry bolls in the morning, keep the pickings separate by quality (first, second, third picking), and dry the seed cotton before ginning.
- Yield: about 1.5-2.0 t/ha of seed cotton (kapas) for irrigated hirsutum and about 0.8-1.2 t/ha for rain-fed cotton; the ginning outturn (lint percentage) is about 33-35 per cent, so the lint yield is about one-third of the seed cotton yield. Staple length varies with species (hirsutum about 25-30 mm, barbadense longer, desi shorter).

13.11 Cotton growth stages - text diagram

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Cotton growth stages (about 150-180 days for hirsutum):
  Germination -> Seedling -> Vegetative growth (squaring)
     (0-10 d)     (10-30 d)      (30-60 d): squares (flower buds) form
                                                |
  Flowering (60-90 d) -> Boll development (90-150 d) -> Boll opening / Picking
     |                        |                              (150-180 d +)
  Flowers -> bolls;       Bolls enlarge; lint        Bolls split open; pick
  square/boll shedding    and seed mature            seed cotton in pickings
  if stressed
     |
  Critical stages for water and pest control: flowering and boll development
  Peak pest attack: squaring to boll formation (pink bollworm, bollworms)
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13.12 Worked example - seed rate and plant population for cotton

Problem: A farmer grows hirsutum cotton at 60 x 30 cm on 1 hectare using delinted seed of 1000-seed weight 100 g, germination 80 per cent and field survival 85 per cent. Calculate the plant population and the seed rate.

- Plants per hectare = 10000 / (0.60 x 0.30) = 10000 / 0.18 = 55,556 plants per hectare (about 55,500).

- Seed to be placed = 55,556 / (0.80 x 0.85) = 55,556 / 0.68 = 81,700 seeds per hectare.
- Seed weight = 81,700 x 100 g / 1000 = 8,170 g = about 8.2 kg per hectare of pure delinted seed.
- The standard recommendation of about 2.5-3 kg/ha of delinted seed assumes higher seed quality and closer management; the calculated figure shows the biological requirement at the stated germination and survival. Where fuzzy (undelinted) seed is used, the rate is about 5-7 kg/ha because the fuzz adds weight and reduces the number of seeds per kilogram. Always check the germination percentage on the seed label.

13.14 Cropping systems, intercropping and fibre quality

Cotton in cropping systems and intercropping

- Cotton is grown as a sole crop and as an intercrop. Because it is a long-duration, wide-spaced crop, it is well suited to intercropping.
- Common intercrops: cotton + black gram, cotton + green gram, cotton + cowpea, cotton + groundnut, cotton + onion and cotton + soybean. The intercrop uses the space between the cotton rows, adds nitrogen, gives extra income and helps control weeds.
- Rotations: cotton-cereal (cotton-sorghum, cotton-maize), cotton-pulse and cotton-rice rotations are used. Cotton is also grown after a rice crop in some tracts.

Fibre quality parameters

- Staple length: the length of the fibre; longer staple gives finer, stronger yarn. Hirsutum cotton has a staple of about 25-30 mm; barbadense cotton is longer; desi cotton is shorter.
- Fibre strength and fineness (micronaire): affect spinning performance and yarn quality.
- Uniformity and maturity: affect the evenness and appearance of the yarn.
- Trash and moisture: clean, dry cotton (low trash and low moisture) gets a better price at the gin and the market.
- The variety, the harvest (timely picking of opened, dry bolls) and the storage all affect fibre quality, and hence the price.

Value addition

- Ginning separates lint from seed; spinning converts lint to yarn; the seed yields oil and cake, and the linters yield cellulose products. Tamil Nadu's textile industry (Coimbatore, Tiruppur, Erode) is a major consumer of the State's cotton and supports a large value chain.

Field calendar and common problems

Crop calendar (operations schedule)

Operation	Time (days)	Remarks
Land preparation	Before sowing	Deep ploughing; ridges and furrows
Sowing	0	2.5-3 kg/ha delinted at 60 x 30 cm
Basal fertiliser	0	Half N, full P and K
Thinning	15-20	One plant/hill
Squaring top-dress N	45-60	Part of remaining N
Flowering top-dress N	75-90	Remaining N
Pest scouting	Squaring to boll formation	Monitor bollworms; pheromone traps
Picking	150-200	Several pickings; dry seed cotton

Common field problems, causes and remedies

Problem	Likely cause	Remedy
Rosetted flowers, damaged lint/seed	Pink bollworm	Timely sowing; destroy residue; traps; rotate Bt
Bored squares and bolls	American/spotted bollworm	IPM; need-based insecticide; Bt refugia
Yellowing, curling, hopper burn	Sucking pests (jassid, whitefly)	Need-based insecticide; conserve predators
Sudden wilting	Fusarium/Verticillium wilt	Resistant variety; rotation; soil health
Square and boll shedding	Moisture stress	Irrigate at flowering and boll development

Objective-type practice questions

Choose the correct answer. Answers are given at the end of the set.

1 Cotton belongs to the family?
2 The chromosome number of Gossypium hirsutum is?
3 The chromosome number of Gossypium arboreum is?
4 The origin of the Asiatic cottons is?
5 Cotton is called the?
6 Ginning outturn of hirsutum cotton is about?
7 The most serious recent pest of cotton is?
8 The standard NPK for irrigated cotton is?
9 The seed rate for delinted hirsutum cotton is?
10 Kapas means?

Answers: 1-Malvaceae; 2-2n=52; 3-2n=26; 4-India; 5-king of fibres; 6-33-35 per cent; 7-pink bollworm; 8-80:40:40; 9-2.5-3 kg/ha; 10-unginned seed cotton

13.13 Exam focus

Key points to remember

- Gossypium spp., family Malvaceae; four cultivated species - G. hirsutum and G. barbadense (2n = 52, New World), G. arboreum and G. herbaceum (2n = 26, Old World/Asiatic). India is the original home of the Asiatic cottons.
- Cotton is the king of fibres and a major cash crop; lint for textiles, seed for oil and cake, linters for cellulose. India is the largest producer of cotton.
- Soil: deep well-drained black soil, pH 6.0-8.0; temperature 21-30 C; water need 500-1000 mm; sensitive to waterlogging, frost and cloudy weather.
- TNAU varieties: MCU 5, MCU 7, MCU 12, MCU 13, SVPR 2/3/4, CO 14, CO 17, K 10, K 11, TCHB 213/451. Bt hybrids dominate the irrigated area.
- Seed rate about 2.5-3 kg/ha (delinted hirsutum) and 15-20 kg/ha (desi); spacing 60 x 30 cm (hirsutum); NPK 80:40:40 (irrigated).
- Major pests: pink bollworm, American bollworm, spotted bollworm and sucking pests. Major diseases: Fusarium and Verticillium wilt, bacterial blight.
- Yield 1.5-2.0 t/ha (irrigated) and 0.8-1.2 t/ha (rain-fed) of seed cotton; ginning outturn 33-35 per cent.

Short-answer questions with model answers

1 Q1. Name the four cultivated species of cotton with their chromosome numbers.

- Gossypium hirsutum (American upland, 2n = 52); G. barbadense (Egyptian/Sea Island, 2n = 52); G. arboreum (Asiatic/desi, 2n = 26); G. herbaceum (Asiatic/desi, 2n = 26). The first two are New World tetraploids and the last two are Old World diploids native to India and Africa.

1 Q2. What is ginning outturn?

- Ginning outturn is the percentage of lint (fibre) obtained from seed cotton (kapas) after ginning. For hirsutum cotton it is about 33-35 per cent, so about one-third of the seed cotton weight is lint and the rest is seed and waste.

1 Q3. Why is pink bollworm a serious problem in Bt cotton?

- Pink bollworm (Pectinophora gossypiella) bores into squares, flowers and bolls and damages lint and seed. It has developed resistance to the Bt toxin in many areas, so Bt cotton alone no longer controls it. Management needs timely sowing, destruction of crop residue and infested bolls, pheromone traps, refugia and need-based insecticide.

Long-answer questions with model-answer outlines

1 Q1. Describe the origin, importance and cultural practices of cotton.

- Outline: (i) the four species with chromosome numbers and their centres of origin (India for desi; America for hirsutum/barbadense); (ii) distribution and India's leading position; (iii) economic importance - lint, oil, cake, linters and the textile value chain; (iv) soil and climate; (v) varieties (MCU 5/7/12/13, SVPR 2/3/4, TCHB, Bt hybrids); (vi) land preparation, seed rate and spacing; (vii) NPK 80:40:40 with splits; (viii) water and weed management; (ix) pest and disease management with emphasis on pink bollworm, bollworms and wilt; (x) harvest in pickings and yield with ginning outturn.

1 Q2. Explain the growth stages of cotton and the management at each critical stage.

- Outline: (i) germination and seedling (protection from damping off and early pests); (ii) vegetative and squaring (nitrogen top-dress, weed control, first pest control); (iii) flowering and boll development (the critical stage - irrigation, pest control for bollworms, nutrient sprays); (iv) boll opening and picking (timely picking by quality, dry seed cotton); (v) the effect of stress at each stage on yield and quality; (vi) a concluding line on matching management to the stage.

Lecture 14 - Jute: Origin, Distribution, Economic Importance, Soil and Climatic Requirements, Varieties, Cultural Practices and Yield

14.1 Origin and distribution

- Common name: Jute, Pat (Hindi), Sanal (Tamil). Botanical name: Corchorus spp. Family: Tiliaceae (placed in Malvaceae in the broader classification).
- Two cultivated species:
- Corchorus olitorius - Tossa jute, 2n = 14; taller, finer fibre; grown mainly in the Ganga delta.
- Corchorus capsularis - White jute, 2n = 14; more tolerant of waterlogging; the traditional jute of low-lying lands.
- Origin: jute originated in the Indo-Burma region (the Indian sub-continent and South-East Asia). C. olitorius also has a centre of diversity in Africa. India (and the erstwhile Bengal region) is the historical home of jute.
- Distribution: the main producers are India and Bangladesh, which together account for the bulk of world jute. Other producers are China, Myanmar, Nepal, Thailand, Vietnam and Brazil. In India the

main states are West Bengal, Bihar, Assam, Odisha, Andhra Pradesh and Uttar Pradesh. In Tamil Nadu jute is grown in small areas, mainly as a fibre and vegetable (the tender leaves are eaten as a green leafy vegetable, 'sanal keerai'), in Cuddalore, Villupuram, Thanjavur and some other districts.

14.2 Economic importance and nutritional value

- Jute is called the 'golden fibre' because of its golden colour and its importance. It is a long, soft, strong, biodegradable and cheap natural fibre, second in importance to cotton among natural fibres.
- Uses: jute fibre is used for sacking (gunny bags), hessian cloth, carpet backing, ropes, twine, matting, geotextiles and eco-friendly packaging. Jute is now used in diversified products - jute geo-textiles, composites, pulp and paper, and as a substitute for synthetic packaging (which gives it a 'green' advantage).
- Jute sticks (the woody core) are used as fuel, for fencing, and in the paper and particle-board industry. The leaves of C. olitorius are eaten as a vegetable and are nutritious. The crop provides rural employment in cultivation, retting, fibre extraction, weaving and the jute industry.
- Nutritional value: jute seed contains about 18-20 per cent protein and about 12-14 per cent oil; the tender leaves are a good source of vitamins A and C, calcium and iron, and are eaten as a leafy vegetable. Jute is grown mainly for fibre, not for food.

Product	Use
Fibre (bast)	Sacking, hessian, ropes, twine, carpet backing, geotextiles
Jute sticks (core)	Fuel, fencing, paper, particle board
Leaves (C. olitorius)	Green leafy vegetable (vitamins A, C, calcium, iron)
Seed	about 18-20 per cent protein, 12-14 per cent oil
Environmental value	Biodegradable, eco-friendly packaging

14.3 Soil and climatic requirements

- Soil: jute grows best on fertile, well-drained alluvial soils of river deltas, which are renewed by silt each year. It also does well on sandy loam and loam. Optimum pH is 6.0-7.0. C. capsularis tolerates waterlogging and low-lying land, while C. olitorius needs better drainage and is grown on higher land.
- Climate: jute is a crop of the warm, humid tropics. It needs high temperature (about 24-37 degrees C), high relative humidity (above 70-80 per cent) and a good rainfall (about 1000-1500 mm), with warm humid weather during growth and fibre formation. It is sensitive to drought and to low temperature.
- Jute is a short-day plant; long days favour vegetative growth (fibre), which is why it is grown in the long, warm, humid summer days of the Ganga delta.

Requirement	Optimum for jute
Soil	Fertile well-drained alluvial/loam
Soil pH	6.0-7.0
Temperature	24-37 C
Rainfall	1000-1500 mm
Relative humidity	above 70-80 per cent
Day length	short-day (long days favour fibre growth)

14.4 Varieties

- JRO 524 - a popular tossa (C. olitorius) variety of medium duration with high fibre yield.
- JRO 8432, JRO 7835 - improved tossa varieties with good fibre yield and quality.
- JRC 321, JRC 212 - white jute (C. capsularis) varieties with good yield and waterlogging tolerance.
- S 19 - a popular white jute variety.
- TNAU has released jute varieties suited to Tamil Nadu conditions; the exact current list should be checked against the latest TNAU Crop Production Guide. National varieties (JRO and JRC series) are also grown where suited.
- Varieties with tolerance to stem rot and other diseases are preferred in disease-prone and high-rainfall areas.

14.5 Land preparation and sowing

- Land preparation: jute needs a fine, deep, well-drained seed-bed because the seed is small. One or two ploughings and several harrowings to a fine tilth; form beds and channels where drainage is needed.
- Sowing time: sown with the pre-monsoon showers, from March to May in the traditional jute tracts (and with the monsoon in June-July where the season is later). Early sowing gives taller plants and finer fibre.
- Seed treatment: treat seed with a fungicide for seed-borne diseases.
- Seed rate: about 5-6 kg/ha for line sowing and 8-10 kg/ha for broadcasting.
- Spacing: about 30 x 10 cm for line sowing, or 25-30 x 8-10 cm; broadcasting is common in the traditional tracts. Thin the seedlings to the correct stand at about 15-20 days.

14.6 Nutrient management

- Standard fertiliser dose: about 60 : 30 : 30 kg N : P2O5 : K2O per hectare (some recommendations use 40-60 : 20-30 : 20-30). Apply FYM or compost 5-10 t/ha.
- Nitrogen is applied in two splits - half basal and half at about 30-35 days after sowing (top-dress). Jute responds to nitrogen, but excess nitrogen encourages lodging and pests.
- Apply lime on acid soils and zinc where deficient.

14.7 Water management

- Jute needs a moist soil throughout its short life. Where rainfall is inadequate, irrigate at establishment and during rapid growth; avoid drought, which reduces plant height and fibre yield.
- Jute is sensitive to waterlogging in its early stage (except C. capsularis, which tolerates it better), so provide drainage. The crop needs moist but not waterlogged soil.

14.8 Weed management

- Jute is a small-seeded, slow-starting crop; keep it weed-free for the first 30-40 days. One or two hand weedings and thinning. A pre-emergence herbicide may be used where labour is scarce.
- Thinning at about 15-20 days to the recommended stand reduces competition and gives uniform, tall plants with good fibre.

14.9 Pest and disease management

Major insect pests

- Jute stem weevil (Apion corchori) - the grub bores into the stem, causing gall-like swellings and breaking the fibre; a serious pest.

- Jute semilooper (Anomis sabulifera) - the caterpillar defoliates the crop.
- Hairy caterpillar (Spilarctia obliqua) and the jute hairy caterpillar; the jute aphid, mealybug, thrips and the jute mite are other pests.
- Management: clean cultivation, timely sowing, hand collection of larvae and egg masses, biological control (Trichogramma, NPV), and need-based insecticide.

Major diseases

- Stem rot (Macrophomina phaseolina) - causes dark lesions on the stem and rotting; favoured by high temperature and moisture stress.
- Root rot (Rhizoctonia), anthracnose (Colletotrichum), powdery mildew and the jute mosaic virus are other diseases.
- Management: resistant varieties, seed treatment, crop rotation, balanced fertiliser and clean cultivation.

14.10 Harvest and fibre extraction

- Jute is grown for fibre, not seed. Harvest the plants for fibre at the small-pod stage (about 100-120 days from sowing), when the flowers have formed small pods but before the pods mature and the fibre becomes coarse. Plants are cut close to the ground.
- Fibre extraction by retting: bundle the cut stems and immerse them in slow-moving, clean water (a pond or a slow stream) for about 10-20 days. Micro-organisms (mainly bacteria) dissolve the pectin that binds the fibre to the woody core, so the fibre separates easily. This process is called retting.
- After retting, strip the fibre from the stems (stripping), wash it in clean water, and dry it in the sun. Bundle the dried fibre for sale.
- Fibre quality depends on the water used for retting, the maturity at harvest and the retting time. Clean, slow-moving water gives the best (golden) fibre.
- Yield: about 2-3 t/ha of dry fibre under good management (rain-fed), and about 1.5-2 t/ha in marginal areas. The stick yield is about 2-3 t/ha.

14.11 Worked example - fibre yield and return

Problem: A farmer grows 1 hectare of jute and obtains 2.5 t/ha of dry fibre. Fibre sells at Rs. 4,500 per quintal. Costs total Rs. 40,000 per hectare. Calculate the gross return, net return and B:C ratio. Also compute the return from sticks at 2.5 t/ha at Rs. 2,000/t.

- Gross return from fibre = 2.5 t/ha = 25 quintals x Rs. 4,500 = Rs. 1,12,500 per hectare.
- Return from sticks = 2.5 t x Rs. 2,000 = Rs. 5,000 per hectare.
- Total gross return = Rs. 1,12,500 + Rs. 5,000 = Rs. 1,17,500 per hectare.
- Net return = Rs. 1,17,500 - Rs. 40,000 = Rs. 77,500 per hectare.
- Benefit-cost ratio = 1,17,500 / 40,000 = about 2.9. The sticks and the eco-friendly demand for jute add to the return.

14.13 Cropping systems and the jute value chain

Jute in cropping systems

- Jute is grown as a sole crop in the traditional tracts and as a short-duration crop in rotation with rice and other crops in Tamil Nadu.
- Rotations: jute-rice, jute-pulse and jute-vegetables are common where jute is grown. In Tamil Nadu jute is often a small-area fibre and leafy-vegetable crop rather than a main crop.

- Intercropping: jute is sometimes intercropped with short-duration pulses or vegetables in the early stage, before the jute canopy closes.

The jute value chain and its future

- The value chain runs from cultivation and retting through to the jute mill (spinning, weaving of sacking and hessian), the diversified-products industry (geotextiles, composites, particle board), and export.
- Jute is promoted as an eco-friendly, biodegradable alternative to synthetic packaging, which is a growing market advantage. Jute geo-textiles are used in soil conservation and road construction.
- Value addition: diversified jute products (bags, mats, handicrafts), jute-reinforced composites, and jute paper add value and employment.

Field calendar and common problems

Crop calendar (operations schedule)

Operation	Time (days)	Remarks
Land preparation	Before sowing	Fine, deep seed-bed; beds and channels
Sowing	0	5-6 kg/ha line; 8-10 kg/ha broadcast
Thinning	15-20	Thin to the correct stand
Basal fertiliser	0	Half N, full P and K
Top-dress N	30-35	Remaining N
Weeding	20 and 35	Keep weed-free early
Harvest for fibre	100-120	At the small-pod stage
Retting	After harvest	10-20 days in clean slow water

Common field problems, causes and remedies

Problem	Likely cause	Remedy
Gall-like swellings, broken fibre	Stem weevil	Clean cultivation; need-based insecticide
Defoliated plants	Semilooper, hairy caterpillar	Hand collection; biological control
Dark stem lesions and rot	Stem rot (Macrophomina)	Rotation; resistant variety; balanced fertiliser
Coarse, dark fibre	Late harvest; poor retting water	Harvest at small-pod stage; use clean water
Weak, thin fibre	Over-retting	Correct retting period (10-20 days)

Objective-type practice questions

Choose the correct answer. Answers are given at the end of the set.

1 Jute belongs to the family?
2 Tossa jute is botanically?
3 White jute is botanically?
4 The origin of jute is?
5 Jute is called the?
6 The process of fibre extraction in jute is?
7 The most serious pest of jute is?

8 Jute is harvested at the?

9 The optimum temperature for jute is?

10 Jute fibre yield is about?

Answers: 1-Tiliaceae; 2-Corchorus olitorius; 3-Corchorus capsularis; 4-Indo-Burma; 5-golden fibre; 6-retting; 7-stem weevil; 8-small-pod stage; 9-24-37 C; 10-2-3 t/ha

14.12 Exam focus

Key points to remember

- Corchorus olitorius (tossa jute) and C. capsularis (white jute), both 2n = 14; family Tiliaceae; origin Indo-Burma; India and Bangladesh are the main producers.
- Jute is the 'golden fibre' - a strong, biodegradable natural fibre used for sacking, hessian, geotextiles and eco-friendly packaging; India and Bangladesh dominate world production.
- Soil: fertile well-drained alluvial/loam, pH 6.0-7.0; climate: 24-37 C, high humidity (above 70-80 per cent), rainfall 1000-1500 mm; a short-day plant.
- Seed rate 5-6 kg/ha (line) and 8-10 kg/ha (broadcast); spacing 30 x 10 cm; NPK about 60:30:30.
- Fibre is extracted by retting - immersing the stems in water for 10-20 days so that bacteria dissolve the pectin.
- Major pests: stem weevil and semilooper. Major diseases: stem rot, root rot, anthracnose.
- Harvest at the small-pod stage (100-120 days); fibre yield 2-3 t/ha; stick yield 2-3 t/ha.

Short-answer questions with model answers

1 Q1. What is retting in jute?

- Retting is the process of extracting jute fibre by immersing bundles of cut stems in slow-moving clean water for about 10-20 days. Bacteria dissolve the pectin that binds the fibre to the woody core, after which the fibre is stripped, washed and dried. Clean water and the right retting period give the best golden fibre.

1 Q2. Name the two cultivated species of jute and one difference between them.

- Corchorus olitorius (tossa jute) and Corchorus capsularis (white jute). Tossa jute is taller and gives finer fibre and is grown on better-drained land; white jute is more tolerant of waterlogging and is grown on low-lying land.

1 Q3. Give the seed rate and fertiliser dose for jute.

- Seed rate about 5-6 kg/ha for line sowing (8-10 kg/ha for broadcasting); fertiliser about 60 : 30 : 30 kg N : P2O5 : K2O per hectare, with nitrogen in two splits.

Long-answer questions with model-answer outlines

1 Q1. Describe the origin, importance, soil and climate, and cultural practices of jute, including fibre extraction.

- Outline: (i) the two species and Indo-Burma origin; (ii) distribution - India and Bangladesh as the main producers; (iii) economic importance - golden fibre, sacking, geotextiles, eco-friendly packaging, sticks and leaves; (iv) soil and climate - alluvial soil, warm humid climate, short-day plant; (v) varieties (JRO 524, JRC 321, S 19); (vi) cultural practices - fine seed-bed, seed rate, spacing, NPK 60:30:30, water and weed management; (vii) pest and disease management; (viii) harvest at the small-pod stage and retting for fibre extraction; (ix) yield of fibre and sticks and a one-line economics.

1 Q2. Explain the fibre extraction process in jute and the factors affecting fibre quality.

- Outline: (i) harvest at the small-pod stage; (ii) bundling and retting in clean slow-moving water for 10-20 days; (iii) the role of bacteria and pectin; (iv) stripping, washing and drying; (v) factors affecting quality - water quality, maturity at harvest, retting time, temperature and cleanliness; (vi) the importance of quality for price; (vii) conclusion on the eco-friendly advantage of jute.

Lecture 15 - Fodder Sorghum and Fodder Cowpea: Origin, Distribution, Economic Importance, Soil and Climatic Requirements, Varieties, Cultural Practices and Yield

Fodder crops are grown to feed livestock, and their value is measured in green fodder yield and quality rather than grain. Fodder sorghum is the backbone of irrigated fodder production in Tamil Nadu, and fodder cowpea is the standard legume fodder grown alone or mixed with cereals. This lecture treats both and then compares them.

PART A - FODDER SORGHUM (Cholam for fodder)

15.1 Origin, distribution and importance

- Common name: Fodder sorghum, Cholam (Tamil, fodder type), Jowar (fodder). Botanical name: Sorghum bicolor (L.) Moench. Family: Poaceae. Chromosome number 2n = 20.
- Origin: Africa (the Ethiopia-Sudan region), the same as grain sorghum; India is a secondary centre. It is grown wherever livestock are kept and irrigation or rainfall is available.
- In Tamil Nadu fodder sorghum is grown throughout the State, especially around dairy centres in Coimbatore, Erode, Tiruppur, Salem, Namakkal, Madurai, Dindigul and the delta districts.
- Importance: fodder sorghum gives a high yield of green fodder in a short time (about 60-75 days for a single cut) and is the main irrigated fodder of the State. It can be grown as a single-cut crop or as a multi-cut crop (ratooned). It is palatable, nutritious and can be preserved as hay or silage. The crop also gives stover for dry fodder.
- Nutritional value: green fodder sorghum contains about 10-12 per cent dry matter, 8-10 per cent crude protein (on a dry-matter basis), and good amounts of carbohydrates and minerals. It is a succulent, palatable fodder. The dry stover is a valuable maintenance fodder.

> Fodder sorghum (and sudan grass) can contain hydrocyanic acid (HCN, prussic acid), which is poisonous. The HCN content is highest in young, quick-growing, drought-stressed or heavily fertilised plants. Graze or feed only after the crop has wilted/dried (for example, after making hay) or after the crop is about 45-50 days old, and never feed wilted regrowth immediately after a drought.

15.2 Soil, climate, varieties and agronomy

- Soil: grows on a wide range of soils, from sandy loam to clay loam; best on deep, fertile, well-drained loams with pH 6.0-8.0. It tolerates poor soils and moderate salinity but not waterlogging.
- Climate: a warm-weather crop; optimum temperature 25-30 degrees C; needs about 400-600 mm of water; drought-tolerant but responds well to irrigation.
- Varieties: CO 27 (a popular multi-cut fodder sorghum), CO 28 (single-cut/multi-cut, high yield), K 12 (dual-purpose grain and fodder), CO(FS) 29 (a fodder sorghum variety), CSV 15 (a national fodder sorghum variety), SSG 59-3 (a sweet, juicy fodder sorghum), and the multi-cut fodder sorghum hybrids (MFSH series) which give several cuts from one sowing.
- Sowing time: throughout the year under irrigation; the main sowings are with the monsoon (June-July) and in summer (February-April).

- Land preparation: one or two ploughings and harrowing to a fine tilth; form ridges and furrows.
- Seed rate: about 25-30 kg/ha for line sowing and 40-50 kg/ha for broadcasting (fodder crops use a dense stand).
- Spacing: 30 x 10 cm or 45 x 15 cm for line sowing; closer spacing gives more, thinner stems and higher green fodder yield.
- Nutrient management: fodder crops need high nitrogen. Apply about 100-150 : 50 : 0 to 50 : 50 kg N : P2O5 : K2O per hectare. Apply half the N and all P and K as basal; top-dress the remaining N after each cut (for multi-cut types). Apply FYM 10 t/ha.
- Water management: irrigate at about 7-10 day intervals; a single-cut crop needs about 5-6 irrigations and a multi-cut crop more. Avoid waterlogging.
- Weed management: one or two hand weedings in the early stage; the dense stand then suppresses weeds.
- Pest and disease management: pests - shoot fly (in the seedling stage), stem borer, aphids, mites and the sugarcane aphid. Diseases - leaf spot, anthracnose, downy mildew and rust. Management: resistant varieties, clean cultivation, timely sowing and need-based control.
- Harvest and yield: for a single-cut crop, harvest at the 50 per cent flowering stage (about 60-75 days) for the best balance of yield and quality; for a multi-cut crop, take the first cut at about 60-65 days and subsequent cuts at 45-50 day intervals. Green fodder yield is about 40-60 t/ha for single-cut varieties and 60-100 t/ha per year for multi-cut types under good management.

PART B - FODDER COWPEA (Lobia for fodder)

15.3 Origin, distribution and importance

- Common name: Fodder cowpea, Lobia (Hindi), Karamani (Tamil, fodder/grain types). Botanical name: Vigna unguiculata (L.) Walp. Family: Fabaceae. Chromosome number 2n = 22.
- Origin: Africa, in the West African savanna (the primary centre of diversity); India is a secondary centre. The same species provides the grain cowpea of Lecture 8 and the fodder cowpea of this lecture; fodder types are selected for leafy growth and multiple cuts.
- Distribution: grown wherever cowpea is grown; in Tamil Nadu it is grown as a fodder and green-manure crop throughout the State.
- Importance: fodder cowpea is the most important legume fodder of Tamil Nadu. It is rich in protein, palatable, succulent and quick-growing (first cut in 50-60 days), and it fixes nitrogen and improves the soil. It is grown as a sole fodder crop and, more commonly, mixed with fodder sorghum, fodder maize or fodder bajra to give a balanced (cereal + legume) fodder.
- Nutritional value: green fodder cowpea contains about 12-15 per cent dry matter and 14-18 per cent crude protein (on a dry-matter basis), higher than the cereal fodders. It is a protein-rich fodder that balances the carbohydrate-rich cereal fodders in a mixed ration.

15.4 Soil, climate, varieties and agronomy

- Soil: grows on a wide range of soils; best on well-drained sandy loam and loam; pH 5.5-7.5; tolerates poor soils and moderate drought but not waterlogging.
- Climate: a warm-weather crop; optimum temperature 25-30 degrees C; needs about 400-600 mm of water; drought-tolerant and quick-growing.
- Varieties: CO 5, CO 6 (TNAU fodder cowpea varieties); Bundel Lobia 1 and UPC 5287 (national fodder cowpea varieties); Russian Giant (a tall, leafy type). Grain-and-fodder dual-purpose types (CO 2, CO 3, CO 4) can also be used for fodder.
- Sowing time: throughout the year under irrigation; the main sowings are with the monsoon and in summer. For a mixed crop, sow the cowpea with the cereal at the same time.

- Land preparation: as for fodder sorghum - a fine, well-drained tilth with ridges and furrows.
- Seed rate: about 25-30 kg/ha for a sole crop and 15-20 kg/ha when mixed with a cereal fodder.
- Spacing: 30 x 10 cm for a sole crop; in mixed cropping, sow cowpea in alternate rows or in the same row with the cereal.
- Nutrient management: about 25 : 50 : 0 to 50 : 50 kg N : P2O5 : K2O per hectare; being a legume, cowpea needs little nitrogen, but a starter dose helps. Apply FYM 5-10 t/ha. Rhizobium seed treatment improves nitrogen fixation.
- Water management: irrigate at about 7-10 day intervals; the crop is sensitive to drought at flowering but tolerant overall. Avoid waterlogging.
- Weed management: keep weed-free for the first 30 days; the crop then covers the ground.
- Pest and disease management: pests - aphids, jassids, thrips and pod borers. Diseases - cowpea mosaic, bacterial blight, powdery mildew and root rot. Management: resistant varieties, seed treatment and need-based control.
- Harvest and yield: harvest for fodder at the 50 per cent flowering to early-pod stage (about 50-65 days), or take a second cut where a multi-cut type is grown. Green fodder yield is about 20-30 t/ha for a sole crop and more in a well-managed mixed crop.

15.5 Fodder cropping systems - text diagram

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Common fodder cropping systems of Tamil Nadu:
  Irrigated (garden land):
     Cumbu Napier / Guinea grass (perennial, multi-cut)   <- year-round base
       + intercropped with fodder cowpea / lucerne (legume)
     or  Fodder sorghum / Fodder maize (single or multi-cut) rotated
         with fodder cowpea
  Seasonal rotation:
     Kharif:  Fodder sorghum + Fodder cowpea (mixed)
     Rabi:    Fodder maize / Fodder bajra
     Summer:  Fodder sorghum / Cumbu Napier
  Rain-fed / dryland:
     Cenchrus (Kollukattai pull) pasture + Stylosanthes (legume)
  Mixed (cereal + legume) cropping is preferred: the cereal gives bulk
  (carbohydrate), the legume gives protein, and the legume fixes nitrogen.
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15.6 Comparison of fodder sorghum and fodder cowpea

Feature	Fodder sorghum	Fodder cowpea
Botanical name	Sorghum bicolor	Vigna unguiculata
Family	Poaceae	Fabaceae
Chromosome (2n)	20	22
Origin	Africa (Ethiopia-Sudan)	Africa (West Africa)
Nature	Cereal fodder (carbohydrate)	Legume fodder (protein)
Duration / first cut	60-75 days (single cut)	50-65 days
Seed rate	25-30 kg/ha (line)	25-30 kg/ha (sole)
Spacing	30 x 10 cm	30 x 10 cm
NPK	100-150 : 50 : 0-50	25-50 : 50 : 0
Green fodder yield	40-60 t/ha (single cut)	20-30 t/ha
Key risk	HCN (prussic acid) poisoning	Waterlogging

15.7 Worked example - fodder requirement and area to be grown

Problem: A dairy farmer has 5 crossbred cows, each needing about 30 kg of green fodder per day. A fodder sorghum crop (single cut) gives 50 t/ha in 70 days. Calculate the daily fodder requirement, the annual requirement, and the area of fodder sorghum needed per cut, assuming the crop is grown in sequence.

- Daily requirement = 5 cows x 30 kg = 150 kg of green fodder per day.
- Annual requirement = 150 kg x 365 days = 54,750 kg = about 54.75 tonnes per year.
- A single-cut crop gives 50 t/ha in 70 days, i.e. 50,000 kg from 1 hectare. For the 70 days, the requirement is 150 x 70 = 10,500 kg.
- Area needed for one crop cycle = 10,500 / 50,000 = 0.21 hectare (about 52 cents).
- If the farmer keeps about 0.25-0.30 hectare under fodder sorghum in rotation, with staggered sowing, the herd's green fodder need can be met. Adding a legume fodder (cowpea) in the rotation improves the protein content of the ration.

15.9 Fodder conservation and the fodder calendar

Conservation of fodder

- Hay: green fodder is dried in the sun to about 15 per cent moisture and stored. Hay is a maintenance fodder; over-drying and rain damage reduce quality.
- Silage: green fodder (maize, sorghum, Cumbu Napier) is chopped, packed in a pit or silo, compressed and sealed so that it ferments to lactic acid, which preserves it. Good silage has a pleasant smell, a green-brown colour and a low pH. Silage makes a succulent fodder available in the dry season.
- For silage, harvest at the dough stage, chop to about 2-3 cm, and seal the pit well to keep out air.

A simple fodder calendar

Season	Fodder crops to grow
Kharif (June-Oct)	Fodder sorghum + cowpea; fodder maize
Rabi (Nov-Feb)	Fodder maize, fodder bajra, berseem/lucerne (where suited)
Summer (Mar-May)	Fodder sorghum, fodder cowpea, fodder maize
Year-round base	Cumbu Napier, Guinea grass (perennial, multi-cut)

- The fodder calendar is planned so that green fodder is available every day of the year; the perennial grasses give the base supply and the seasonal crops fill the gaps. Silage and hay bridge the lean season.

Field calendar and common problems

Crop calendar (operations schedule)

Operation	Time	Remarks
Fodder sorghum sowing	Monsoon or summer	25-30 kg/ha at 30 x 10 cm
Cowpea sowing (mixed)	With the cereal	15-20 kg/ha in mixed crop
Basal fertiliser	At sowing	Half N, full P
Top-dress N	After each cut	Remaining N (multi-cut types)
First cut	60-75 days	At 50 per cent flowering

Operation	Time	Remarks
Subsequent cuts	Every 45-50 days	Multi-cut fodder sorghum
Conservation	Surplus season	Hay or silage for the dry season

Common field problems, causes and remedies

Problem	Likely cause	Remedy
Sudden death of cattle after feeding	HCN poisoning in young/stressed sorghum	Feed after wilting/drying; avoid young growth
Poor fodder quality	Harvested too late (over-mature)	Harvest at 50 per cent flowering
Low yield	Low nitrogen; no irrigation	High N; regular irrigation
Bloat in cattle	Legume-rich fresh fodder	Feed after wilting; mix with dry fodder
Weeds in fodder	Slow early growth	Early weeding; dense stand

Objective-type practice questions

Choose the correct answer. Answers are given at the end of the set.

1 Fodder sorghum belongs to the family?
2 Fodder cowpea belongs to the family?
3 HCN (prussic acid) poisoning occurs in?
4 The main legume fodder of Tamil Nadu is?
5 A mixed fodder crop is grown to give?
6 Fodder crops are harvested at?
7 Green fodder yield of fodder sorghum is about?
8 Fodder is conserved as?
9 Fodder cowpea is rich in?
10 The seed rate for fodder sorghum is?

Answers: 1-Poaceae; 2-Fabaceae; 3-fodder sorghum; 4-fodder cowpea; 5-a balanced cereal + legume ration; 6-50 per cent flowering; 7-40-60 t/ha; 8-hay and silage; 9-protein; 10-25-30 kg/ha

15.8 Exam focus

Key points to remember

- Fodder sorghum: Sorghum bicolor, 2n = 20, origin Africa; high-yielding cereal fodder; single-cut (60-75 days) or multi-cut; varieties CO 27, CO 28, K 12, CO(FS) 29, CSV 15, SSG 59-3, MFSH hybrids; seed 25-30 kg/ha; NPK 100-150:50:0-50; green fodder 40-60 t/ha.
- Fodder cowpea: Vigna unguiculata, 2n = 22, origin West Africa; the main legume fodder; protein-rich; first cut 50-65 days; varieties CO 5, CO 6, Bundel Lobia 1, UPC 5287; seed 25-30 kg/ha; NPK 25-50:50:0; green fodder 20-30 t/ha.
- Fodder crops are judged by green fodder yield and quality (protein, digestibility), not grain yield.
- Mixed cereal + legume fodder cropping gives a balanced ration and fixes nitrogen.
- HCN (prussic acid) poisoning is the main risk in fodder sorghum; feed after wilting/drying and avoid young, drought-stressed growth.
- Harvest fodder at the 50 per cent flowering stage for the best balance of yield and quality.

Short-answer questions with model answers

1 Q1. What is HCN poisoning in fodder sorghum and how is it avoided?

- Fodder sorghum can contain hydrocyanic acid (prussic acid), which is poisonous to livestock. The content is highest in young, quick-growing, drought-stressed or heavily fertilised plants. It is avoided by feeding the crop only after it has wilted/dried (or after making hay) and by not feeding fresh regrowth immediately after a drought or frost; the crop is usually safe after about 45-50 days.

1 Q2. Why is fodder cowpea grown mixed with fodder sorghum or maize?

- Fodder cowpea is a legume rich in protein, while cereal fodders are rich in carbohydrate and low in protein. A mixed crop gives a balanced ration, improves fodder quality, and the legume fixes nitrogen, which benefits the cereal. The mixed crop also yields more total fodder per unit area.

1 Q3. Give the seed rate and green fodder yield of fodder sorghum and fodder cowpea.

- Fodder sorghum: seed 25-30 kg/ha (line), green fodder 40-60 t/ha (single cut). Fodder cowpea: seed 25-30 kg/ha (sole), green fodder 20-30 t/ha.

Long-answer questions with model-answer outlines

1 Q1. Describe the importance and cultural practices of fodder sorghum.

- Outline: (i) origin and importance as the main irrigated fodder; (ii) nutritional value and the HCN risk; (iii) soil and climate; (iv) varieties - CO 27, CO 28, K 12, CO(FS) 29, SSG 59-3, MFSH hybrids; (v) land preparation, seed rate 25-30 kg/ha, spacing 30 x 10 cm; (vi) NPK 100-150:50:0-50 with N split after each cut; (vii) irrigation; (viii) weed, pest and disease management; (ix) harvest at 50 per cent flowering and green fodder yield; (x) a line on mixed cropping with cowpea.

1 Q2. Explain fodder cropping systems and the role of legume fodders.

- Outline: (i) the need for year-round green fodder; (ii) perennial multi-cut grasses (Cumbu Napier, Guinea) as the base; (iii) seasonal cereal fodders (sorghum, maize, bajra); (iv) legume fodders (cowpea, lucerne) for protein and nitrogen fixation; (v) mixed cereal + legume cropping and the balance it gives; (vi) rain-fed pasture systems (Cenchrus + Stylosanthes); (vii) a worked example of computing fodder area for a herd; (viii) conclusion on balanced ration and soil fertility.

Lecture 16 - Fodder Maize and Cenchrus (Kollukattai Pull): Origin, Distribution, Economic Importance, Soil and Climatic Requirements, Varieties, Cultural Practices and Yield

Fodder maize is the most succulent and highest-yielding annual cereal fodder for stall-fed cattle, and Cenchrus (Kollukattai pull) is the standard drought-hardy perennial pasture grass for the dry tracts. Together they represent the two ends of fodder production - intensive irrigated fodder and extensive rain-fed pasture. This lecture treats both.

PART A - FODDER MAIZE

16.1 Origin, distribution and importance

- Common name: Fodder maize, Makka cholam (fodder type). Botanical name: Zea mays L. Family: Poaceae. Chromosome number 2n = 20.
- Origin: Central America (southern Mexico and Guatemala), the same as grain maize; the fodder types are selections and hybrids of Z. mays bred for tall, leafy, high-biomass growth.

- Distribution: grown wherever maize is grown and livestock are kept; in Tamil Nadu fodder maize is grown around dairy centres in Coimbatore, Erode, Tiruppur, Salem, Namakkal, Madurai, Dindigul and the delta.
- Importance: fodder maize is the most succulent and palatable of the cereal fodders and gives a very high yield of green fodder in a short time (about 60-75 days). It is rich in soluble carbohydrates, low in fibre (when harvested at the right stage) and highly digestible, which makes it the preferred fodder for high-yielding crossbred cows and buffaloes. It is also the best crop for making silage, because of its high fermentable carbohydrate content.
- Nutritional value: green fodder maize contains about 12-15 per cent dry matter, 7-9 per cent crude protein (on a dry-matter basis), and high total digestible nutrients; it is an energy-rich, palatable fodder. Maize silage is a standard conserved fodder.

16.2 Soil, climate, varieties and agronomy

- Soil: grows best on deep, fertile, well-drained loam and sandy loam soils rich in organic matter; pH 5.5-7.5; sensitive to waterlogging and salinity.
- Climate: a warm-weather crop; optimum temperature 21-27 degrees C; needs about 500-800 mm of water; sensitive to frost and waterlogging.
- Varieties: African Tall (a tall, high-yielding fodder maize variety, the most popular); CO 6 (a TNAU variety that is dual-purpose for grain and fodder); APFM 8 (a fodder maize variety); Ganga Safed 2 and other fodder maize varieties/hybrids; and private fodder maize hybrids. Sweet corn and baby corn types are not fodder types.
- Sowing time: throughout the year under irrigation; main sowings with the monsoon (June-July) and in summer (February-April).
- Land preparation: one or two ploughings and harrowing to a fine tilth; form ridges and furrows.
- Seed rate: about 40-50 kg/ha (fodder maize uses a dense stand for high biomass).
- Spacing: 30 x 15 cm or 45 x 15 cm for a dense, leafy stand.
- Nutrient management: fodder maize needs high nitrogen. Apply about 100-150 : 50-60 : 40 kg N : P2O5 : K2O per hectare (or up to 200 kg N/ha for very high yields). Apply half the N and all P and K as basal, and top-dress the rest at the knee-high stage. Apply FYM 10-12.5 t/ha.
- Water management: irrigate at about 7-10 day intervals; the crop needs about 5-7 irrigations. Avoid waterlogging.
- Weed management: keep weed-free for the first 30-40 days; one or two hand weedings or a pre-emergence herbicide (atrazine).
- Pest and disease management: the fall armyworm is now the most serious pest of fodder maize, followed by the stem borer. Diseases include turcicum leaf blight and downy mildew. Management: IPM with emphasis on scouting and control of fall armyworm.
- Harvest and yield: harvest for green fodder at the milk to early-dough stage (about 60-75 days), when the fodder is most succulent and digestible; for silage, harvest at the dough stage. Green fodder yield is about 40-60 t/ha under good management.

PART B - CENCHRUS (Kollukattai Pull / Buffel Grass)

16.3 Origin, distribution and importance

- Common name: Cenchrus, Buffel grass, Anjan grass, Kollukattai pull (Tamil). Botanical name: Cenchrus ciliaris L. (syn. Pennisetum ciliare). Family: Poaceae. Chromosome number varies with type (2n = 32, 36, 40, 44, 54; commonly 2n = 36 or 40).
- A closely related species, Cenchrus setigerus (birdwood grass), is also grown as a pasture grass.

- Origin: Cenchrus originated in the arid and semi-arid regions of Africa and India (the tropical and subtropical drylands). It is native to the dry tracts of Africa, Arabia and India.
- Distribution: grown as a pasture grass in the drylands of India (Rajasthan, Gujarat, Madhya Pradesh, Maharashtra, Karnataka, Andhra Pradesh, Telangana and Tamil Nadu), Africa, Australia and the Americas.
- In Tamil Nadu Cenchrus (Kollukattai pull) is grown in the dry tracts of Ramanathapuram, Virudhunagar, Tirunelveli, Thoothukudi, Madurai, Sivaganga, Dharmapuri and Krishnagiri, mainly for rain-fed pasture and for soil conservation.
- Importance: Cenchrus is a deep-rooted, drought-hardy, perennial grass that gives repeated cuts and stays green in dry spells. It is the standard pasture grass for the drylands and is used for grazing, for cut-and-carry fodder and for soil conservation (it binds the soil and checks erosion). It is palatable and nutritious when young.
- Nutritional value: green Cenchrus fodder contains about 25-35 per cent dry matter (it is more fibrous than fodder maize) and about 7-10 per cent crude protein (on a dry-matter basis), with good mineral content. It is a valuable maintenance fodder for dryland livestock.

16.4 Soil, climate, varieties and agronomy

- Soil: grows on a wide range of soils, including light, sandy, gravelly and shallow soils, and tolerates low fertility and drought; best on well-drained sandy loam with pH 6.5-8.5. It is tolerant of salinity and alkalinity and is very sensitive to waterlogging.
- Climate: a crop of the hot, dry tropics. It tolerates high temperature (up to about 40-45 degrees C) and low rainfall (about 300-500 mm) and is extremely drought-hardy. It is a short-day plant and is photo-insensitive in many types. It goes dormant in severe drought and revives with rain.
- Varieties: CO 1 (a TNAU Cenchrus variety); CAZRI 75 (from the Central Arid Zone Research Institute); IGFRI 3108 (from the Indian Grassland and Fodder Research Institute); Bichpuri; and other national varieties. Varieties differ in growth habit, leafiness and drought tolerance.
- Sowing time: with the onset of the monsoon (June-July) for rain-fed pasture; it can be sown under irrigation too.
- Land preparation: plough and harrow to a fine tilth; on sloping land, sow on the contour or in trenches for moisture conservation.
- Seed rate and planting: about 4-5 kg/ha of seed (the seed is light and fluffy; use a carrier for even sowing) or about 20,000-30,000 rooted slips per hectare. Spacing about 45 x 30 cm or 30 x 30 cm.
- Nutrient management: rain-fed pasture receives about 40 : 20 : 0 kg N : P2O5 : K2O per hectare, with a top-dress of N after each cut where irrigated. Apply FYM 5 t/ha at establishment.
- Water management: mostly rain-fed; where irrigated, light irrigations after each cut. The crop is very drought-tolerant and survives long dry spells.
- Weed management: keep the pasture weed-free during establishment; after establishment the grass forms a dense sod and suppresses weeds.
- Pest and disease management: Cenchrus is relatively free of serious pests and diseases; occasional grasshoppers and leaf diseases occur. Management is largely through good establishment and grazing management.
- Harvest and yield: take the first cut at about 60-75 days (or allow light grazing after establishment), then cuts at about 30-45 day intervals in the growing season. Green fodder yield is about 20-40 t/ha per year in multiple cuts under rain-fed/limited-irrigation conditions; under good irrigation it can be higher. The pasture lasts several years.

16.5 Cenchrus pasture establishment - text diagram

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Cenchrus (Kollukattai pull) dryland pasture layout:
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Contour / trench sowing on sloping land for moisture conservation:
[trench]   ~~~  ~~~  ~~~  ~~~  ~~~   <- Cenchrus rows (45 x 30 cm)
[trench]   ~~~  ~~~  ~~~  ~~~  ~~~
[trench]   ~~~  ~~~  ~~~  ~~~  ~~~   Add a legume (Stylosanthes) in
[trench]   ~~~  ~~~  ~~~  ~~~  ~~~   alternate rows for protein + N
[trench]   ~~~  ~~~  ~~~  ~~~  ~~~
Perennial: lasts several years; repeated cuts/grazing; survives drought
by going dormant and reviving with the monsoon rains.
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16.6 Comparison of fodder maize and Cenchrus

Feature	Fodder maize	Cenchrus (Kollukattai pull)
Botanical name	Zea mays	Cenchrus ciliaris
Family	Poaceae	Poaceae
Origin	Central America	Africa/India (arid tropics)
Nature	Annual, succulent fodder	Perennial pasture grass
Duration / first cut	60-75 days (single cut)	60-75 days, then repeated cuts
Seed rate	40-50 kg/ha	4-5 kg/ha seed or 20,000-30,000 slips/ha
Spacing	30 x 15 cm	45 x 30 cm
NPK	100-150 : 50-60 : 40	40 : 20 : 0
Green fodder yield	40-60 t/ha	20-40 t/ha per year (multi-cut)
Drought tolerance	Low (needs irrigation)	Very high (dryland pasture)
Best use	Silage and stall feeding	Grazing, cut-and-carry, soil conservation

16.7 Worked example - silage requirement and area

Problem: A farmer with 10 crossbred cows wants to conserve silage for 120 days of the dry season. Each cow needs about 20 kg of silage per day. Fodder maize gives 50 t/ha. Calculate the silage needed and the area of fodder maize to grow.

- Daily silage need = 10 cows x 20 kg = 200 kg per day.
- Total silage for 120 days = 200 x 120 = 24,000 kg = 24 tonnes.
- Fodder maize gives about 50 t/ha of green fodder, and the recovery of good silage is about 70-80 per cent of the green fodder weight, i.e. about 35-40 t of silage per hectare.
- Area of fodder maize needed = 24 / 37.5 = about 0.64 hectare (about 1.6 acres).
- With staggered sowing (for example, one sowing every 3-4 weeks), the farmer can keep supplying green fodder and also fill the silage pit. This is the standard approach for year-round fodder supply.

16.9 Pasture management and fodder conservation

Pasture management

- A pasture is a managed area of grass and legumes for grazing. Good pasture management involves the right species mix (a grass plus a legume, such as Cenchrus + Stylosanthes), controlled grazing (rotational or deferred grazing so that plants recover), and periodic fertiliser and weed management.
- Overgrazing weakens the pasture and leads to soil erosion; under-grazing allows the grass to become coarse and unpalatable. Rotational grazing with rest periods keeps the pasture productive.
- On dryland and degraded land, Cenchrus and other hardy grasses are used for pasture development and soil conservation; contour sowing and trenching conserve moisture.

Fodder conservation

- Surplus green fodder in the monsoon is conserved as hay (sun-dried) or silage (fermented) for the dry season. Fodder maize and sorghum are the main silage crops; Cenchrus and other grasses can be made into hay.
- Fodder banks (community fodder reserves) are promoted in dryland areas to protect livestock during drought. Conservation turns a seasonal surplus into a year-round supply and stabilises the livestock economy.

Field calendar and common problems

Crop calendar (operations schedule)

Operation	Time	Remarks
Fodder maize sowing	Monsoon or summer	40-50 kg/ha at 30 x 15 cm
Cenchrus sowing	Monsoon	4-5 kg/ha seed or 20,000-30,000 slips/ha
Basal fertiliser	At sowing	Maize 100-150:50-60:40; Cenchrus 40:20:0
Top-dress N	Knee-high (maize); after each cut (Cenchrus)	For high biomass
Harvest (maize)	60-75 days	Milk to early-dough stage
Harvest (Cenchrus)	First cut 60-75 d; then 30-45 d	Repeated cuts for years
Conservation	Surplus season	Silage (maize) and hay (Cenchrus)

Common field problems, causes and remedies

Problem	Likely cause	Remedy
Ragged whorls in fodder maize	Fall armyworm	Scout; Bt/NPV; whorl insecticide
Poor Cenchrus establishment	Poor seed quality; crusting; drought	Use good slips/seed; sow with monsoon
Coarse, unpalatable pasture	Over-mature grass; overgrazing	Harvest/graz e at the right stage; rotational grazing
Low biomass	Low nitrogen; no top-dress	Top-dress N after each cut
Silage spoilage	Air entry; wet fodder; poor sealing	Chop, compact and seal the pit well

Objective-type practice questions

Choose the correct answer. Answers are given at the end of the set.

1 Fodder maize belongs to the family?
2 Cenchrus is botanically?
3 Cenchrus is also called?
4 Cenchrus is a?
5 Fodder maize is harvested at the?
6 The best crop for silage is?
7 The seed rate for fodder maize is?
8 The origin of Cenchrus is?
9 Green fodder yield of fodder maize is about?
10 The major pest of fodder maize is?

Answers: 1-Poaceae; 2-Cenchrus ciliaris; 3-buffel grass / Kollukattai pull; 4-perennial pasture grass; 5-milk to early-dough stage; 6-fodder maize; 7-40-50 kg/ha; 8-Africa/India (arid tropics); 9-40-60 t/ha; 10-fall armyworm

16.8 Exam focus

Key points to remember

- Fodder maize: Zea mays, 2n = 20, origin Central America; the most succulent and palatable cereal fodder and the best silage crop; varieties African Tall, CO 6, APFM 8; seed 40-50 kg/ha; spacing 30 x 15 cm; NPK 100-150:50-60:40; green fodder 40-60 t/ha in 60-75 days.
- Cenchrus (Kollukattai pull): Cenchrus ciliaris, family Poaceae, origin Africa/India (arid tropics); a drought-hardy perennial pasture grass; varieties CO 1, CAZRI 75, IGFRI 3108; seed 4-5 kg/ha or 20,000-30,000 slips/ha; spacing 45 x 30 cm; NPK 40:20:0; green fodder 20-40 t/ha per year.
- Fodder maize is harvested at the milk to early-dough stage for the best quality; Cenchrus gives repeated cuts and lasts several years.
- Cenchrus is used for dryland pasture, grazing, cut-and-carry fodder and soil conservation; it tolerates salinity, alkalinity and long dry spells.
- Both are Poaceae (grasses); fodder maize is an annual high-input crop, Cenchrus a perennial low-input pasture grass.

Short-answer questions with model answers

1 Q1. Why is fodder maize the best crop for silage?

- Fodder maize has a high yield of succulent green fodder, a high content of fermentable soluble carbohydrates, moderate moisture and good digestibility. These properties favour good lactic acid fermentation, so the fodder can be conserved as high-quality silage with low losses. This makes maize the standard silage crop.

1 Q2. Give the botanical name and one special feature of Cenchrus.

- Cenchrus ciliaris (buffel grass / anjan grass / Kollukattai pull), family Poaceae. Its special feature is extreme drought hardiness - it is deep-rooted, survives long dry spells by going dormant, revives with rain, and is a perennial pasture grass that gives repeated cuts for several years.

1 Q3. At what stage is fodder maize harvested and why?

- Fodder maize is harvested at the milk to early-dough stage (about 60-75 days) because at this stage the fodder is most succulent, low in fibre and highly digestible, giving the best balance of yield and quality. For silage, harvest at the dough stage.

Long-answer questions with model-answer outlines

1 Q1. Describe the importance and cultural practices of fodder maize.

- Outline: (i) origin and importance as the most succulent cereal fodder and the best silage crop; (ii) nutritional value; (iii) soil and climate; (iv) varieties - African Tall, CO 6, APFM 8; (v) land preparation, seed rate 40-50 kg/ha, spacing 30 x 15 cm; (vi) NPK 100-150:50-60:40 with split N; (vii) irrigation; (viii) weed, pest (fall armyworm) and disease management; (ix) harvest at the milk-dough stage and green fodder yield; (x) a worked example of computing the area for silage.

1 Q2. Write about Cenchrus as a dryland pasture grass.

- Outline: (i) botanical identity and origin in the arid tropics; (ii) drought hardiness and perennial habit; (iii) uses - grazing, cut-and-carry fodder, soil conservation; (iv) nutritional value; (v) soil and climate; (vi) varieties - CO 1, CAZRI 75, IGFRI 3108; (vii) establishment - seed rate/slips, spacing,

contour/trench sowing; (viii) fertiliser (40:20:0), water and grazing management; (ix) green fodder yield and longevity; (x) the role of legume (Stylosanthes) association for protein and nitrogen.

Lecture 17 - Cumbu Napier and Guinea Grass; Rice Ecosystems, Climate, Weather, Seasons and Varieties of Tamil Nadu

This final lecture has two parts. Part A covers the two perennial multi-cut fodder grasses - Cumbu Napier hybrid and Guinea grass - which form the year-round base of irrigated fodder production. Part B studies the rice ecosystems of the world and of Tamil Nadu, and the climate, weather, seasons and varieties of Tamil Nadu.

PART A - CUMBU NAPIER HYBRID AND GUINEA GRASS

17.1 Cumbu Napier hybrid

Origin, distribution and importance

- Common name: Cumbu Napier hybrid, Hybrid Napier, CO grass. Botanical name: Pennisetum glaucum (L.) R. Br. x Pennisetum purpureum Schumach. (a hybrid of pearl millet and napier grass). Family: Poaceae.
- Origin: the hybrid was developed by crossing pearl millet (Cumbu, African origin) with napier grass (elephant grass, African origin). It is a man-made hybrid, first produced in India, and combines the high tillering and quick establishment of pearl millet with the tall, vigorous, perennial growth of napier grass.
- Distribution: widely grown throughout India and in other tropical countries as an irrigated fodder grass. In Tamil Nadu it is the most important perennial fodder, grown around dairy centres in Coimbatore, Erode, Tiruppur, Salem, Namakkal, Madurai, Dindigul and the delta.
- Importance: Cumbu Napier is a perennial, multi-cut fodder grass that gives 5-6 cuts a year for 3-4 years from one planting. It gives the highest green fodder yield per unit area of any fodder crop (about 200-300 t/ha per year under good management), is palatable and nutritious, and is the backbone of intensive dairy farming. It is succulent and can also be used for silage.
- Nutritional value: green Cumbu Napier contains about 15-20 per cent dry matter and 8-10 per cent crude protein (on a dry-matter basis), with good digestibility when harvested young. It is an energy-rich fodder.

Varieties

- CO 3, CO 4, CO 5 - TNAU Cumbu Napier hybrids of high yield and quality; CO 5 is a popular high-yielding multi-cut hybrid.
- BN 2 - a popular Cumbu Napier hybrid (from the national programme).
- IGFRI 3 (from the Indian Grassland and Fodder Research Institute) and APBN 1 (from Andhra Pradesh) - high-yielding hybrids.
- Suguna and Sampoorna - high-yielding hybrids with good quality and, in some cases, tolerance to pests/diseases.

Agronomy

- Land preparation: plough and harrow to a fine tilth; form ridges and furrows. The crop is planted once and stands for several years, so prepare the field deeply and enrich it with organic matter.
- Planting material and rate: planted with stem cuttings (setts) or rooted slips, about 20,000-30,000 slips per hectare. Each sett has 2-3 nodes.

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- Spacing: 60 x 45 cm or 75 x 45 cm; plant the setts in furrows and cover them.
- Planting time: with the onset of the monsoon (June-July) or under irrigation at any time when water is assured; planting in the rainy season gives the best establishment.
- Nutrient management: fodder grasses need very high nitrogen. Apply about 150-200 : 50 : 50 kg N : P2O5 : K2O per hectare per year. Apply half the N and all P and K as basal at planting, and top-dress the remaining N in equal splits after each cut. Apply FYM 10-20 t/ha at planting.
- Water management: irrigate at about 7-10 day intervals; a cut of green fodder removes a large amount of water and nutrients, so regular irrigation is essential. Avoid waterlogging.
- Weed management: keep the crop weed-free during establishment (the first 60-90 days); after establishment the dense clumps suppress weeds. One or two hand weedings or intercultivation between rows.
- Harvest and yield: take the first cut at about 60-75 days after planting, then cuts at about 45-60 day intervals (5-6 cuts a year). Harvest when the crop is about 1-1.5 m tall and before flowering for the best quality. Green fodder yield is about 200-300 t/ha per year under good management.
- Pest and disease management: the crop is relatively free of serious pests; occasional stem borers, aphids, leaf diseases (leaf spot, rust) and root rot occur. Management: clean cultivation, balanced fertiliser and need-based control. The crop is ratooned or replanted after 3-4 years when yields decline.

17.2 Guinea grass

Origin, distribution and importance

- Common name: Guinea grass, Hamil grass, Guinea pull. Botanical name: Megathyrsus maximus (Jacq.) B. K. Simon & S. W. L. Jacobs (syn. Panicum maximum Jacq.). Family: Poaceae.
- Origin: Africa (tropical Africa is the centre of origin and diversity). It spread to the tropics worldwide.
- Distribution: grown throughout the tropical and subtropical world; in India and Tamil Nadu it is a standard irrigated pasture and cut-and-carry fodder grass.
- Importance: Guinea grass is a perennial, multi-cut fodder grass that gives several cuts a year for several years. It is leafy, palatable and nutritious, gives a high green fodder yield (about 150-200 t/ha per year), and is used for cut-and-carry feeding and for pasture. It is easy to establish from seed or slips and is shade-tolerant.
- Nutritional value: green Guinea grass contains about 18-22 per cent dry matter and 8-11 per cent crude protein (on a dry-matter basis); it is a good-quality fodder when harvested young.

Varieties

- CO 1, CO 2 - TNAU Guinea grass varieties of high yield and quality.
- CO(Gg) 3 - a TNAU Guinea grass variety.
- IGFRI 2 (from the Indian Grassland and Fodder Research Institute) and Hamil - popular varieties grown in India.

Agronomy

- Land preparation: plough and harrow to a fine tilth; form ridges and furrows; enrich the soil with organic matter as the crop stands for several years.
- Planting material and rate: about 4-5 kg/ha of seed (the seed is light; use a carrier), or about 20,000-30,000 rooted slips per hectare.
- Spacing: 45 x 30 cm for line planting of slips or seed; for seed sowing, sow in lines and thin to the stand.
- Planting time: with the onset of the monsoon (June-July) or under irrigation when water is assured.

- Nutrient management: apply about 100-150 : 50 : 50 kg N : P2O5 : K2O per hectare per year, with half the N and all P and K as basal and the rest top-dressed after each cut. Apply FYM 10 t/ha at planting.
- Water management: irrigate at about 7-10 day intervals; regular irrigation maintains high yields. Avoid waterlogging.
- Weed management: keep the crop weed-free during establishment; afterwards the grass competes well with weeds.
- Harvest and yield: take the first cut at about 60-75 days, then cuts at about 30-45 day intervals; harvest before flowering for good quality. Green fodder yield is about 150-200 t/ha per year. The crop lasts several years; rejuvenate by intercultivation and fertiliser.
- Pest and disease management: occasional grasshoppers, aphids, leaf spots and rust; manage by clean cultivation and balanced fertiliser.

17.3 Comparison of Cumbu Napier and Guinea grass

Feature	Cumbu Napier hybrid	Guinea grass
Botanical name	Pennisetum glaucum x P. purpureum	Megathyrsus maximus
Family	Poaceae	Poaceae
Origin	Hybrid of two African species	Tropical Africa
Nature	Perennial, multi-cut grass	Perennial, multi-cut grass
Planting material	Stem cuttings/slips	Seed or rooted slips
Rate	20,000-30,000 slips/ha	4-5 kg seed or 20,000-30,000 slips/ha
Spacing	60-75 x 45 cm	45 x 30 cm
NPK (per year)	150-200 : 50 : 50	100-150 : 50 : 50
Cuts per year	5-6	4-6
Green fodder yield	200-300 t/ha/year	150-200 t/ha/year
Longevity	3-4 years	Several years

PART B - RICE ECOSYSTEMS, CLIMATE, WEATHER, SEASONS AND VARIETIES OF TAMIL NADU

17.4 Rice ecosystems

Rice is grown in a wider range of environments than any other cereal - from dry uplands to deep water. Rice ecosystems are classified by the water regime and the topography. Recognising the ecosystem explains why the same crop is managed so differently in different places.

Classification of rice ecosystems

- Irrigated lowland (irrigated rice): the largest and most productive ecosystem. Water is assured by canals, tanks or wells, and the crop is grown under controlled standing water. Yields are the highest (5-8 t/ha). Most of Tamil Nadu's rice (the Cauvery delta, tank-fed and well-irrigated areas) is irrigated lowland rice.
- Rain-fed lowland: water comes only from rainfall and the crop is grown in low-lying fields where water accumulates. The crop may face both drought and submergence. Yields are moderate (2-3.5 t/ha).
- Rain-fed upland: rice is grown on well-drained, higher land without standing water (aerobic rice). The crop depends entirely on rainfall and is subject to drought. Yields are low (1-2 t/ha). The Kar and rain-fed crops of some tracts are of this type.

- Deepwater and floating rice: grown where the water depth exceeds about 50 cm, sometimes up to 3-5 m, during the season. Traditional tall, floating varieties are used. Yields are low. This ecosystem is not important in Tamil Nadu.
- Tidal wetland (coastal saline rice): rice grown in coastal areas affected by tides and salinity. Salt-tolerant varieties (for example, CO 43 in Tamil Nadu) are grown. The coastal rice of Nagapattinam and Ramanathapuram is of this type.

Rice ecosystem - text diagram

```
Rice ecosystems by water regime and topography:
  UPLAND (no standing water, aerobic, rain-fed)      -> low yield, drought risk
     |
  LOWLAND (standing water in low-lying fields)
     |--- Irrigated lowland (assured water, controlled depth) -> highest yield
     |--- Rain-fed lowland (rain-fed, drought + submergence)  -> moderate yield
     |
  DEEPWATER / FLOATING (water 50 cm to several metres)      -> traditional tall types
     |
  TIDAL WETLAND (coastal, saline, tide-affected)            -> salt-tolerant varieties
  Water depth increases downwards; yield potential is highest in irrigated lowland.
```

17.5 Climate of Tamil Nadu

- Tamil Nadu lies in the tropical monsoon climatic zone of southern India. The State has a tropical climate with high temperature and moderate rainfall, modified by altitude in the hill areas (the Nilgiris, the Western Ghats) and by the sea in the coastal districts.
- Temperature: the plains have a mean temperature of about 28-30 degrees C; summer (March-May) is hottest (often 35-40 degrees C in the interior, such as Vellore, Tiruchirappalli and Madurai); winter (December-January) is mild (about 20-25 degrees C). The hill stations are much cooler.
- Rainfall: the average annual rainfall of Tamil Nadu is about 900-1000 mm, but it varies widely (from about 600 mm in the dry interior to over 2000 mm in the hills). The State depends mainly on the north-east monsoon (October-December), which brings about 60 per cent of the annual rainfall; the south-west monsoon (June-September) brings less rain to most of the State but more to the Western Ghats. This is unlike most of India, where the south-west monsoon dominates.
- Humidity: high in the coastal districts (above 70-80 per cent) and lower in the interior.

17.6 Weather and its elements

- Weather is the state of the atmosphere at a place at a given time; climate is the average weather over a long period (usually 30 years or more). Weather elements include rainfall, temperature (maximum, minimum, mean), humidity, wind (speed and direction), sunshine (bright sunshine hours) and evaporation.
- These elements affect crop growth: temperature controls germination, growth and flowering; rainfall and irrigation supply water; humidity affects transpiration and disease; sunshine drives photosynthesis; wind affects pollination, transpiration and lodging; evaporation determines the irrigation requirement.
- Weather-based crop management (for example, adjusting sowing dates to rainfall onset, applying fertiliser before rain, and protecting the crop during heat waves or cyclones) is part of modern agronomy. Weather forecasting and advisories help farmers take timely decisions.

17.7 Seasons of Tamil Nadu

Tamil Nadu has a complex set of agricultural seasons, linked to the monsoon and to canal releases. The named seasons are used in official cropping programmes and in the delta.

Season (Tamil name)	Months	Description
Sornavari (Chithirai)	April-May	Early crop using the SW monsoon and canal water; delta and tank-fed
Kar (Vaikasi)	May-June	Rain-fed crop of the tank-fed and upland tracts
Kuruvai (Aani)	June-July	Short-duration delta crop with Cauvery water (ADT 43)
Samba (Aadi)	August	The main delta season; medium/long duration (White Ponni, BPT 5204)
Thaladi (Purattasi)	September-October	Late crop after Kuruvai; nursery in the Samba field
Late Samba	October-November	Long-duration Samba harvested the next year
Navarai (Margazhi)	December-January	Winter/spring crop with tank and well water

The delta's classic sequence is Kuruvai (short rice) -> Thaladi (medium rice) -> rice-fallow pulse (black gram or green gram). This is the most intensive cropping pattern of the State. The main rainfall season for Tamil Nadu is the north-east monsoon (October-December).

17.8 Varieties of Tamil Nadu (rice)

- Short duration (about 105-120 days): ADT 43, ADT 45, ADT 48, ADT 49, ADT 50, ADT 51, ADT 52, ASD 16, ASD 18, ASD 19, ASD 20, CO 47, CO 48, CO 50, CO 51, TKM 9/11/13, TPS 3/5.
- Medium duration (about 120-135 days): ADT 38, ADT 39, ADT 46 (BPH tolerant), ADT 47, ASD 17, CO 49, and others.
- Long duration (above 135 days): Improved White Ponni, White Ponni, BPT 5204 (Samba Mahsuri), CR 1009, CO 43 (salt tolerant, about 135 days).
- Special-purpose varieties: CO 43 (salt/alkali tolerance), ADT 46 (BPH tolerance), ASD 16 (blast tolerance), CO 51 (drought tolerance).

17.9 Worked example - matching variety to season and ecosystem

Problem: A farmer in the Cauvery delta has assured Cauvery water in June (Kuruvai) and again in August (Samba), and a small plot of coastal saline soil. Suggest a suitable variety for each situation and give the reason.

- Kuruvai (June, short window before the Samba crop): choose a short-duration variety such as ADT 43 (about 110-115 days) or ADT 45 (about 105-110 days), so the crop matures before the Samba transplanting and the canal turn ends.
- Samba (August, main season with a longer window): choose a medium or long duration variety such as Improved White Ponni or BPT 5204 (about 135-140 days) for the premium market, or a medium-duration variety (ADT 39, ADT 47) for earlier harvest.
- Coastal saline plot: choose CO 43, which is tolerant to salinity and alkalinity, and manage with gypsum and leaching.
- Rain-fed/upland plot: choose a short-duration, drought-tolerant variety such as CO 51 or a rain-fed-adapted type (TKM series).
- The principle: match the variety's duration to the length of the water window, and match special tolerances (salt, pest, drought) to the problem of the field.

17.11 Cropping systems of Tamil Nadu and weather-based advisories

Major cropping systems of Tamil Nadu

Region / water source	Typical cropping system
Cauvery delta (assured water)	Kuruvai rice - Thaladi rice - rice-fallow pulse
Delta (limited water)	Rice (Samba) - rice-fallow pulse - dry crop
Tank-fed / rain-fed upland	Rice - groundnut / pulses; millet - pulse
Garden land (well irrigated)	Rice - vegetables; banana - pulses; maize - vegetables
Dry tracts (Ramanathapuram, Virudhunagar)	Cotton, millets, pulses, groundnut, castor; Cenchrus pasture
Hilly tracts (Nilgiris)	Finger millet, potato, vegetables, tea

Weather-based agro-advisories

- Weather-based advisories tell farmers when to sow, when to apply fertiliser and when to protect the crop, based on the forecast. For example, sowing is advised with the onset of the monsoon; fertiliser is applied before forecast rain; and sprays are avoided before rain.
- The main weather risks in Tamil Nadu are the delayed onset or failure of the monsoon, cyclones and heavy rain in the north-east monsoon season (which cause flooding and crop damage), and heat waves in summer. Drought and flood management, and the choice of short-duration and stress-tolerant varieties, are the main responses.
- District-level agro-advisories (from TNAU and the India Meteorological Department) and mobile-based advisories help farmers take timely decisions.

Field calendar and common problems

Crop calendar (operations schedule)

Operation	Time	Remarks
Cumbu Napier planting	Monsoon	20,000-30,000 slips/ha at 60-75 x 45 cm
Guinea grass planting	Monsoon	4-5 kg seed or 20,000-30,000 slips/ha
Basal fertiliser	At planting	Half N, full P and K
Top-dress N	After each cut	Remaining N in splits
First cut	60-75 days	Then every 30-60 days
Rejuvenation	Every 3-4 years	Intercultivate; replant if yield falls
Rice season planning	Per season	Match variety duration to the water window

Common field problems, causes and remedies

Problem	Likely cause	Remedy
Declining grass yield	Ageing crop; low N; soil compaction	Intercultivate; top-dress N; replant after 3-4 years
Waterlogging in grass	Poor drainage	Form ridges and furrows; drain
Wrong rice variety for the season	Duration not matched to the water window	Choose short/medium/long as per the season
Crop failure in a dry spell	Rain-fed upland; no irrigation	Drought-tolerant variety; moisture conservation
Flood damage in NE monsoon	Heavy rain; poor drainage	Drain; choose submergence-tolerant types

Objective-type practice questions

Choose the correct answer. Answers are given at the end of the set.

1. Cumbu Napier hybrid belongs to the family?
2. Cumbu Napier is a hybrid of?
3. Guinea grass is botanically?
4. The origin of Guinea grass is?
5. The highest-yielding perennial fodder grass is?
6. Rice ecosystem with the highest yield potential is?
7. Aerobic rice is grown?
8. Tamil Nadu's main rainfall comes from the?
9. The average annual rainfall of Tamil Nadu is about?
10. Salt-tolerant rice for coastal areas is?

Answers: 1-Poaceae; 2-pearl millet x napier grass; 3-Megathyrsus maximus; 4-tropical Africa; 5-Cumbu Napier hybrid; 6-irrigated lowland; 7-without standing water (upland-like); 8-north-east monsoon; 9-900-1000 mm; 10-CO 43

17.10 Exam focus

Key points to remember

- Cumbu Napier hybrid: Pennisetum glaucum x P. purpureum, family Poaceae; a perennial multi-cut grass; varieties CO 3, CO 4, CO 5, BN 2, IGFRI 3, APBN 1; planted with 20,000-30,000 slips/ha; NPK 150-200:50:50 per year; green fodder 200-300 t/ha/year.
- Guinea grass: Megathyrsus maximus, family Poaceae, origin tropical Africa; a perennial multi-cut grass; varieties CO 1, CO 2, CO(Gg) 3, IGFRI 2; seed 4-5 kg/ha or slips; NPK 100-150:50:50; green fodder 150-200 t/ha/year.
- Both are Poaceae fodder grasses that give several cuts a year for several years and need high nitrogen and regular irrigation.
- Rice ecosystems: irrigated lowland (highest yield), rain-fed lowland, rain-fed upland (aerobic), deepwater/floating, and tidal wetland (coastal saline).
- Tamil Nadu climate: tropical monsoon; average rainfall about 900-1000 mm, of which about 60 per cent comes from the north-east monsoon (October-December).
- Weather elements: rainfall, temperature, humidity, wind, sunshine and evaporation; climate is the average weather over a long period.
- Tamil Nadu seasons: Sornavari, Kar, Kuruvai, Samba, Thaladi, Late Samba and Navarai; the delta sequence is Kuruvai-Thaladi-rice-fallow pulse.
- Rice varieties by duration: short (ADT 43, ADT 45, ASD 16), medium (ADT 39, ADT 46), long (Improved White Ponni, BPT 5204), with special-purpose types (CO 43 for salt, ADT 46 for BPH).

Short-answer questions with model answers

1. Q1. Classify the rice ecosystems of the world.

- Rice ecosystems are classified by the water regime and topography into: irrigated lowland (assured water, highest yield); rain-fed lowland (rain-fed, drought and submergence risk); rain-fed upland (aerobic, no standing water, low yield); deepwater/floating (water deeper than about 50 cm, traditional tall types); and tidal wetland (coastal, saline, tide-affected, salt-tolerant varieties).

1. Q2. Why does Tamil Nadu depend mainly on the north-east monsoon?

- Tamil Nadu lies in the rain-shadow of the Western Ghats during the south-west monsoon (June-September), so it receives less rain then. The north-east monsoon (October-December) brings moist air from the Bay of Bengal across the State and supplies about 60 per cent of the annual rainfall, so the State depends mainly on it. The main rice seasons (Samba, Thaladi) are aligned with this monsoon.

1 Q3. Give the botanical names of Cumbu Napier hybrid and Guinea grass.

- Cumbu Napier hybrid: Pennisetum glaucum x Pennisetum purpureum. Guinea grass: Megathyrsus maximus (syn. Panicum maximum). Both belong to the family Poaceae.

Long-answer questions with model-answer outlines

1 Q1. Describe the perennial multi-cut fodder grasses of Tamil Nadu and their management.

- Outline: (i) the need for a year-round fodder base; (ii) Cumbu Napier hybrid - origin (hybrid of pearl millet and napier grass), varieties, planting material and rate, spacing, NPK 150-200:50:50, cuts and yield 200-300 t/ha/year; (iii) Guinea grass - origin (tropical Africa), varieties, seed/slip planting, NPK 100-150:50:50, cuts and yield 150-200 t/ha/year; (iv) common management - land preparation, irrigation, weed control during establishment, top-dressing after each cut, longevity and rejuvenation; (v) a comparison table; (vi) the role of legume association and mixed fodder systems.

1 Q2. Explain the rice ecosystems and the climate, weather and seasons of Tamil Nadu.

- Outline: (i) classification of rice ecosystems with the water regime and yield potential of each; (ii) the ecosystem diagram; (iii) the climate of Tamil Nadu - tropical monsoon, temperature range, rainfall (about 900-1000 mm) and the dominant north-east monsoon; (iv) weather and its elements, and the difference between weather and climate; (v) the seasons of Tamil Nadu with the table; (vi) the delta cropping sequence (Kuruvai-Thaladi-pulse); (vii) matching varieties to seasons and ecosystems with the worked example; (viii) conclusion on the role of climate and season in crop planning.

Appendix I - Glossary of Terms

Term	Meaning
Agronomy	The science and practice of crop production and soil management; the branch of agriculture dealing with field crop production.
Kharif	The monsoon season crop, sown with the onset of the south-west monsoon (June-July) and harvested in autumn.
Rabi	The winter/post-monsoon season crop, sown from October and harvested in spring.
Samba	The main rice season of the Tamil Nadu delta, sown in August with medium and long duration varieties.
Kuruvai	The short-duration delta rice season of June-July, grown with Cauvery water.
Thaladi	The late delta rice season of September-October, following the Kuruvai crop.
Navarai	The winter/spring rice season of December-January, grown with tank and well water.
Rice-fallow pulse	A short-duration pulse (black gram/green gram) grown on residual moisture after the rice harvest, with little or no irrigation.
Puddling	Churning of the flooded soil for transplanted rice to destroy structure, create a plough pan and reduce percolation.
Plough pan	A compacted layer below the plough depth that reduces water percolation in paddy fields.
Nursery	A small bed where rice seedlings are raised before transplanting to the main field.

Term	Meaning
Tillering	Production of side shoots (tillers) from the base of a cereal plant; a key yield component.
Panicle	The branched seed head of rice and other cereals (also called ear or head in other crops).
Dead heart	Drying of the central shoot of a tiller or plant caused by stem borer larvae, pullable easily.
White ear head	An empty, white panicle caused by stem borer damage below the panicle after flowering.
Hopper burn	Circular patches of dried, brown plants in rice caused by heavy brown plant hopper attack.
Khaira	Zinc deficiency in rice, marked by rusty brown spots and stunted growth.
SRI	System of Rice Intensification - young seedlings, single seedling per hill, wide spacing, alternate wetting and drying, mechanical weeding.
Heterosis	Hybrid vigour; the superior performance of an F1 hybrid over its parents.
CMS line (A line)	Cytoplasmic male sterile line, the female parent in hybrid rice seed production.
Maintainer (B line)	The line that maintains the sterility of the A line when crossed with it.
Restorer (R line)	The line that restores fertility in the F1 hybrid in hybrid rice.
Pegging	The downward growth of the groundnut gynophore (peg) into the soil, where the pod develops.
Gynophore	The stalk-like peg of groundnut that carries the fertilised ovule into the soil.
Gypsum	Calcium sulphate; applied to groundnut at pegging to supply calcium and sulphur.
Retting	Soaking of jute stems in water so that bacteria dissolve the pectin and the fibre separates.
Ginning outturn	The percentage of lint obtained from seed cotton (kapas) after ginning; about 33-35 per cent in hirsutum cotton.
Lint	The cotton fibre; the commercially valuable part of the cotton boll.
Kapas	Unginned seed cotton.
Gossypol	A toxic pigment in cotton seed glands; the meal must be detoxified before feeding.
Ricin	A toxic protein in castor seed; makes raw castor cake unsafe as feed.
Ricinoleic acid	The unique fatty acid of castor oil (about 85-90 per cent), giving it industrial value.
HCN (prussic acid)	Hydrocyanic acid; a poison that may be present in young or stressed fodder sorghum.
Silage	Green fodder conserved by controlled fermentation (usually in a pit or silo).
Hay	Green fodder conserved by drying.
Multi-cut fodder	A perennial or ratoon fodder crop that gives several harvests a year (for example, Cumbu Napier, Guinea grass).
Cenchrus	Buffel grass (Kollukattai pull); a drought-hardy perennial pasture grass.
Rhizobium	A bacterium that fixes atmospheric nitrogen in the root nodules of legumes; used as a seed inoculant.
Azospirillum	A free-living bacterium that fixes nitrogen and is used as a bio-fertiliser.
Phosphobacteria	Bacteria that solubilise soil phosphorus and are used as a bio-fertiliser.

Term	Meaning
MYMV	Mung bean yellow mosaic virus, transmitted by the whitefly; a major disease of green gram and black gram.
Phyllody	A mycoplasma disease of sesame in which floral parts become leafy and no seed forms.
Tikka leaf spot	The most important leaf disease of groundnut, caused by Cercospora species.
Bud necrosis	A serious viral disease of groundnut transmitted by thrips.
Sterility mosaic	A viral disease of pigeon pea transmitted by the eriophyid mite; causes sterile, bushy plants.
Wilt	A disease (often Fusarium) causing sudden wilting and death of the plant.
IPM	Integrated Pest Management - combining resistant varieties, cultural, mechanical, biological and need-based chemical control.
IWM	Integrated Weed Management - combining cultural, mechanical and need-based chemical weed control.
ETL	Economic Threshold Level - the pest population at which control measures become economically justified.
AWD	Alternate Wetting and Drying - a water-saving rice irrigation practice.
LCC	Leaf Colour Chart - a tool to decide the need for nitrogen top-dressing in rice.
SSNM	Site-Specific Nutrient Management - matching fertiliser to the yield target and soil supply.
Yield gap	The difference between the potential (or attainable) yield and the actual yield obtained by farmers.
C4 plant	A plant using the C4 photosynthetic pathway, with high water-use efficiency and heat tolerance (maize, sorghum, millets).
Photo-insensitive	A plant whose flowering does not depend on day length; can be grown in several seasons.
Test weight	The weight of 1000 grains (or seeds); used to compute seed rate.
1000-grain weight	Same as test weight.

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Appendix II - Useful Formulas

- Plant population per hectare = 10000 (m2) / (row spacing in m x plant spacing in m).
- Seed rate (kg/ha) = (plant population x 1000-grain weight in g) / (germination fraction x field-survival fraction x 1,000,000). Add a practical margin for the recommended rate.
- Fertiliser quantity (kg) = nutrient dose (kg/ha) x area (ha) / nutrient fraction in the fertiliser.
- Nutrient fractions: urea 0.46 N; single super phosphate (SSP) 0.16 P_2O_5; muriate of potash (MOP) 0.60 K_2O; di-ammonium phosphate (DAP) 0.46 N and 0.46 P_2O_5; rock phosphate about 0.20 P_2O_5.
- Gross return = yield x price per unit.
- Net return = gross return - cost of cultivation.
- Benefit-cost ratio = gross return / cost of cultivation.
- Yield gap = potential yield - actual yield.
- Yield gap index (per cent) = (yield gap / potential yield) x 100.
- Rice equivalent yield of a pulse = pulse yield x (price of pulse / price of rice).
- Ginning outturn (per cent) = (lint weight / seed cotton weight) x 100.
- Shelling percentage (groundnut) = (kernel weight / pod weight) x 100.

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Appendix III - Abbreviations

Abbreviation	Full form
NPK	Nitrogen, Phosphorus (P2O5), Potassium (K2O)
FYM	Farm Yard Manure
DAP	Di-Ammonium Phosphate
MOP	Muriate of Potash
SSP	Single Super Phosphate
TNAU	Tamil Nadu Agricultural University
ICAR	Indian Council of Agricultural Research
IARI	Indian Agricultural Research Institute
ICRISAT	International Crops Research Institute for the Semi-Arid Tropics
IRRI	International Rice Research Institute
IGFRI	Indian Grassland and Fodder Research Institute
CAZRI	Central Arid Zone Research Institute
BPH	Brown Plant Hopper
MYMV	Mung bean Yellow Mosaic Virus
QPM	Quality Protein Maize
Bt	Bacillus thuringiensis (insect-resistant transgenic cotton)
CMS	Cytoplasmic Male Sterile
GA3	Gibberellic Acid
SRI	System of Rice Intensification
AWD	Alternate Wetting and Drying
IPM	Integrated Pest Management
IWM	Integrated Weed Management
ETL	Economic Threshold Level
t/ha	tonnes per hectare
kg/ha	kilograms per hectare

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Appendix IV - Consolidated Package of Practices (All Crops)

Crop	Seed rate	Spacing	NPK (kg/ha)	Duration	Yield (t/ha)	Key pest
Rice	40-60 kg/ha	20 x 10 cm	150:50:50	105-140 d	5-6 (irrig.)	BPH, stem borer
Maize	20-25 kg/ha	60 x 20 cm	250:75:75	85-110 d	6-9	Fall armyworm
Sorghum	10 kg/ha	45 x 15 cm	100:50:50	105-110 d	3-4	Shoot fly
Pearl millet	4 kg/ha	45 x 15 cm	80:40:20	85-90 d	2-2.5	Shoot fly
Finger millet	10-20 kg/ha	22.5 x 10 cm	60:30:30	100-115 d	2.5-3	Blast
Pigeon pea	15 kg/ha	60 x 30 cm	25:50:25	120-180 d	1.0-1.5	Pod borer
Green gram	15-20 kg/ha	30 x 10 cm	25:50:25	60-75 d	0.8-1.2	Whitefly (MYMV)
Black gram	15-20 kg/ha	30 x 10 cm	25:50:25	60-70 d	0.7-1.0	Whitefly (MYMV)
Cowpea	20-25 kg/ha	45 x 15 cm	25:50:25	70-90 d	0.8-1.2	Aphids
Groundnut	100 kg kernels/ha	30 x 10 cm	25:50:75	100-140 d	2-3	Leaf miner
Sesame	4-5 kg/ha	30 x 15 cm	35:23:23	80-90 d	0.4-0.6	Leaf webber
Soybean	70-75 kg/ha	45 x 5 cm	30:60:40	85-100 d	2-2.5	Girdle beetle
Castor	10-12 kg/ha	90 x 60 cm	40:50:30	150-180 d	1.0-1.5	Semilooper
Cotton	2.5-3 kg/ha (delinted)	60 x 30 cm	80:40:40	150-180 d	1.5-2 (kapas)	Pink bollworm
Jute	5-6 kg/ha	30 x 10 cm	60:30:30	100-120 d	2-3 (fibre)	Stem weevil
Fodder sorghum	25-30 kg/ha	30 x 10 cm	100-150:50:0-50	60-75 d	40-60 (green)	Shoot fly
Fodder cowpea	25-30 kg/ha	30 x 10 cm	25-50:50:0	50-65 d	20-30 (green)	Aphids
Fodder maize	40-50 kg/ha	30 x 15 cm	100-150:50-60:40	60-75 d	40-60 (green)	Fall armyworm
Cenchrus	4-5 kg/ha	45 x 30 cm	40:20:0	perennial	20-40/yr (green)	Minor
Cumbu Napier	20,000-30,000 slips/ha	60-75 x 45 cm	150-200:50:50	perennial	200-300/yr (green)	Minor
Guinea grass	4-5 kg/ha or slips	45 x 30 cm	100-150:50:50	perennial	150-200/yr (green)	Minor

The figures are the standard textbook and TNAU ranges used in this material; rain-fed yields are lower and actual results vary with soil, season and management. Follow the latest TNAU Crop Production Guide for field use.

Appendix V - Model End-Semester Question Paper

The following paper follows the usual end-semester pattern for AGR 201 and covers all seventeen lectures.

Section A - Short answers (2 marks each, answer any ten)

1. Give the botanical name and family of rice.
2. State the six principles of SRI.
3. Name the three lines used in hybrid rice seed production.
4. Give the seed rate and spacing for irrigated hybrid maize.
5. Which millet is the richest source of calcium and by how much?
6. Name two diseases of pigeon pea.
7. Define rice-fallow pulses and name two crops.
8. Why is gypsum applied to groundnut?
9. What is phyllody in sesame?
10. What is ricinoleic acid and why is it important?
11. Name the four cultivated species of cotton.
12. What is retting in jute?
13. What is HCN poisoning and in which fodder crop does it occur?
14. Give the botanical name of Cenchrus and one special feature.
15. Name the main rainfall season of Tamil Nadu.

Section B - Short notes (5 marks each, answer any five)

1. System of Rice Intensification.
2. Hybrid rice seed production.
3. Millets as nutri-cereals.
4. Rice fallow pulse agronomy.
5. Integrated Pest Management in cotton.
6. Fodder cropping systems of Tamil Nadu.
7. Rice ecosystems of the world.
8. Cropping systems of the Cauvery delta.

Section C - Long answers (10 marks each, answer any three)

1. Describe the origin, importance and complete package of practices of rice.
2. Write the package of practices for irrigated hybrid maize with a numerical on fertiliser for a given area.
3. Compare sorghum and pearl millet under origin, botany, agronomy and yield.
4. Explain the agronomy of the kharif pulses with a comparison table.
5. Describe the origin, importance and cultural practices of groundnut, with the economics.
6. Explain the SRI method with its principles, layout and a worked example.
7. Describe the perennial multi-cut fodder grasses of Tamil Nadu and their management.
8. Explain the rice ecosystems and the climate, weather and seasons of Tamil Nadu.

Section D - Numericals (compulsory, 10 marks)

1 A farmer grows rice at 20 x 10 cm with 2 seedlings per hill. Given a 1000-grain weight of 22 g, 90 per cent germination and 90 per cent survival, calculate the plant population and the seed rate. Then compute the urea, SSP and MOP needed to apply 150:50:50 kg NPK per hectare on 1 hectare, and find the net return if the yield is 5.5 t/ha at Rs. 2,300 per quintal with a cost of cultivation of Rs. 46,000 per hectare.

Appendix VI - Quick Revision: One-line Facts

These one-line facts cover the whole syllabus and are useful for last-minute revision and for objective-type questions.

- Rice: Oryza sativa, Poaceae, 2n = 24; indica (tropics) and japonica (temperate) groups.
- Rice origin: Indo-Burma (indica) and China (japonica); Vavilov Indian and Chinese centres.
- India is the largest exporter and the second largest producer of rice in the world.
- Milled rice: about 78-80 per cent carbohydrate, 6-7 per cent protein, about 345 kcal per 100 g.
- Rice soil: clay to clay loam, pH 5.5-6.5 (tolerant 4.5-8.0); it is the most salt-tolerant cereal.
- Rice: optimum 30-32 C day, 20-22 C night; water need 1000-1500 mm; low temperature at flowering causes sterility.
- Standard rice NPK: 150:50:50 kg/ha, with nitrogen in three splits.
- Zinc deficiency in rice is called khaira; remedy is ZnSO4 25 kg/ha.
- Hopper burn is caused by the brown plant hopper; dead heart and white ear head by the stem borer.
- Tamil Nadu rice seasons: Kuruvai, Samba, Thaladi, Navarai, Sornavari, Kar, Late Samba.
- The delta cropping sequence is Kuruvai rice - Thaladi rice - rice-fallow pulse.
- SRI: 8-12 day seedlings, one per hill, 25 x 25 cm, seed 5-8 kg/ha, alternate wetting and drying.
- Hybrid rice: F1 with about 15-20 per cent heterosis; A (CMS), B (maintainer) and R (restorer) lines.
- Maize: Zea mays, Poaceae, 2n = 20; origin Central America; ancestor teosinte; 'queen of cereals'.
- Maize grain: about 9-10 per cent protein, 4-5 per cent fat, about 342 kcal per 100 g.
- Maize NPK: 250:75:75 kg/ha; seed 20 kg/ha hybrid at 60 x 20 cm; critical stages tasselling and silking.
- Fall armyworm is the most serious recent pest of maize; turcicum leaf blight and downy mildew are major diseases.
- Sorghum: Sorghum bicolor, 2n = 20, origin Africa; dual purpose (grain + stover).
- Pearl millet: Pennisetum glaucum, 2n = 14, origin Africa (Sahel); the hardiest, most drought-tolerant cereal.
- Sorghum NPK 100:50:50 (irrigated); pearl millet NPK 80:40:20 (irrigated); both 40:20:0 rain-fed.
- Sorghum TNAU variety CO 30 (105-110 d); pearl millet TNAU variety CO 9 (85-90 d).
- Finger millet: Eleusine coracana, 2n = 36, origin Africa; the richest cereal in calcium (300-350 mg/100 g).
- Minor millets: kodo (Paspalum scrobiculatum), little (Panicum sumatrense), foxtail (Setaria italica), proso (Panicum miliaceum), barnyard (Echinochloa frumentacea).
- Foxtail millet has 2n = 18; barnyard millet has 2n = 54; finger millet 2n = 36.
- Millets are now promoted as nutri-cereals (gluten-free, low glycemic index, high fibre and minerals).
- Pigeon pea: Cajanus cajan, Fabaceae, 2n = 22, origin peninsular India; protein 20-22 per cent.
- Pigeon pea NPK 25:50:25; seed 15 kg/ha at 60 x 30 cm; major pest pod borer; major diseases wilt and sterility mosaic.
- Green gram: Vigna radiata; black gram: Vigna mungo; cowpea: Vigna unguiculata; all 2n = 22, Fabaceae.
- Green gram and black gram originate in India; cowpea in West Africa.
- MYMV (yellow mosaic) is the key disease of green gram and black gram, spread by the whitefly.
- Rice-fallow pulses: 60-65 day black gram/green gram on residual moisture; seed 25-30 kg/ha; yield 300-500 kg/ha.
- Groundnut: Arachis hypogaea, Fabaceae, 2n = 40, origin South America; kernel 44-50 per cent oil, 25-30 per cent protein.
- Groundnut NPK 25:50:75; gypsum 400 kg/ha at pegging for calcium; bunch and spreading types.

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- Tikka leaf spot is the most important groundnut disease; bud necrosis is transmitted by thrips.
- Sesame: Sesamum indicum, Pedaliaceae, 2n = 26, origin India; 'queen of oilseeds'; 45-50 per cent oil.
- Phyllody (mycoplasma) is the key sesame disease, spread by leaf hoppers.
- Soybean: Glycine max, Fabaceae, 2n = 40, origin China; about 40 per cent protein, 18-20 per cent oil.
- Castor: Ricinus communis, Euphorbiaceae, 2n = 20, origin Africa; oil rich in ricinoleic acid (85-90 per cent).
- Castor seed contains the toxin ricin; the semilooper is the major pest; wilt is the major disease.
- Cotton: Gossypium spp., Malvaceae; hirsutum and barbadense 2n = 52 (New World), arboreum and herbaceum 2n = 26 (Old World).
- India is the original home of the Asiatic cottons; cotton is the 'king of fibres'.
- Cotton NPK 80:40:40 (irrigated); ginning outturn 33-35 per cent; pink bollworm is the major pest.
- Jute: Corchorus olitorius (tossa) and C. capsularis (white), 2n = 14; the 'golden fibre'; fibre extracted by retting.
- Jute is harvested at the small-pod stage (100-120 days); fibre yield 2-3 t/ha.
- Fodder sorghum: Sorghum bicolor; fodder cowpea: Vigna unguiculata; harvested at 50 per cent flowering.
- HCN (prussic acid) poisoning occurs in fodder sorghum; feed after wilting/drying.
- Fodder maize: the best silage crop; harvest at the milk to early-dough stage; green fodder 40-60 t/ha.
- Cenchrus ciliaris (buffel grass, Kollukattai pull): a drought-hardy perennial pasture grass.
- Cumbu Napier hybrid (Pennisetum glaucum x P. purpureum): highest green fodder yield, 200-300 t/ha per year.
- Guinea grass (Megathyrsus maximus): a perennial multi-cut grass, origin tropical Africa, 150-200 t/ha per year.
- Rice ecosystems: irrigated lowland (highest yield), rain-fed lowland, rain-fed upland, deepwater/floating, tidal wetland.
- Tamil Nadu's main rainfall comes from the north-east monsoon (October-December), about 60 per cent of the annual total.
- Weather elements: rainfall, temperature, humidity, wind, sunshine and evaporation; climate is the average weather over about 30 years.
- Plant population per hectare = 10000 / (row spacing x plant spacing in metres).
- Fertiliser quantity = nutrient dose x area / nutrient fraction (urea 0.46 N; SSP 0.16 P2O5; MOP 0.60 K2O).
- Net return = gross return - cost of cultivation; B:C ratio = gross return / cost of cultivation.
- Yield gap = potential yield - actual yield; yield gap index = (yield gap / potential yield) x 100.

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Appendix VII - Practice Numericals with Solutions

Work through each problem before reading the solution. Numerical questions in AGR 201 are almost always drawn from these types.

Problem 1 - Seed rate and plant population (maize)

Hybrid maize at 60 x 20 cm; 1000-grain weight 250 g; germination 90 per cent; survival 90 per cent. Find the plant population and the seed rate.

- Plants/ha = 10000 / (0.60 x 0.20) = 83,333 plants/ha.
- Seed to place = 83,333 / (0.90 x 0.90) = 1,02,880 seeds/ha.
- Seed weight = 1,02,880 x 250 / 1000 = 25,720 g = about 25.7 kg/ha.
- The recommended rate (about 20 kg/ha) assumes better seed and closer management; the calculated value shows the biological requirement at the stated germination and survival.

Problem 2 - Fertiliser quantity (rice)

Apply 150:50:50 kg NPK per hectare to 0.4 hectare of rice. Find the urea, SSP and MOP required.

- For 0.4 ha: N = 60 kg; P2O5 = 20 kg; K2O = 20 kg.
- Urea = 60 / 0.46 = about 130 kg; SSP = 20 / 0.16 = 125 kg; MOP = 20 / 0.60 = about 33 kg.
- Split the urea: 1/3 basal (43 kg), 1/3 at tillering (43 kg), 1/3 at panicle initiation (44 kg).

Problem 3 - Cost of cultivation and net return (maize)

1 hectare of irrigated hybrid maize yields 7 t/ha at Rs. 2,200 per quintal; cost of cultivation Rs. 55,000. Find gross return, net return and B:C ratio.

- Gross return = 70 quintals x Rs. 2,200 = Rs. 1,54,000.
- Net return = Rs. 1,54,000 - Rs. 55,000 = Rs. 99,000 per hectare.
- B:C ratio = 1,54,000 / 55,000 = about 2.8.

Problem 4 - Yield gap

The potential yield of a groundnut variety is 3.0 t/ha and the farmer obtains 1.8 t/ha. Find the yield gap and the yield gap index.

- Yield gap = 3.0 - 1.8 = 1.2 t/ha.
- Yield gap index = (1.2 / 3.0) x 100 = 40 per cent.
- The gap is closed by a better variety, timely sowing, balanced fertiliser (including gypsum at pegging), correct water management and timely pest control.

Problem 5 - SRI spacing and plant population

SRI at 30 x 30 cm with one seedling per hill on 1 hectare. Find the number of hills and the plant population, and compare with 20 x 10 cm with 2 seedlings per hill.

- SRI hills/ha = 10000 / (0.30 x 0.30) = 1,11,111 hills/ha; plants = 1,11,111 (one per hill).
- Conventional hills/ha = 10000 / (0.20 x 0.10) = 5,00,000 hills; plants = 10,00,000.
- SRI uses far fewer hills and seedlings, but each hill tillers profusely (30-50 tillers), so the productive tiller number can equal or exceed that of the conventional crop.

Problem 6 - Fodder area for a herd

8 cows need 25 kg green fodder each per day. Fodder sorghum gives 50 t/ha in 70 days. Find the daily and 70-day requirement and the area per crop cycle.

- Daily need = 8 x 25 = 200 kg.
- Need for 70 days = 200 x 70 = 14,000 kg = 14 tonnes.
- Area = 14,000 / 50,000 = 0.28 hectare (about 70 cents) per 70-day cycle.
- With staggered sowing and a perennial grass base (Cumbu Napier), the herd can be fed green fodder all year.

Problem 7 - Ginning outturn and lint yield (cotton)

A farmer gets 1.8 t/ha of seed cotton (kapas) with a ginning outturn of 34 per cent. Find the lint yield and the seed yield (approximately).

- Lint = 1.8 x 0.34 = 0.612 t/ha = about 612 kg/ha of lint.
- Seed and waste = 1.8 - 0.612 = 1.188 t/ha; of this, the seed is roughly two-thirds and waste about one-third.
- Lint price is much higher than seed price, so a higher ginning outturn raises the value of the crop.

References

The following standard texts and official guides are the sources for the figures and practices in this material. Students should consult the latest editions, because recommendations (especially fertiliser doses, varieties and plant protection chemicals) are revised from time to time.

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- International Rice Research Institute (IRRI). Rice Knowledge Bank and Rice Production Manual.
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- Indian Grassland and Fodder Research Institute (IGFRI). Fodder crop production and pasture management publications.

The fertiliser doses, yields, costs and prices used in the worked examples are standard textbook ranges and indicative values for study; actual field figures vary with locality, season, soil and market prices. For field use, always follow the latest TNAU Crop Production Guide, the TNAU Plant Protection Guide and the pesticide label.